Conceptual and Geometrical Problems in Quantum Gravity

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ABSTRACT

A summary is given of some of the conceptual and geometrical problems that arise in various attempts to construct a quantum theory of gravity. Particular emphasis is placed on the canonical quantisation of general relativity and the problem of 'time'.

1. Introduction

1.1 Preamble

These notes are based on a course of lectures entitled "Conceptual and Geometrical Problems in Quantum Gravity" given at the 1991 Schladming Winter School in Theoretical Physics. The majority of students at the school had a background in elementary particle physics and this is reflected in the exposition in which aspects of the subject that derive from general relativity are more detailed than those that come from particle physics. Particular emphasis is placed on the canonical quantisation programme and the problems it engenders.

It must be emphasised that this is *not* meant to be a full-blown review of quantum gravity, not even of the conceptual and geometrical aspects of the subject. It is merely an account of some of the problems that arise in this area, and the treatment is necessarily cursory in a number of places. So far as references are concerned, I have

concentrated mainly on more recent papers and, although some of the "classics" in the field are cited, the list—both modern and ancient—is very incomplete. Most of the papers mentioned are ones that I happened to find useful when preparing the notes. My genuine apologies to authors who feel unduly neglected by this process.

1.2 What is the Problem of Quantum Gravity?

Techniques of differential geometry are widely used in modern theoretical physics and there are many areas about whose geometrical aspects one might write. However, quantum gravity differs from most other branches of physics by the way in which it raises severe problems of a conceptual nature concerning the status of our conventional ideas on space, time and matter. Furthermore, because of the specific mathematical structure of general relativity, these conceptual problems are frequently linked to the more technical question of the applicability of standard geometrical ideas to the quantum theory. It is therefore quite appropriate to discuss geometrical and conceptual issues within the framework of a single course of lectures.

But what is the problem of quantum gravity? And how significant are the conceptual problems which arise? There is no easy answer to these questions, and certainly no consensus among the physicists who work in the field. The minimal requirement for a successful theory is that it should reproduce conventional classical general relativity and normal quantum theory in their appropriate domains. The key quantities in this respect are the *Planck length* $L_P = (\frac{G\hbar}{c^3})^{\frac{1}{2}} \approx 10^{-35}m$, (corresponding to a Planck energy of $10^{28}eV$) and the associated *Planck time* $t_P = L_p/c \approx 10^{-42}s$, where G is Newton's constant and c is the speed of light. Dimensional arguments suggest that quantum gravity effects will become important at these scales and, conversely, that standard general relativity and quantum theory should apply at distances and times significantly greater than these minute quantities.

However, even if there is agreement about the physics at large distances and times, there is still a wide variety of opinions about what happens at the Planck length itself and, in particular, the extent to which the conceptual and structural frameworks of general relativity and quantum theory are still applicable.

The conservative view is that nothing special happens at the Planck length. The opposite, iconoclastic, view is that fundamental changes are needed leading to a radical

reassessment of the traditional ideas of space, time and quantum matter. For example, there is a recurrent idea that the use of *continuum* concepts (in particular, differential geometry) is inapplicable in this domain. Another example is Penrose's long-held view that quantum theory becomes *non-linear* at scales of the Planck length, and in such a way as to provide a natural mechanism for explaining the much-debated quantum mechanical problem of the reduction of the state vector (Penrose 1986).

One difficulty in making theoretical judgements on the nature of Planck length physics is that the agreed minimal requirement—that a theory of quantum gravity should reproduce classical general relativity at large scales—does not seem to place many restrictions on the overall theory. For example, within the context of the particle-physics approach to quantum gravity (see §3), there are a series of, more or less, rigorous theorems which show that *any* Lorentz-invariant theory of a spin-2 graviton coupled to a conserved energy-momentum tensor will necessarily yield the same low-energy scattering results as those obtained from the tree graphs of a weak-field perturbative expansion of the Einstein Lagrangian (*e.g.*, Gupta 1955, Feynman 1963, Weinberg 1964a,b 1965, Boulware and Deser 1975). [1] These theorems do not mean that it is easy to construct a quantum theory of gravity (it isn't!) but they do imply that not much can be learnt from the requirement that the low-energy limit must be general relativity. One result of this lack of *a priori* information about what is wanted of the theory is a notable absence of any comprehensive axiomatisation of the subject. In particular, there is nothing like the Wightman scheme, or the approaches based on C^*-algebras which have played such an important role in clarifying the content of normal quantum field theory. Indeed, it is by no means obvious that we would recognise a proper theory of quantum gravity even if it was presented to us on a revelatory plate!

1.3 Great Expectations

This uncertainty about the aims of quantum gravity has lead to a proliferation of approaches to the subject and of expectations of how the theory might be used. To a considerable extent these depend on the general techniques of the community to which a researcher belongs: in particular, there tend to be significant differences between the

[1] This result is often quoted, but there are loop holes. In particular, Wald (1986) demonstrated the existence of a consistent theory of a massless, interacting spin-2 field which is *not* generally covariant.

methods adopted by an elementary particle physicist (or quantum field theorist) and those employed by someone whose primary training has been in general relativity.

1.3.1 The Approach of a Particle Physicist

The emphasis here is mainly on techniques drawn from conventional, Poincaré-group based quantum field theory. Thus the key concept is *special* relativity and the idea of "gravitons" (quanta of the gravitational field) propagating in a fixed Minkowski spacetime. The minimal expectation of a quantum gravity theory (over and above the general requirements mentioned above) is that it should produce scattering amplitudes for gravitons and other particles that are free of irremovable divergences; thus the theory must be renormalisable or, better still, genuinely finite. The maximal expectation would be that such a theory is part of a general Grand Unified Theory with the gravitational sector playing an essential role.

The main technical approach in the early studies was an expansion of the spacetime metric tensor $g_{\mu\nu}(x)$ around the Minkowski metric $\eta_{\mu\nu}$ in the form $g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x)$ where $\kappa = (16\pi G/c^2)^{\frac{1}{2}}$ is the (dimensional) coupling constant. The Lorentz tensor field $h_{\mu\nu}(x)$ can be quantised using any of the standard techniques drawn from the relativistic quantum field theory of gauge systems. Such an expansion (to which we will return in §3) is anathema to most general relativists, but that has never upset the particle physics community to any noticeable degree!

More recently, the efforts of elementary particle physicists have been directed towards theories of superstrings and other extended objects. Most calculations in this area also employ the language of gravitons, although it should be emphasised that this is not an inevitable consequence of string theory: it is just that suitable non-perturbative methods have not yet been developed.

The main problem with the particle physics approach is how to do full justice to the concepts of space and time engendered by the classical theory of general relativity. For example, a perturbative expansion around any background metric [2] leaves open the question of the "real" causal structure of the theory; there is certainly no reason why it should be the same as that of the background space. Similarly, it is difficult to handle cosmological issues within such a framework. Simply using a standard classical

[2] In the modern background-field methods this need not just be Minkowski space.

cosmology (*e.g.*, the Robertson-Walker metric) as the background is not sufficient if one wishes to study, for example, the influence of quantum gravity on the initial spacetime singularity.

1.3.2 The Approach of a General Relativist

General relativists usually approach the problem of quantum gravity from a very different perspective, with the emphasis being placed on the geometrical features of the theory and the role of spacetime structure. In particular, there is a great reluctance to employ any quantisation scheme which is based on a special background like, for example, Minkowski spacetime. The assumption is that if such a space does have a special role, it should appear naturally as part of the structure of the theory itself and not just be put in by hand. The question of background structure is very important and is one to which we will return in §2.

A general relativist's minimal expectation of a quantum theory of gravity is likely to include the possibility of solving the problems posed by spacetime singularities, black holes, and similar features of the classical theory. In particular, one expects to be able to recover Hawking's famous results on the quantum production of particles by a classical black hole, and to extend them to answer questions concerning the back reaction of the created particles and the final state of the evaporating hole. A more ambitious goal is to apply the theory to cosmological issues and, in particular, to study the creation of the universe as a quantum theoretical process.

Various technical approaches have been employed over the years but there has always been a tendency to adopt a "back to basics" philosophy with the emphasis being placed on quantum theory in general rather than on the more specialised techniques of relativistic quantum field theory. The problems of particular interest are mainly those relating to the quantum status of the spacetime concepts of classical general relativity, and the extent to which conventional quantum theoretical ideas can be applied. In general, relativists tend to be more concerned with the conceptual issues which arise in quantum gravity than are their cousins who approach the subject from a background of elementary particle physics.

In addition to the sociological issues that generate such different approaches to the subject there is one question which, until it is answered satisfactorily, is guaranteed to maintain the uncertainty about what one is trying to do in quantum gravity:

> Is the central problem of quantum gravity one of (i) physics, (ii) mathematics, or (iii) philosophy?

In particular, how severe are the conceptual difficulties? Is it possible that one needs to get to grips with them *before* any serious technical development can be made?

This central question is considered in chapter 2 which lists some of the prima facie conceptual problems which arise in quantum gravity. Chapter 3 is a brief account of the particle physics approach, while in chapter 4 there is a discussion of the canonical quantisation of gravity which places more emphasis on the geometrical aspects of the subject. In chapter 5 we consider one of the central issues—the problem of time—and the final chapter is a short introduction to some of the special problems of quantum cosmology.

2. Prima Facie Problems in Quantum Gravity

2.1 General Remarks

In this chapter we will consider some of the prima facie problems that can arise in various general approaches to constructing a quantum theory of gravity. Some problems of this type stem from incompatibilities in the way the two theories are normally interpreted. Others are more technical in nature and arise when a structural feature of one of the theories is particularly difficult to maintain in the presence of the other.

One of the broadest problems of all is the extent to which a quantum theory of gravity can be expected to maintain:

1. the picture of spacetime afforded by general relativity;
2. the interpretative and structural frameworks of conventional quantum theory.

This question is central, but it is too broad to succumb to a frontal assault and is better served by a more circumspective approach. We will start therefore by considering two problems which, although very important, are more specific. They are

1. the role of the spacetime diffeomorphism group Diff($\mathcal{M}$);
2. the significance of background structure.

2.2 Spacetime Diffeomorphisms

The equations of general relativity are covariant with respect to the group Diff($\mathcal{M}$) of diffeomorphisms of the spacetime manifold $\mathcal{M}$. We will restrict our attention to diffeomorphisms with compact support, *i.e.,* those that are equal to the unit map outside some closed and bounded region of $\mathcal{M}$. Thus, for example, a Poincaré -group transformation of Minkowski spacetime is not deemed to belong to Diff($\mathcal{M}$). This restriction is imposed because the role of transformations with a non-trivial action in the asymptotic regions of $\mathcal{M}$ is quite different from those that act trivially.

It must be emphasised that Diff($\mathcal{M}$) means the group of active point transformations of $\mathcal{M}$. This should not be confused with the pseudo-group which describes the relations between overlapping pairs of coordinate charts (although of course there is a connection between them).

In some respects, the role of the diffeomorphism group in both classical and quantum general relativity is analogous to that of the gauge group in Yang-Mills theory. For example, in both cases the groups are associated with field variables that are non-dynamical.

However, in another, and very important, sense the two groups are quite different. Yang-Mills transformations occur at a fixed spacetime point whereas the diffeomorphism group moves points around. This is responsible for the famous Einstein "hole" problem (Stachel 1987, Earman and Norton 1986, Rovelli 1989) with its conclusion that invariance under Diff($\mathcal{M}$) means that the individual mathematical points in $\mathcal{M}$ have no intrinsic physical significance. This is closely related to the question of what constitutes an "observable" in general relativity—a surprisingly contentious issue that has generated much debate over the years and which is of particular interest in the context of quantum gravity.

For example, although the Riemann scalar curvature $R(x) := g^{\mu\nu}(x)R_{\mu\nu}(x)$ is a scalar function on $\mathcal{M}$, its value at any particular point $x \in \mathcal{M}$ can *not* be regarded as an observable. In the quantum case, this follows from the observation that if $U(f)$ is a unitary representation of $f \in \mathrm{Diff}(\mathcal{M})$, the action on the quantised metric field should lead to the transformation law

$$U(f)R(x)U(f)^{-1} = R\big(f^{-1}(x)\big) \tag{2.2.1}$$

provided that $R(x)$ can be defined as a proper operator function of $g_{\mu\nu}(x)$ and its derivatives. Thus if a physical observable is defined to be one that *commutes* with the action of the "gauge group" (as is the case in Yang-Mills theory), then $R(x)$ is evidently not such an object.

One way of constructing genuine invariants is to integrate scalar functions of $g_{\mu\nu}(x)$ over the entire spacetime such as

$$\int_{\mathcal{M}} R(x)\big(\det g(x)\big)^{\frac{1}{2}}\, d^4x; \quad \int_{\mathcal{M}} R^{\mu\nu}(x)R_{\mu\nu}(x)\big(\det g(x)\big)^{\frac{1}{2}}\, d^4x. \tag{2.2.2}$$

Note that these are highly *non-local* and therefore any quantisation scheme based on expressions of this type will be very different from a conventional quantum field theory. [3] Another, quite different, approach to the problem of constructing invariants has a long history in classical general relativity and has played an important role in much recent work on quantum gravity. The central idea is that although, for example, $R(x)$ is not an observable, the expression $R(X)$ *is* if X refers to the point on the manifold occupied by an actual physical particle. In other words, we must locate ourselves on the spacetime manifold with the aid of a "material reference system". Thus simple general relativity is incomplete as a physical theory because it ignores both the equations of motion of the reference system and the effects of its energy-momentum tensor on the gravitational field. This idea of using "physical" coordinates is very important and we will return to it in §4 and §5 in the contexts of the canonical quantisation of gravity and the problem of time.

[3] The nearest analogue would perhaps be a quantisation of Yang-Mills theory using the gauge-invariant, but non-local, Wilson loop variables as the basic quantum objects.

2.3 Background Structure

A key feature of any approach to quantum gravity is the background structure it assumes and the way it is used. This "background" can take various forms. Sometimes it involves a choice of a particular mathematical element in the theory. On other occasions it refers to the conceptual or interpretative framework that is assumed *a priori*.

An example of the former is any theory that employs a fixed manifold to represent space or spacetime, or in which a particular spacetime metric (*e.g.*, the Minkowski metric) is assigned a central role. [4] The conceptual background may include part, or all, of the interpretative structure of conventional quantum theory. This in turn may assume the existence of a background classical world within which measurements of observables are made. Indeed, it can be argued that the conventional Copenhagen interpretation of quantum theory presupposes as part of its background a *fixed* spacetime (in both the topological and the metrical sense) and that it is therefore intrinsically incompatible with the idea of quantum gravity! The existence of such a background is also frequently associated with a division of a part of the universe into a 'system' and an 'observer'—a split that is particularly difficult to maintain in the context of quantum cosmology.

There are several reasons why it is important to understand clearly the precise background structure that is assumed in any particular approach to quantum gravity. For example, in the case of a background manifold or metric it can be argued that an object of this type constitutes a special choice for what could, or should, be determined dynamically. This is closely related to what Smolin (1991) calls an "ideal" element. At best, such a situation is likely to yield only an approximation to the complete theory, and it could be extremely misleading.

A more insidious problem arises if background structure exists that is not consciously recognised as such. This may be part of a conceptual or interpretative framework, or it may be more technical: like the use of special reference frames to label spacetime points and thus (superficially) to avoid the Einstein hole problem. A property of all background structure is that once inserted into a theory it tends to stay

[4] In saying that the Planck length or Planck time have definite values, an implicit structure of this sort is needed to define the units in which length or time are expressed.

there. This does no harm if it is recognised for what it is, but if its initial presence is not made explicit the structure may be "rediscovered", accompanied perhaps by shouts of great triumph. For example, some of the claims that spacetime "emerges" from string theory look somewhat shaky once it is realised that much of the structure (*e.g.*, the manifold nature of spacetime) is imposed *a priori* in the very formulation of the theory.

2.4 How Should the Construction of a Quantum Theory of Gravity Be Approached?

Let us see how these issues arise in the context of the various general ways in which one might approach the problem of constructing a quantum theory of gravity. The discussion here is very brief and many of the topics will be taken up later in more detail.

2.4.1 Quantise General Relativity

Many of the most famous applications of quantum theory start with a classical theory which is then 'quantised'. Note that "quantise" is a verb: it is something that is done to a classical theory, presumably with the expectation of improving it in some way. Certainly this is how elementary quantum theory is taught: we 'quantise' the hydrogen atom, the simple harmonic oscillator *etc.* Even quantum chromodynamics [5] is based on the idea of quantising a given classical Lagrangian system.

In the case of general relativity we do not know if it is possible to quantise the simple Einstein action plus whatever additional fields we fancy, or whether a consistent theory requires a very special selection of matter. This latter idea has been a constant source for the hope that a quantum theory of gravity will lead to a unique grand unified theory of all the fundamental forces. The last major programme of this sort was N=8 supergravity.

One big advantage of quantising a given classical theory is that it is guaranteed that the theory can be recovered in an appropriate classical limit. In particular, the desire

[5] This is the first, and probably the only, connection between my course and the other lectures at the Schladming Winter School!

to reproduce general relativity at low energies should be easy to meet if one starts with the Einstein action $S = 1/\kappa^2 \int R(g)(\det g)^{\frac{1}{2}}\, d^4x$ and then quantises the metric using operator methods, functional integrals or whatever else can be drawn from the armoury of quantum field theory. This approach is also helpful in interpreting the quantum results in so far as the theory is understood to deal with small "quantum fluctuations" about some classical solution to the field equations.

Unfortunately, it is not a trivial matter to turn the metric tensor $g_{\mu\nu}(x)$ into an operator $\widehat{g}_{\mu\nu}(x)$, or to construct a functional integral over fields of this type. One major problem is that many of the techniques of conventional quantum field theory are simply not applicable.

The most obvious example is the entire edifice of Poincaré -group based quantum field theory which can only be invoked if a background Minkowski spacetime is employed from the outset. Without such a structure, even the notion of a 'graviton' becomes very problematic. Selecting any other background only makes matters worse since the theory of quantum fields in a general curved space has many ambiguous features.

Problems with the diffeomorphism group

Another major difficulty arises if one takes seriously the idea that the quantum theory should reflect the diffeomorphism group invariance of classical general relativity (*e.g.*, Rovelli and Smolin 1990). In a normal quantum field theory, the two-point function for a scalar field ϕ has the typical behaviour

$$W(x,y) := \langle 0|\, \widehat{\phi}(x)\widehat{\phi}(y)\, |0\rangle \sim \frac{\text{const.}}{(x-y)^2} \tag{2.4.1}$$

as the spacetime points x and y approach each other (note that the distance between x and y is measured using the background Minkowski metric). The fact that $W(x,y)$ depends only on the difference vector $x - y$ is a direct consequence of the invariance of the vacuum state $|0\rangle$ under the action of the translation subgroup of the Poincaré -group.

On the other hand, if $|0\rangle$ is Diff($\mathcal{M}$) invariant, and if $\widehat{\phi}(x)$ transforms as in (2.2.1), then

$$W(x,y) := W\big(f(x), f(y)\big) \tag{2.4.2}$$

for all $f \in$ Diff($\mathcal{M}$). However, for any two pairs of points (x,y) and (x',y') that are sufficiently close to each other (for example, if all four points lie in a single coordinate

chart) there is a diffeomorphism f such that $x' = f(x)$ and $y' = f(y)$. It follows that $W(x,y)$ is actually a *constant* for all points y in a sufficiently small neighbourhood of x.

In assessing the significance of this behaviour it should be noted that it is related to the fact that the value $\phi(x)$ of the scalar field ϕ at a point $x \in \mathcal{M}$ is not an observable in a Diff($\mathcal{M}$)-invariant theory. Nevertheless, it does show that the questions of short-distance behaviour and ultraviolet divergences are likely to be very different in quantum gravity than in a normal quantum field theory. Similarly, the concept of regularisation is difficult to implement without the explicit introduction of a background metric with which to give a numerical measure of the "nearness" of one spacetime point to another.

The question of Diff($\mathcal{M}$) invariance also produces problems in functional integral approaches to quantisation. One might try to construct a theory of quantum gravity by using functional integrals analogous to those employed in normal quantum field theory to produce the vacuum expectation values of the time-ordered product of a string of fields. For example, one might look at

$$G(x_1, x_2 \ldots x_n) = \int \mathcal{D}[g]\, R(x_1)R(x_2)\ldots R(x_n)e^{i\int_{\mathcal{M}} R(g)\sqrt{g}\, d^4x}. \qquad (2.4.3)$$

However, it is difficult to see what a "time-ordered" product could mean in the absence of any background metric to provide a preferred notion of spacelike and timelike. [6] But even if such a background *is* provided, the idea of time-ordering still seems inconsistent with Diff($\mathcal{M}$) invariance since whether a pair of points is timelike or spacelike can be changed by the action of a diffeomorphism. [7] A related problem appears in a spacetime operator approach to quantisation. In a conventional quantum field theory, a scalar field $\widehat{\phi}$ obeys the basic microcausality condition

$$[\,\widehat{\phi}(x), \widehat{\phi}(y)\,] = 0 \qquad (2.4.4)$$

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[6] This particular problem can be superficially avoided by using a "Euclidean" formalism in which the functional integral is over metrics with a Riemannian, rather than Lorentzian, signature. The problem then is to relate these Riemannian n-point functions to physical amplitudes.

[7] Of course, if the background metric is transformed at the same time, then it is tautological that nothing changes. But this merely reflects the absence of any intrinsic physical meaning for the mathematical points in the spacetime manifold. The situation is very different if the points $x_1, x_2 \ldots x_n$ are labelled by the positions of *physical* particles, or the values of physical fields.

if x and y are spacelike separated. But what does this mean in a quantum theory of gravity? As emphasised by Fredenhagen and Haag (1987), for most [8] pairs of points $x, y \in \mathcal{M}$ there will exist at least one Lorentzian metric with respect to which they are *not* spacelike separated, and hence, in so far as all metrics are "virtually present" (or summed over in a functional integral) the right hand side of (2.4.4) will never vanish. Thus one of the bedrocks of conventional quantum field theory is removed at a stroke!

The same argument throws doubts on the use of canonical commutation relations (with respect to what metric is the hypersurface meant to be spacelike?) and on the whole C^*-algebra approach to quantum field theory.

Use of a Minkowski background metric

In the light of the above, it is not surprising that much of the work on quantum gravity has involved the introduction of a background Minkowski space with the intent of using conventional, Poincaré -group based quantum field theory. As we will see briefly in §3, this leads naturally to a perturbative approach based on writing the metric tensor as $g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x)$ and then quantising $h_{\mu\nu}(x)$ as a normal field with "gravitons" as quanta.

This is closely related to the possibility of using the background metric $\eta_{\mu\nu}$ to construct a Gaussian measure with the weight factor

$$\exp\left(-\int_{\mathcal{M}} \eta^{\mu\nu}\eta^{\alpha\beta}h_{\mu\alpha}(x)h_{\nu\beta}(x)\,d^4x\right). \tag{2.4.5}$$

On the other hand, the analogous construction using just the metric $g_{\mu\nu}(x)$ (*i.e.*, replacing both η and h with g) leads only to a constant under the spacetime integral!

Many advantages (and disadvantages) accrue from the introduction of such a background structure, but further discussion is deferred to §3 where a short survey is given of the approach.

A minimum length in quantum gravity?

Another problem with the use of local quantum fields is the possible existence of a natural "minimum" length/time related to the Planck units. An example of a recent suggestion of this sort is the work of Padmanabhan who argued that

1. geodesic distance is intrinsically bounded from below in quantum gravity (Padmanabhan 1985a,b);

[8] The qualification arises from the possible existence of global obstructions to the construction.

2. the uncertainty relations, plus the existence of the Schwarschild radius, impose lower bounds on measurements of both space and time (Padmanabhan 1987).

The latter is reminiscent of some of the recent studies on "quantum clocks" which play an important role in the analysis of time in the canonical quantisation of gravity. We will return to this topic in §5.

Another example is the suggestion by Greensite (1990) that a minimum length may arise in the context of a lattice approach to quantum gravity. And of course there are several indications that string theory may lead to a natural minimum length (§3.3).

Note that the idea of a minimum length can have several different meanings. One possibility is that a length can be *measured* only to an accuracy of the Planck value (presumably specified with respect to some background metric?) but that the underlying model of a continuum spacetime is nethertheless still valid. A more iconoclastic view is that the existence of a minimum length/time signals the *breakdown* of the continuum picture itself.

Ideas like this cast further shadows over the idea of trying to construct a theory of quantum gravity by quantising the point fields of classical general relativity. In particular, if time cannot be measured, or defined, to an accuracy greater than the Planck time, one needs to rethink the significance of "equal-time" commutators and the general quantum mechanical idea of a complete commuting algebra of "simultaneously" measurable observables.

The breakdown of the simple continuum picture would be even more serious. It suggests that the Diff($\mathcal{M}$) invariance of the theory is only a coarse-grained feature—an idea that is difficult to implement in practice, partly because of the absence of any finite-dimensional approximations to the diffeomorphism group. This also causes serious problems in the Regge calculus approach to quantum gravity, and contrasts strongly with the situation vis-a-vis the gauge group in Yang-Mills theory.

Quantum topology too?

But once the idea of dropping the classical continuum picture of spacetime has been entertained, a number of related possibilities come to mind. In particular, general relativity really involves a pair $(\mathcal{M}, g)$ where $\mathcal{M}$ is a manifold and g is a metric, and therefore, if a 'hat' is to be placed on the g, why should not the same be done to $\mathcal{M}$?

The intriguing idea that the manifold itself should be quantised goes back to the seminal ideas of Wheeler (1964). However, it is far from clear what the symbol $\widehat{\mathcal{M}}$ is supposed to mean. The mathematical framework of general relativity has a hierarchical form that can be written symbolically as

$$\text{set} \rightarrow \text{topology} \rightarrow \text{differential structure} \rightarrow \text{Lorentzian metric} \qquad (2.4.6)$$

in which each step represents a structure superimposed on the previous one in the chain. *A priori*, a 'hat' might be inserted at any of these stages. The least adventurous approach is to keep $\mathcal{M}$ fixed and just quantise the metric. But there are other possibilities. For example:

1. One might remain within the category of differential geometry but let the manifolds become part of the quantum structure. This idea was developed at length by Hawking (1978,1979) and collaborators in the context of his "Euclidean" quantum gravity programme which is based on the use of path integrals over Riemannian, rather than Lorentzian, metrics. Thus a typical quantity would be something like

$$Z = \sum_{\mathcal{M}} \int_{\mathrm{Riem}(\mathcal{M})} \mathcal{D}[g]\, e^{-\int_{\mathcal{M}} R(g)\sqrt{g}\, d^4x} \qquad (2.4.7)$$

   where the integral is over the set $\mathrm{Riem}(\mathcal{M})$ of all Riemannian metrics on $\mathcal{M}$ and the sum [9] is over all four-manifolds $\mathcal{M}$. A recent application of this idea has been the theory of wormholes and their possible implications for determining some of the basic coupling constants (*e.g.*, the cosmological constant) of nature (Coleman 1988, Hawking 1988, Hawking and LaFlamme 1988).

   The use of topologically complicated manifolds $\mathcal{M}$ raises intriguing possibilities, one of which is that information may be lost if, for example, particles fall into the event horizon of a (virtual) black hole. An observer external to the black hole would interpret this as a transition from a pure quantum state to one that is mixed, leading to what Hawking called the "$-matrix": the pure→mixed analogue of a normal S-matrix (Hawking 1982, 1984).

2. A less ambitious project might be to keep the topology of $\mathcal{M}$ fixed but to allow quantum fluctuations in the different differential structures that are compatible with this topology. It is known that four-dimensional manifolds (including

[9] In practice it is very difficult to know what this means, not least because it is not possible to classify four-manifolds in any algorithmic way.

Minkowski space) do admit such exotic structures although, to the best of my knowledge, no one has yet proposed a scheme for quantising them.

3. A much more adventurous approach is to quantise at the topology stage. This means constructing a theory in which the basic point set $\mathcal{M}$ is fixed but where the topology (in the technical sense of a collection of open sets) is quantised. This involves quantum fluctuations over a much wider class of topological space than those that can carry the structure of a differentiable manifold (*e.g.,* Isham 1989, 1990, 1991). A more modest version of this scheme is to restrict attention to those topological spaces whose topology can be obtained from a distance function (Alvarez 1988, Alvarez *et al* 1990, Isham *et al* 1990).

4. Perhaps the most radical idea of all is to place the 'hat' on the first entry in the chain in (2.4.6) and to quantise the set of spacetime points. This might be interpreted in various ways.
   a) It could mean that the *number* of points in the set $\mathcal{M}$ is subject to quantum fluctuations. This is technically feasible in the topological quantisation scheme mentioned above and is particularly relevant in models of spacetime that employ a finite number of spacetime points in any bounded region.
   b) A more radical proposal is to interpret the "quantisation of points" as saying that the very concept of a spacetime point loses its meaning.

   This might perhaps be suggested by the ideas of a minimum length mentioned earlier. Another motivation stems from the long-standing suggestion of Penrose (1975) who, in the context of his twistor programme, advocates a projective picture in which the null-lines in spacetime are the primary entities and with a spacetime 'point' defined to be the intersection of a family of such lines. If the quantisation of gravity leads to quantum fluctuations in the null-cones, then the null-lines will be spread out and the "intersection" will no longer correspond to a single point.

Ideas on quantising sets, topologies and manifolds can be (and have been) developed in a number of directions and there is a substantial gnostic underworld of people (including myself) who work in this area. However, these ideas tend to be very speculative and rather difficult to relate to conventional approaches to quantum gravity. For this reason I have elected not to develop the theme further in these notes and I will con-

centrate instead on the more "respectable" conceptual and geometrical problems in quantum gravity.

However, whether an approach be speculative or mainstream, it is clear that the idea of "quantising" classical general relativity opens up a number of important issues. In particular, a major question in any approach of this type will always be:

> How are the classical concepts of space and time *modified* in the transition *to* the quantum theory?

2.4.2 Quantise Some Other Theory That Can Give General Relativity as the Low-Energy Limit

I have spent some time discussing the issues that arise in "quantising" the classical theory of general relativity since this is the approach to quantum gravity that has received most attention over the years. Another possibility is to remain with the idea of quantising a classical system but start with something other than general relativity. The key step here is to find a system whose quantum theory is well-defined and which yields classical general relativity as a low-energy limit, even though that is not the starting point.

A good example of a scheme of this type is the "induced gravity" programme which was popular in the early 1980s. In this approach, the action principle includes the usual coupling of matter fields to the metric tensor, but *not* the Einstein-Hilbert action of pure gravity. The crucial idea is that the latter will arise as a counterterm from the quantised matter fields although the gravitational field itself is *not* quantised.

A much more sophisticated example is the theory of closed superstrings in which the graviton occurs as just one of an infinite number of particles associated with the quantised string (§3.2).

Note that theories of this type still involve an act of "quantisation", but the classical theory that is quantised is not general relativity. This is particularly clear in the Polyakov approach to string theory in which, genus by genus in the target space, one is faced with the quantisation of a conventional, two-dimensional non-linear σ-model.

The central philosophy of such approaches is that general relativity is only a low-energy "phenomenological" theory; rather as the standard four-dimensional non-linear σ-model can be regarded as an effective Lagrangian for calculating the low-energy results of QCD. Consequently, it is *wrong* to assign any fundamental operator status to the metric tensor. If correct, this position means that much past (and current) work in quantum gravity was (and is) very misguided!

The possibility also arises that if the metric tensor is only a "phenomenological" concept, then perhaps the same is true of the spacetime manifold? This raises the intriguing question of where such an "effective" manifold could come from. In practice, almost all theories of this type introduce the spacetime manifold as part of the background structure, even though the metric tensor is a derived concept.

In any event, it is clear that a key question in this general approach to the problem of quantum gravity will always be

> How are the classical concepts of space and time *recovered* in the transition *from* the quantum theory?

2.4.3 "General-Relativise" Quantum Theory

The verb "general-relativise" will not be found in any of the standard dictionaries, but the broad idea should be clear enough. Rather than starting with classical general relativity which is then quantised, one begins instead with normal quantum theory and studies the extent to which it can be made compatible with the ideas of general relativity.

This approach is intriguing but it has not been developed much other than the substantial effort that has been devoted to quantum field theory in a fixed spacetime.[10] An exception is the interesting paper by Fredenhagen and Haag (1987) who tackle the problem of making a quantum theory invariant under spacetime diffeomorphisms (see also Bannier 1988).

[10] This has been a very active area of research and one which raises its own conceptual and geometrical problems. However, for reasons of space, in these notes I will only discuss quantum gravity proper.

2.4.4 General Relativity and Quantum Gravity Both Emerge From Something Quite Different

In this perception of the problem of quantum gravity, the relation between the sought-for theory and general relativity (and/or quantum theory) is deemed to be analogous to the relation between quantum theory and classical mechanics. Thus what is called for is a radically new theoretical and conceptual structure which—although reducing to normal quantum theory and general relativity in their appropriate domains—is nevertheless quite different from both.

It is this exciting possibility that entices many physicists to work in quantum gravity in the first place. But it is not easy to effect a fundamental transformation in the foundations of physics, especially in the absence of any experimental or observational evidence to point in the right direction!

2.4.5 The Semi-Classical Option

At the opposite extreme is the idea, dating back to Moller (1962), that perhaps it is not necessary to quantise the gravitational field, but only the matter to which it couples. This "semi-classical" approach can, and has, been interpreted in several different ways, one of which is the induced gravity programme alluded to earlier.

However, the original idea was to study the system of coupled equations

$$G_{\mu\nu}(g) = {}_t\langle\psi|\, T_{\mu\nu}(\widehat{matter}, g)\,|\psi\rangle_t \tag{2.4.8}$$

and

$$i\hbar\frac{d}{dt}\,|\psi\rangle_t = H(\widehat{matter}, g)\,|\psi\rangle_t \tag{2.4.9}$$

in which the source of the gravitational field is the expectation value in some special state $|\psi\rangle_t$ of the energy-momentum of the quantised matter.

There was renewal of interest in this approach in the early 1980s, although the results were not conclusive (for a survey see Kibble 1981). One problem is that if the matter is chosen to be a quantised field, the right hand side of (2.4.8) can only be defined after the quantum energy-momentum tensor has been regularised and renormalised (Randjbar-Daemi *et al* 1980)—a procedure that is ambiguous in the presence of

the non-trivial metric field. This process introduces counter-terms that involve higher powers of the Riemann curvature (as does the induced gravity scheme).

The two equations (2.4.8-9) are strongly cross-coupled and the effective equations for the metric tensor $g_{\mu\nu}$ are far more non-linear than those of normal general relativity. This has lead to claims that the system is intrinsically unstable (Horowitz and Wald 1978, Horowitz 1980, Suen 1989a,b). However, these calculations were mainly perturbative in nature and have been challenged recently by Simon (1990); see also Fulling (1990). The question is an important one and should be studied further.

From the perspective of conventional quantum theory, the coupled equations (2.4.8-9) are rather peculiar. In particular, the effective equation for $|\psi\rangle$ is *non-linear*, and hence the superposition principle is lost. Whether or not this is a problem depends on one's attitude towards conventional quantum theory. For example:

1. The theory is difficult to interpret in the absence of the superposition principle.
2. What is the status of different states $|\psi\rangle$ that satisfy the cross-coupled equations (2.4.8-9)? Are we supposed to select one particular state via a theory of quantum cosmology?
3. On the other hand, it has frequently been suggested that linear quantum theory is only an approximation to a deeper non-linear theory and that a non-linear term could provide a physical mechanism for state-vector reduction. Perhaps the semi-classical approach to quantum gravity is the source of this non-linearity?

One might wonder if the whole idea of quantising everything but the gravitational field is simply inconsistent—perhaps along the lines of the famous Bohr-Rosenfeld (1933) argument which showed that the electromagnetic field *has* to be quantised if it is to couple consistently to the current generated by quantised matter. However, Rosenfeld (1963) himself pointed out that there is no direct analogue in the gravitational case since the proof for electromagnetism involves taking to infinity the ratio e/m of the charge e to the inertial mass m of a test particle—a procedure that is impossible in the gravitational case since the analogue of e is the gravitational mass whose ratio to the inertial mass is fixed by the equivalence principle. There have been several attempts since then (*e.g.*, Eppley and Hannah 1977, Page and Geilker 1981) to clarify the situation but it is still somewhat unclear.

The semi-classical approach has reappeared in recent years in the guise of a Born-Oppenheimer/WKB approximation to quantum gravity. Once again, the quantum matter effectively couples to a classical gravitational field but with the difference that, unlike (2.4.8-9), this is only intended to be an *approximation* to the, as yet unknown, full theory of quantum gravity. We will return later to the WKB approximation in the context of the problem of 'time' in canonical quantum gravity (§5).

2.5 Other Prima Facie Questions in Quantum Gravity

There are many other questions that arise when one considers carefully what is involved in trying to construct a quantum theory of gravity. For example, what role is played by *complex* metrics, or those that are *degenerate*? Both possibilities have arisen several times in recent work on quantum gravity, and they pose non-trivial problems for the geometrical interpretation of the theory.

But perhaps the most substantial of the issues that we have not yet addressed is the whole question of quantum cosmology and, in particular, the problems that arise when attempts are made to apply the interpretative framework of quantum theory to the universe in its entirety.

1. The conventional Copenhagen interpretation of quantum theory places great emphasis on the role of "measurement" and "probability", often interpreted in a relative frequency sense. But who can stand outside the entire universe to measure it? And what is an ensemble of universes?
2. Theories of the quantum creation of the universe (QCU) aspire to construct a *unique* quantum state in terms of some simple quantum boundary conditions near the big-bang. Is such an idea compatible with the standard uses of quantum theory?
3. As long as we do not probe too deeply, the world around us is distinctly classical. How did such a world emerge from a totally quantum mechanical origination region?
4. QCU theories involve the idea of the "beginning" of time. Is such a concept compatible with either general relativity or conventional quantum theory?
5. Is there a potential Gödel-type paradox in proposing a theory that, if correct, must presumably predict itself in some way?

These questions are very non-trivial and we will not attempt to address them until the final section of the notes.

3. The Particle Physics Approach

3.1 Quantisation of General Relativity

In this section we will consider some of the approaches to quantum gravity that derive from the community of elementary particle physicists. The discussion is brief and is focused on the conceptual and geometrical issues that arise in schemes of this type. More detailed accounts of the technical material can be found in the recent review by Alvarez (1989), two of my own reviews (Isham 1985, 1987), and in many of the older accounts of quantum gravity (*e.g.,* the various articles in Isham *et al* 1975, 1981).

3.1.1 The Graviton

In the particle physics approach to quantum gravity, the basic entity is the *graviton*—the quanta of the gravitational field. This particle is perceived as propagating on a background Minkowski spacetime, and—like all elementary particles—is associated with a specific representation of the Poincaré group which is labelled by its mass m and spin s. The possible values of m and s are sharply limited by the functions which the graviton is to serve:

1. The t-channel exchange of a particle of mass m can give rise to a static force of the form e^{-mr}/r^2 where r is the distance between the two particles. Thus the graviton must be massless to secure the usual gravitational inverse-square law.
2. The spin of a graviton cannot be half-integral because the Pauli exclusion principle makes it impossible to construct a classical-sized field from a coherent superposition of fermions.
3. Weinberg (1964, 1965) showed that a particle whose spin is greater than two will not produce a static force, and hence the only options are $s = 0, 1$ or 2. However, $s = 1$ gives a repulsive force between like particles (*e.g.,* the spin-1 photon serves this function in electromagnetism), and therefore only $s = 0$ or $s = 2$ is possible.

Zero spin is associated with a scalar field $\phi(x)$ whereas $s = 2$ comes from a (symmetric) Lorentz tensor field $h_{\mu\nu}(x)$. The obvious implication is that $s = 0$ and $s = 2$ correspond respectively to Newtonian gravity and general relativity.

The general theory of relativistic (*i.e.*, special relativity) wave equations shows that a free, massless spin-2 field $h_{\mu\nu}$ must satisfy the field equation

$$h_{\mu\nu,\alpha}{}^{\alpha} - h_{\mu}{}^{\alpha}{}_{,\alpha\nu} - h_{\nu}{}^{\alpha}{}_{,g\mu} + h^{\rho}{}_{\rho,\mu\nu} + \eta_{\mu\nu}(h^{g\beta}{}_{,\alpha\beta} - h^{\alpha}{}_{\alpha,}{}^{\beta}{}_{\beta}) = 0 \tag{3.1.1}$$

up to an arbitrary redefinition of the form

$$h_{\mu\nu} \to h_{\mu\nu} + \lambda\eta_{\mu\nu}h^{\alpha}{}_{\alpha} \tag{3.1.2}$$

with $\lambda \neq 1gver4$ (to avoid the new field being traceless). Note that all indices in these expressions are raised and lowered with the Minkowski metric $\eta_{\mu\nu}$.

An important property of the field equations (3.1.1) is their invariance under the gauge transformations

$$h_{\mu\nu} \to h_{\mu\nu} + \xi_{\nu,\mu} + \xi_{\mu,\nu} \tag{3.1.3}$$

where $\xi_{\mu}(x)$ is an arbitrary Lorentz vector field. This invariance is a consequence of the massless nature of the graviton and is necessary to project out the lower spin ghosts which are otherwise associated with the tensor field $h_{\mu\nu}$.

3.1.2 The Derivation From General Relativity

The connection of the above with general relativity is as follows. If we take the Einstein-Hilbert action

$$S[g] = \frac{1}{\kappa^2}\int R(g(x))\big(-\det g(x)\big)^{\frac{1}{2}}\, d^4x \tag{3.1.4}$$

and write the metric tensor as

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \kappa h_{\mu\nu}(x), \tag{3.1.5}$$

then the lowest order (in κ) contribution to the field equations for $h_{\mu\nu}(x)$ is precisely (3.1.1). Furthermore, the redefinition (3.1.2) simply corresponds to using $g_{\mu\nu}(-\det g)^{\lambda}$ as the basic field rather than $g_{\mu\nu}$— a change that should have no effect on on-shell S-matrix elements.

The gauge transformation (3.1.3) also has a simple geometrical interpretation: it is just the effect induced on $h_{\mu\nu}$ (via (3.1.5) by an infinitesimal diffeomorphism of Minkowski space generated by the vector field ξ.

As was remarked in §2, several significant advantages are gained by introducing a background Minkowski metric and hence reducing the problem of quantum gravity to one of conventional quantum field theory. For example:

1. The background metric provides a reference framework in which to discuss "short-distance" behaviour, operator-product expansions, regularisation, and related topics.
2. The background metric gives a fixed causal structure which can be used to define
   (a) microcausal spacetime commutation relations for spacetime fields;
   (b) equal-time commutation relations for canonical fields;
   (c) a good notion of "time-ordering" for use in a functional integral or general LSZ formalism.
3. Some of the prima facie problems with Diff($\mathcal{M}$) invariance are converted into standard problems for gauge-invariant quantum field systems. These can be treated in ways that are familiar from Yang-Mills theory (gauge-fixing, ghost particles, BRS techniques *etc*).
4. The underlying Minkowski structure can be used to give a sensible definition of physical observables that is related to the *asymptotic* behaviour of the fields. The crucial point is that the gauge group generators ξ have compact support and therefore do not affect the fields asymptotically. The use of asymptotic observables played an important role in Bryce DeWitt's seminal investigations into the structure of spacetime-covariant quantum gravity (DeWitt 1965, 1967a,b).

3.1.3 Geometric and Conceptual Objections

Unfortunately there are several objections to the introduction of this background structure which throw the entire scheme into some doubt:

1. There is no reason to suppose that the causal structure of the Minkowski metric is the physically correct one. Indeed, it has often been suggested that a non-perturbative treatment could lead to light cones that are displaced from those of

the background Minkowskian space. The status of the initial microcausal structure is then uncertain.

2. The Minkowskian background also fixes the *topology* of spacetime to be that of a trivial vector space. It is thus particularly difficult to discuss cosmological problems, spacetime singularities, black holes and event horizons, or any other features of classical general relativity that involve non-trivial topological structure. For example, it would not be easy to construct a theory of the quantum creation of the universe in terms of a few gravitons propagating in Minkowski space!
3. The expansion (3.1.5) is a poor one when viewed from the geometrical perspective of classical general relativity. In particular, the object on the left hand side will only be a genuine metric tensor (*i.e.*, an invertible matrix with signature (-1,1,1,1)) for small values of the field $h_{\mu\nu}(x)$. However, naive quantisation methods involve, for example, integrating over *all* values of $h_{\mu\nu}$. In effect, rather than quantising on the space PRiem($\mathcal{M}$) of pseudo-Riemannian metrics, we are only quantising in the tangent space to the particular metric $\eta_{\mu\nu}$.

This latter problem is reminiscent of the use of normal coordinates in a non-linear σ-model and suggests that the quantum theory cannot be globally correct. Even if a parametrisation is chosen for which the values of $h_{\mu\nu}$ are unrestricted (*e.g.*, $g_{\mu\nu}(x) = \big(\exp h(x)\big)_{\mu\nu}$), it is very unclear if, or how, one should "patch" together the quantisations on the different tangent spaces of PRiem($\mathcal{M}$) (assuming it is possible to do the quantisation around any background metric other than $\eta_{\mu\nu}$). [11] To see how non-trivial this problem can be, consider the quantisation of a system whose phase space $\mathcal{P}$ is the (compact) two-dimensional sphere S^2. The Lie group $SU(2)$ acts transitively and symplectically on this space, and there are good arguments for saying that a correct quantisation of the system involves finding an irreducible representation of this group. In other words, $SU(2)$ plays the same role for $\mathcal{P} = S^2$ as does the Weyl-Heisenberg group for $\mathcal{P} = \mathbb{R}^2$. However, note that each such representation of $SU(2)$ has a *finite* dimension whereas the non-trivial representations of the Weyl-Heisenberg group are all infinite dimensional.

[11] Of course, background field methods are apparently designed to perform a quantisation around any background. However, this "quantisation" is rather formal and cannot be made rigorous in terms of a unique Hilbert space and identification of, say, particle observables. This is the unsolved (and probably insoluble) general problem of quantum field theory in a curved background.

Now imagine performing a "tangent-space" quantisation on S^2. Each tangent space of S^2 is a copy of $\mathbb{R}^2$ and is therefore naturally quantised using the Weyl-Heisenberg group and its infinite dimensional representation. Thus one is faced with the problem of patching together a collection of infinite dimensional quantum theories (one for each tangent plane on S^2) in such a way as to reproduce the finite-dimensional state space of the globally-correct quantisation. This is not a trivial task!

3.1.4 Non-Renormalisability

In spite of these reservations there is no doubt that, had this approach to quantising gravity been successful, the resulting theory would have been of intense interest. However, as is well-known, the theory is not renormalisable and hence fails, even in its own terms. This result is not very surprising. If the expansion (3.1.5) is inserted into the Einstein-Hilbert action (3.1.4), the ensuing Lagrangian for $h_{\mu\nu}(x)$ has interactions that are non-polynomial, derivatively-coupled, and with a dimensional coupling constant κ—each of which is a recipe for non-renormalisability for a quantum field theory in four spacetime dimensions.

Early power-counting estimates indicated that the theory should indeed be non-renormalisable, and this was rendered more plausible by a number of explicit calculations made in the mid 1970s of the one-loop graphs for gravity plus various matter fields (for purely kinematical reasons, the pure gravity one-loop graphs are finite on-shell). But it was not until quite recently that lengthy calculations by Goroff and Sagnotti (1985, 1986) of the two-loop contributions finally disposed of the faint remaining hope that gravity might have been well-behaved, notwithstanding all the evidence to the contrary.

Tweak the theory

Various schemes have been suggested whereby this sad state of affairs might be averted. A conservative view is that the infinite quantities can be controlled by "tweaking" the theory slightly using some, not too drastic, modification of conventional general relativity.

One suggestion which has received persistent attention over the years is to use an "$R + R^2$" Lagrangian, *i.e.*, augment the original Einstein-Hilbert action with terms

involving the square of the Riemann curvature. The linearised theory becomes fourth-order in derivatives and, consequently, the free propagator changes from a $1/p^2$ form to something like

$$\frac{1}{p^2(\alpha + p^2)}. \tag{3.1.6}$$

This faster fall-off in momentum reduces the severity of the ultraviolet divergences and leads to a theory that is renormalisable (Stelle 1977) and, in some versions (Fradkin and Tseytlin 1981, 1982), even asymptotically free (for further details see the review articles cited earlier). Unfortunately, the momentum behaviour in (3.1.6) also produces ghosts and non-unitarity, and the community at large has not been persuaded to accept the suggestion (Tomboulis 1984) that these are merely an artifact of an inappropriate form of perturbation theory.

The theory of supergravity was another major programme aimed at removing the ultraviolet divergences with the hope that the infinities produced by the additional fermi loops would cancel those of the bosonic graviton loops. Some infinities can indeed be removed in this way, but the discovery of supersymmetric counterterms (at 3 loops for $N = 1$, and at 8 loops for $N = 8$) blighted this particular approach to constructing a fundamental theory of quantum gravity. [12] One of the attractive feature of this approach is that, had it worked, it would have yielded a definite prediction for the fundamental matter Lagrangian which must be attached to general relativity. This could have provided a basis for a true grand unified field theory.

Non-perturbative approaches

Another frequent response to the non-renormalisability of quantum gravity has been the claim that the problem stems from the use of weak-field perturbation theory. This statement has several interpretations, one of which is that it is still feasible to use a Minkowski background and associated quantum field theory but this must be augmented with some sort of non-perturbative [13] summation of the terms that arise. The various attempts/suggestions that have been made along these lines include:

[12] To the best of my knowledge, there are no explicit calculations showing that these counterterms do actually occur. So a miracle could still happen, although the odds must be strongly against it in the absence of any good theoretical reason why the coefficients of the counterterms should vanish.

[13] In every branch of physics, almost any calculation that is feasible involves *some* sort of approximation method, and the same is doubtless true of quantum gravity. Thus "non-perturbative" does not mean literally "without the use of any sort of perturbative technique" but only that weak-field expansions like (3.1.5) are to be avoided.

1. Trying to sum infinite sets of Feynman graphs to give a finite result. This has often been seen as an implementation of the old idea (Pauli 1967) that ultraviolet divergences can be removed by including quantum gravity effects which lead to a smearing of the lightcone and an effective cut-off at the Planck length.
2. Various sorts of $1/N$ expansion.
3. Large coupling constant expansions in the inverse of κ.
4. Weinberg's idea of *asymptotic* safety in which the renormalised coupling constants are forced to lie in a finite-dimensional subspace of the infinite-dimensional space of all possible values (Weinberg 1979).
5. Klauder's work on techniques for handling non-renormalisable quantum field theories in general (*e.g.*, Klauder 1979).

For our purposes, the most relevant property of approaches of this type is that they still use a Minkowskian background structure. They are therefore subject to the objections raised earlier, with the caveat that some of these may be addressed (albeit indirectly) by the non-perturbative nature of the methods.

However, another interpretation of the failure of weak-field perturbation theory is that the entire programme is wrong, including the use of a Minkowski background. Some of the more interesting suggestions in this direction (and still involving the idea of "quantising" the classical theory of general relativity) are as follows.

1. There have been a number of attempts to set up lattice versions of quantum gravity. Some of these use the type of lattice employed in Yang-Mills theory; others employ simplicial approximations and Regge Calculus. A number of technical difficulties arise in the process (especially in regard to invariance under diffeomorphisms) but perhaps the most serious worry is the absence of any *a priori* reason for supposing the continuum limit to be well behaved.

   However, serious attempts of this sort are still made, one of the most striking of which is DeWitt's ongoing programme for evaluating a Lorentzian functional integral version of quantum gravity with the aid of large scale computing techniques.
2. The canonical theory of quantum gravity has also been studied at length using techniques which avoid weak-field perturbation theory. These include minisuperspace approximations and various schemes involving the Ashtekar variables.

The work on canonical quantum gravity is very important but, since it cannot be said to fall under the category of a "particle physics approach", discussion is deferred to later (§4). On the other hand, string theory certainly does come under this heading, and undoubtedly constitutes one of the more important current approaches to the problem of quantum gravity.

3.2 String Theory

The idea that a quantum theory of gravity can be constructed using closed strings emerged in the early 1970s from studies of the zero-slope limit of the dual resonance model (Scherk and Schwarz 1974, Yoneya 1974) and is frequently cited as one of the main reasons for studying string theory. It is therefore surprising that only a small fraction of the huge literature on superstring theory has been devoted directly to the types of issue that are of interest in quantum gravity. However, even this residue is still quite large and I will restrict myself here to some of the main ideas and, in particular, to those that relate directly to the issue of geometrical and/or conceptual problems in quantum gravity. A very useful review of some of these matters is Horowitz (1989).

3.2.1 The Polyakov Action

In its current formulation, string theory is an example of the second class of approach discussed in §2, *i.e.*, it still involves a "quantisation" of a given classical system but the system that is quantised is not general relativity. Instead we have the Polyakov action

$$S[q,X] := \frac{1}{4\pi\alpha'}\int_W q^{ij}(\sigma)\partial_i X^\mu(\sigma)\partial_j X^\nu(\sigma) g_{\mu\nu}(X(\sigma))\big(\det q(\sigma)\big)^{\frac{1}{2}}\, d^2\sigma \qquad (3.2.1)$$

where q_{ij} is a metric on the two-dimensional manifold W (known as the "world-sheet"), $X : W \to \mathcal{M}$ are the string fields which map W into the spacetime manifold $\mathcal{M}$, and $g_{\mu\nu}$ is a background metric on $\mathcal{M}$. The constant α is related to the string tension and is assumed to be of the order of the Planck length.

The classical system is invariant under the conformal transformations $q_{ij}(\sigma) \mapsto F(\sigma)q_{ij}(\sigma)$, $F(\sigma) > 0$, but this will be lost when the two-dimensional fields q_{ij} and X^μ are quantised unless the metric $g_{\mu\nu}$ satisfies an equation which is effectively the

vanishing of the trace of the energy-momentum tensor of the two-dimensional quantum field theory. These equations have the form

$$0 = R_{\mu\nu} + \frac{1}{2}\alpha' R_{\mu\alpha\beta\gamma} R_{\nu}{}^{\alpha\beta\gamma} + \ldots \tag{3.2.2}$$

where ... refers to higher-order powers and derivatives of the Riemann curvature. Other background spacetime fields may also be introduced (in particular, massless dilaton and 2-form fields) and generate modified forms of (3.2.2).

Conformal invariance also requires the dimension of $\mathcal{M}$ to be equal to some critical value which depends on the background fields that are present as well any two-dimensional spinor fields that may be added to the system (*e.g.*, in a superconformal theory).

The field equations (3.2.2) are the string theory replacement for the classical equations of general relativity. Note that, in so far as $g_{\mu\nu}$ is determined by a solution to a set of dynamical equations, it is not strictly correct to think of it as part of the background structure. On the other hand, it does function in this way whenever calculations are made of quantum fluctuations around this metric.

3.2.2 Exact Solutions and Spacetime Singularities

A number of attempts have been made to simplify the field equations (3.2.2) and/or to construct exact solutions. For example:

1. The number of extra terms (*i.e.*, over and above conventional general relativity) can be drastically reduced by imposing invariance under world-sheet supersymmetry transformations. The presence of one supersymmetry generator removes the R^2 and R^3 terms, and two such generators means there are no higher-order terms at all. The latter condition can only be satisfied if the metric $g_{\mu\nu}$ is complex and Kähler. An example is the well known Calabi-Yau space solutions of Candelas *et al* (1985).
2. Exact solutions can be found if the target space $\mathcal{M}$ is a group manifold (Witten 1984).
3. Plane wave solutions have been found Güven (1987) and Amati and Klimcik (1989).
4. Solutions of a cosmological type have been found by Antoniadis *et al* (1988, 1989).

From the viewpoint of traditional general relativity, some of the most interesting work is that of Horowitz and Steif who have discussed a number of exact solutions that contain genuine spacetime singularities (Horowitz and Steif 1990a, Horowitz 1990a,b).

An important question in this context is precisely what is meant by a "singularity". The usual definition in general relativity involves the properties of geodesics, which can be thought of as the paths of hypothetical test particles. However, this seems inappropriate in a theory in which the basic entities are *loops*, and Horowitz and Steif have developed a definition of a singularity which refers to the properties of a first-quantised string propagating in the spacetime, rather than a point particle.

3.3 A Minimum Length in String Theory?

From one perspective, the existence of singular solutions to the string-induced equations for $g_{\mu\nu}$ is perhaps disappointing since one of the hopes for quantum gravity is that it will remove the spacetime singularities that are such an endemic feature of classical general relativity. The string result may seem all the more surprising because of the persistent claim that there is a minimum length in the theory; indeed, that is often said to be the basic reason why the quantum amplitudes should be free of the ultraviolet divergences that plague conventional quantum field theory. The idea of a minimum length comes from a variety of sources:

1. The string is basically an extended object whose internal quantum fluctuations stop it degenerating to a point.
2. In some of the compactification schemes, the scattering amplitudes are invariant under the transformation $R \to \frac{1}{R}$ where R is the radius of the internal space in units of the Planck length. This suggests the existence of a minimum value for this radius.

   However, it is not clear if this is property is generic or if is only a feature of some special solutions. Horowitz and Steif (1990c) have recently shown that a compactified form of the plane-wave solutions mentioned above is *not* invariant under this transformation, although of course this does not rule out the existence of a more generalised form of duality operation.
3. There have been a series of calculations of high energy scattering amplitudes in string theory. Some of these are at large s and fixed angle (Gross and Mende 1987,

1988) while others are at small t (*i.e.*, large distance) and large s (Amati *et al* 1987, 1988, 1989). The results suggest the existence of a minimum length that can be reached by the string, and which has been associated with some sort of generalised uncertainty relation in the form (Veneziano 1989a)

$$\Delta x > \frac{\hbar}{\Delta p} + \alpha' \Delta p \tag{3.3.3}$$

See also Veneziano (1989b), Fabbrichesi and Veneziano (1989) and Konish *et al* (1990).

It should be noted that, as with previous discussions about a minimum length, it is not clear whether the implication is that strings can only *probe* to a minimum length (but leaving intact the notion of a spacetime continuum) or whether the phenomenon signals the breakdown of the entire continuum picture. This issue cannot be decided from within the Polyakov approach since the presence of a continuum manifold $\mathcal{M}$ is part of the background structure.

The central difficulty is the same as that which arises when trying to assess the significance of the existence of singular solutions to the effective field equations. Namely, the next step in the string calculations involves the computation of quantum fluctuations around the classical background solution, and in that sense we are back once more to the weak-field scenario of the old approaches to quantum gravity. It is true that many of the high-energy calculations cited above involve non-perturbative methods for summing the various contributions but, nevertheless, what is lacking at the moment is any complete, non-perturbative alternative to the Polyakov approach which could be used to tackle the types of problem that are of real interest in quantum gravity (spacetime singularities, quantum topology, quantum cosmology *etc*).

For example, there have been several suggestions that there may be a phase change in the theory at the Planck length, with the graviton becoming a meaningless concept. But this is a very difficult thing to study using the current formulation of string theory.

The perturbative results are important and encouraging, but in some respects the situation is similar to what would have been the case had the original particle physics approach been successful. In particular, many of the challenges faced by modern string theory are reflected in the very non-trivial question:

> What would have been the next stage of the theoretical development if ordinary quantum gravity had been perturbatively ultraviolet finite?

One can only hope that the on-going attempts to face some of these challenges will meet with success in the years to come.

4. The Canonical Quantisation of Gravity

4.1 Introductory Remarks

4.1.1 Advantages of the Canonical Approach

The development of the canonical quantisation of gravity has been mainly at the hands of physicists with backgrounds in general relativity. As a result, the techniques employed differ considerably from those discussed in the previous chapter and are likely to be relatively unfamiliar to a particle physics audience. My account of this material will therefore be more detailed than was the case for the "graviton" approaches.

There are several reasons why the canonical theory may be preferred to approaches based on more conventional quantum field theory. For example,

1. Many of these "conventional" approaches use formal functional integral techniques in which a number of serious difficulties are swept under the carpet as problems to do with the integration measure, the contour of integration *etc.* On the other hand, canonical quantisation is usually discussed in an operator-based framework, with the advantage that problems appear in a more explicit (and honest!) way.
2. It is easier to develop techniques in canonical quantisation that do not depend on a background metric. This offers new possibilities for non-perturbative approaches to quantum gravity.
3. For related reasons, canonical quantisation is better for discussing quantum cosmology, spacetime singularities and similar topics.
4. Canonical methods place more emphasis on the geometrical structure of general relativity. In particular, it is easier to address the issue of the extent to which such structure is, or should be, maintained in the quantum theory.

5. Many of the deep conceptual problems in quantum gravity become more transparent in a canonical approach. This applies in particular to the problems of 'time' and the applicability of the interpretative framework of conventional quantum theory.

It should be emphasised that some of these advantages arise only in comparison with weak-field perturbation theory, and this does *not* mean that a canonical scheme is intrinsically superior to one in which spacetime fields are used from the outset (*i.e.*, the so-called "covariant" approaches). For example, there exist *bona fide* approaches to quantum gravity that involve the calculation of a functional integral like

$$Z = \int \mathcal{D}[g]\, e^{iS(g)} \tag{4.1.1}$$

using methods other than weak-field perturbation theory. Indeed, many of the non-perturbative techniques that have been developed in canonical quantum gravity have a fairly obvious translation into a framework of this type.

On the other hand, there *are* fundamental differences between the canonical and the covariant approaches to general relativity—especially in regard to the ultimate status of the concepts of space and time—and it is necessary to commit oneself, at least partially, to one position or the other. However, there is no conclusive technical or conceptual argument to decide the point and it is largely a question of personal taste.

4.1.2 Prima Facie Problems with a Canonical Theory of Gravity

A canonical approach to a standard field theory has three important ingredients:

1. Canonical variables are defined on spacelike hypersurfaces. All the points on such a surface are at "equal-time" and the dynamical equations specify how the canonical variables evolve from one equal-time hypersurface to another.
2. The quantised canonical variables obey canonical commutation relations on these surfaces. In particular, if ϕ is the canonical field variable then

$$[\,\hat{\phi}(x), \hat{\phi}(x')\,] = 0 \tag{4.1.2}$$

for all points x, x' in such a surface.

3. If the theory has a gauge group then some of the potential canonical variables are manifestly non-dynamical and can be removed before quantisation takes place.

The gauge group of general relativity is Diff($\mathcal{M}$) and one of the important steps in the canonical construction (see below) is to show that, roughly speaking, the four components $g_{0\mu}$, $\mu = 0 \ldots 3$ of the metric tensor are gauge variables and have no dynamical evolution. Thus the canonical configuration variable in general relativity is the metric g_{ab} where $a, b = 1, 2, 3$. However, there are several obvious difficulties in trying to apply the ideas of canonical quantisation to g_{ab} and its conjugate:

1. A time parameter in general relativity is one of the coordinates on the spacetime $\mathcal{M}$ and, like all coordinates, it may only be defined in a local region. This could mean that
   a) the time variable t is only defined for some finite range of values for t; or
   b) the subset of points t=const may not be a complete three-dimensional submanifold of $\mathcal{M}$.
2. In a normal field theory, the idea of an "equal-time" hypersurface is synonymous with one that is "spacelike" But the situation in quantum gravity is different. Whether or not a particular hypersurface is spacelike depends on the Lorentzian metric, but this is precisely what is being quantised! This harks back to the discussion of microcausality in §2.4 (cf (2.4.4)) and implies that the meaning, and validity, of the gravitational analogue of (4.1.2) needs some thought.

4.2 Quantum Field Theory in a Curved Background

4.2.1 The Canonical Formalism

Let us start by considering briefly the quantisation of a scalar field ϕ propagating in a background Lorentzian metric $g_{\mu\nu}$ on a spacetime manifold $\mathcal{M}$. The action for the system is

$$S[\phi] = -\int_{\mathcal{M}} \left(\frac{1}{2} g^{\mu\nu}(x) \partial_\mu \phi(x) \partial_\nu \phi(x) + V(\phi(x)) \right) \left(-\det g(x) \right)^{\frac{1}{2}} d^4x \qquad (4.2.1)$$

where $V(\phi)$ is an interaction potential for ϕ (which could include a mass term).

This system has a well-posed classical Cauchy problem only if $(\mathcal{M}, g)$ is a *globally-hyperbolic* pair (Hawking and Ellis 1973). In particular, this means that $\mathcal{M}$ is topologically equivalent to the Cartesian product $\Sigma \times \mathbb{R}$ where Σ is some three-manifold

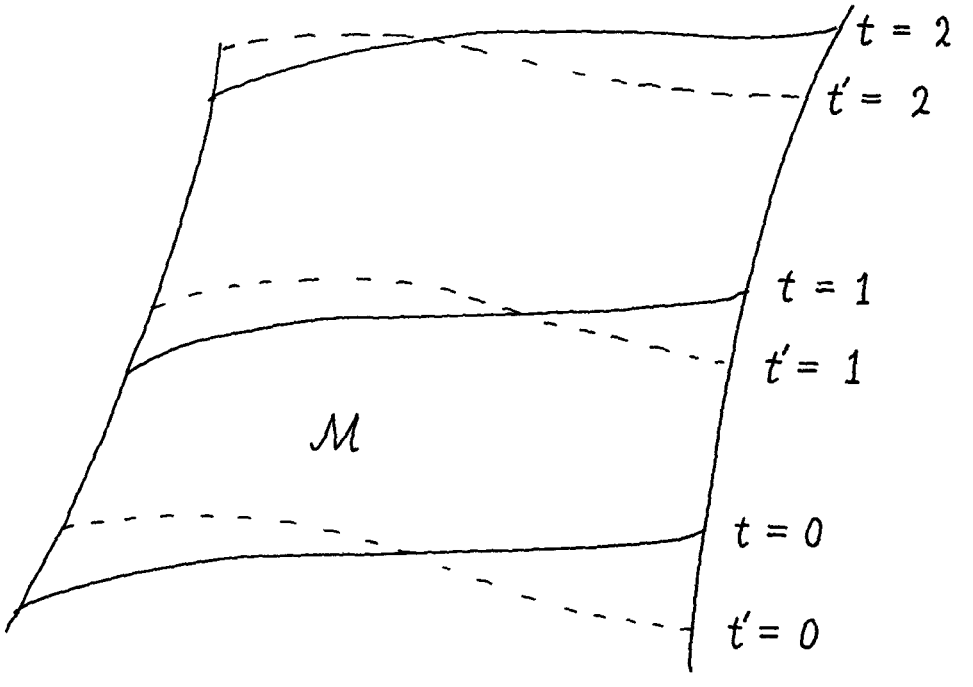


Fig. 4.1. Different global times correspond to different foliations

and $\mathbb{R}$ is a global time direction. We will assume that $(\mathcal{M}, g)$ is such that a global time structure of this sort exists. Technically, we can say that $\mathcal{M}$ has been *foliated* into a one-parameter family of hypersurfaces that are spacelike with respect to the background Lorentzian metric $g_{\mu\nu}$. Note that different choices of such a global time correspond to different foliations (Figure 1.1).

The canonical variables are defined on Σ and consist of the scalar field ϕ and its conjugate variable π (a scalar density) which is essentially the time-derivative of ϕ. Classically, these constitute a well-defined set of Cauchy data and satisfy the Poisson bracket relations

$$\{\phi(x), \phi(x')\} = 0 \tag{4.2.2}$$

$$\{\pi(x), \pi(x')\} = 0 \tag{4.2.3}$$

$$\{\phi(x), \pi(x')\} = \delta(x, x'). \tag{4.2.4}$$

The Dirac δ-function is defined such that the smeared fields satisfy

$$\{\phi(f), \pi(h)\} = \int_{\Sigma} fh \, d^3x \tag{4.2.5}$$

where the test-functions f and h are respectively a scalar density, and a scalar, on the three-manifold Σ.

The formal canonical quantisation of this system follows the usual rules of replacing Poisson brackets with operator commutators so that

$$[\widehat{\phi}(x), \widehat{\phi}(x')] = 0 \tag{4.2.6}$$

$$[\widehat{\pi}(x), \widehat{\pi}(x')] = 0 \tag{4.2.7}$$

$$[\widehat{\phi}(x), \widehat{\pi}(x')] = i\hbar\delta(x, x') \tag{4.2.8}$$

which can be made rigorous by smearing and exponentiating in the usual way.

4.2.2 Representations of the Canonical Algebra

Most treatments of conventional quantum field theory use Fock space representations of the canonical field algebra. These representations are based on the idea of "quanta" of the quantum field and are particularly useful in situations where weak-field perturbative methods are appropriate. However, we are striving to escape from this strait-jacket and it is more productive to develop an approach to the representations of (4.2.6-8) that is not based directly on Fock space.

The starting point is the familiar canonical commutation relations of the standard quantum mechanics of a particle moving in three dimensions. These are

$$[\widehat{x}_i, \widehat{x}_j] = [\widehat{\pi}_i, \widehat{\pi}_j] = 0 \tag{4.2.9}$$

$$[\widehat{x}_i, \widehat{\pi}_j] = i\hbar\delta_{ij} \tag{4.2.10}$$

where $i, j = 1, 2, 3$. Modulo certain mathematical niceties, all irreducible, self-adjoint representations of this algebra are unitarily equivalent to the familiar wave-function representation

$$(\widehat{x}_i\psi)(\underline{x}) := x_i\psi(\underline{x}) \tag{4.2.11}$$

$$(\widehat{\pi}_j\psi)(\underline{x}) := -i\hbar\partial\psi ver\partial x_j(\underline{x}) \tag{4.2.12}$$

on the Hilbert space $L^2(\mathbb{R}^3, d^3x)$ of all functions that are square-integrable with respect to the volume element d^3x. Thus the inner product is

$$\langle\psi_1|\psi_2\rangle = \int_{\mathbb{R}^3} \psi_1^*(x)\psi_2(x)\, d^3x \tag{4.2.13}$$

and the interpretation of a normalised state ψ is that, if a measurement is made of the position of the particle, the probability of finding it in some integrable region $B \subset \mathbb{R}^3$ is

$$\text{Prob}(x \in B; \psi) = \int_B |\psi(x)|^2 \, d^3x. \tag{4.2.14}$$

This suggests that it might be possible to take the state space of the quantum field theory to be a set of functionals Ψ on the space $\mathcal{Q}$ of all classical fields, with the canonical operators defined as

$$\big(\widehat{\phi}(x)\Psi\big)[\phi] := \phi(x)\Psi[\phi] \tag{4.2.15}$$

$$\big(\widehat{\pi}(x)\Psi\big)[\phi] := -i\hbar \frac{\delta \Psi}{\delta \phi(x)}[\phi]. \tag{4.2.16}$$

Thus the inner product would be

$$\langle \Psi_1 | \Psi_2 \rangle = \int_{\mathcal{Q}} \Psi_1^*[\phi] \Psi_2[\phi] \, d\mu[\phi] \tag{4.2.17}$$

and

$$\text{Prob}(\phi \in B; \Psi) = \int_{\mathcal{Q}} |\Psi[\phi]|^2 \, d\mu[\phi] \tag{4.2.18}$$

is the probability that a measurement of the field on Σ will find it in the subset B of $\mathcal{Q}$.

This analysis can be made rigorous after smearing the fields with functions from an appropriate test function space, and gives the following results:

1. Equation (4.2.16) only yields a hermitian operator if the measure $d\mu$ on $\mathcal{Q}$ is an infinite-dimensional version of the Lebesgue measure d^3x. However, no such measure exists. Rather, one needs to employ the infinite-dimensional analogue of a weighted measure, with (4.2.16) being replaced by

$$\big(\widehat{\pi}(x)\Psi\big)[\phi] := -i\hbar \frac{\delta \Psi}{\delta \phi(x)}[\phi] + i\rho(x)\Psi[\phi] \tag{4.2.19}$$

   where $\rho(x)$ is a function that compensates for the weight factor in the measure $d\mu$.

2. The support of a state functional Ψ is typically on *distributions* rather than smooth functions. Thus the Hilbert space is actually $\mathcal{L}^2(\mathcal{Q}', d\mu)$ where $\mathcal{Q}'$ is the topological dual of the infinite dimensional topological vector space $\mathcal{Q}$.

3. There are an infinite number of unitarily inequivalent representations of the canonical commutation relations (4.2.6-8). Each can be written on a Hilbert space of the

form $\mathcal{L}^2(\mathcal{Q}', d\mu)$ with the measure $d\mu$ determining which representation is which. For example, the familiar Fock space representations are associated with Gaussian measures on $\mathcal{Q}'$.

One of the major problems in quantum field theory on a curved background is knowing what representations to choose and what their physical interpretations are.

One consequence of the appearance of the space of distributions $\mathcal{Q}$ is that the value $\widehat{\phi}(x)$ of the field at any particular point $x \in \Sigma$ is likely to be ill-defined. For example, the operator $\widehat{\phi^2}(x)$ acts on $\Psi[\phi]$ as

$$(\widehat{\phi^2}(x)\Psi)[\phi] = (\phi(x))^2\Psi[\phi] \tag{4.2.20}$$

and the right hand side may not exist if Ψ is non-vanishing on fields ϕ that are distributions.

Another source of divergences is the appearance of expressions like $(\pi(x))^2$ in, for example, the hamiltonian of the system. This becomes the operator $-\hbar^2 \frac{\delta^2}{\delta\phi(x)\delta\phi(x)}$ which is singular on many functionals Ψ. As we will see, both types of divergence can appear in the canonical quantisation of gravity.

4.3 The Canonical Structure of General Relativity

4.3.1 Introduction of the Foliation

Let us now consider how these ideas extend to the canonical formalism of general relativity itself. The early history of this subject is well-known and culminated in the investigations of Arnowitt, Deser and Misner (*e.g.*, Arnowitt *et al* 1962). However, these original studies involved the selection of a specific coordinate system on the spacetime manifold, and some of the global issues were thereby obscured. A more geometrical, global, approach was developed by Kuchar in a series of papers (Kuchar 1972, 1976a,b,c, 1977) and this will be adopted here. A good general review of canonical quantisation from this perspective is Kuchar (1981).

The starting point is a four-dimensional manifold $\mathcal{M}$ (a model for spacetime) and a three-dimensional manifold Σ which we assume to be compact [14] and which will

[14] This is the usual situation in cosmology. If the space is asymptotically flat then some of the expressions to be used will need to be augmented by surface terms.

play the role of physical three-space. Furthermore, the topology of $\mathcal{M}$ is assumed to be such that it can be foliated by a one-parameter family of copies of Σ. Thus there exists a family of embeddings $X_t : \Sigma \to \mathcal{M}$, with $t \in \mathbb{R}$ and such that the induced map

$$X : \Sigma \times \mathbb{R} \to \mathcal{M}; \quad X(x,t) := X_t(x) \tag{4.3.1}$$

is a diffeomorphism of $\Sigma \times \mathbb{R}$ with $\mathcal{M}$. The space of all embeddings of Σ in $\mathcal{M}$ is denoted $\mathrm{Emb}(\Sigma, \mathcal{M})$ and can be regarded as an infinite-dimensional manifold. [15]

Since X is a diffeomorphism from $\Sigma \times \mathbb{R}$ to $\mathcal{M}$, its inverse $X^{-1} : \mathcal{M} \to \Sigma \times \mathbb{R}$ is also a diffeomorphism and can be written in the form

$$X^{-1}(y) = \big(S(y), T(y)\big) \in \Sigma \times \mathbb{R} \tag{4.3.2}$$

where $S : \mathcal{M} \to \Sigma$ and $T : \mathcal{M} \to \mathbb{R}$. The map T is a global *time function* [16] and gives the natural time coordinate associated with the foliation since $T(X_t(x)) = t$ for all $x \in \Sigma$.

Note that the existence of such a foliation imposes a substantial *a priori* limitation on the possible topologies for the background manifold $\mathcal{M}$. One consequence is that the topology of Σ is also a fixed part of the background structure. In particular, it is not possible to discuss the idea of quantum topology fluctuations of Σ. On the other hand, nothing has been said about any background metric on $\mathcal{M}$, and none is intended.

As we have seen, the foliation provides a natural choice of time variable and we will use this in what follows. However, the foliation is not unique and there exist many other diffeomorphisms of $\Sigma \times \mathbb{R}$ with $\mathcal{M}$. Each corresponds to a possible choice of time function, as illustrated by the two time slicings in Figure 1.1.

If we wanted to be more mathematically precise we should probably restrict our attention to those foliations whose leaves (*i.e.*, equal-time slices) can be deformed smoothly into the leaves of the given foliation X. Similarly, the set of Lorentzian metrics g that might be placed on $\mathcal{M}$ should be limited to those for which there exists at least one foliation whose leaves are all spacelike with respect to g.

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[15] $\mathrm{Emb}(\Sigma, \mathcal{M})$ is a subset of the space $C^\infty(\Sigma, \mathcal{M})$ of *all* smooth maps from Σ to $\mathcal{M}$. The latter can be equipped with the structure of an infinite-dimensional manifold in several well-known ways and $\mathrm{Emb}(\Sigma, \mathcal{M})$ becomes an open subset. As such, it inherits a differential structure, and this is the one we will use.

[16] Strictly speaking, one should not talk about a 'time' function until $\mathcal{M}$ has been equipped with a Lorentzian metric such that each subspace $X_t(\Sigma)$ is spacelike. Therefore, for the moment, our use of the word is topological rather than metrical.

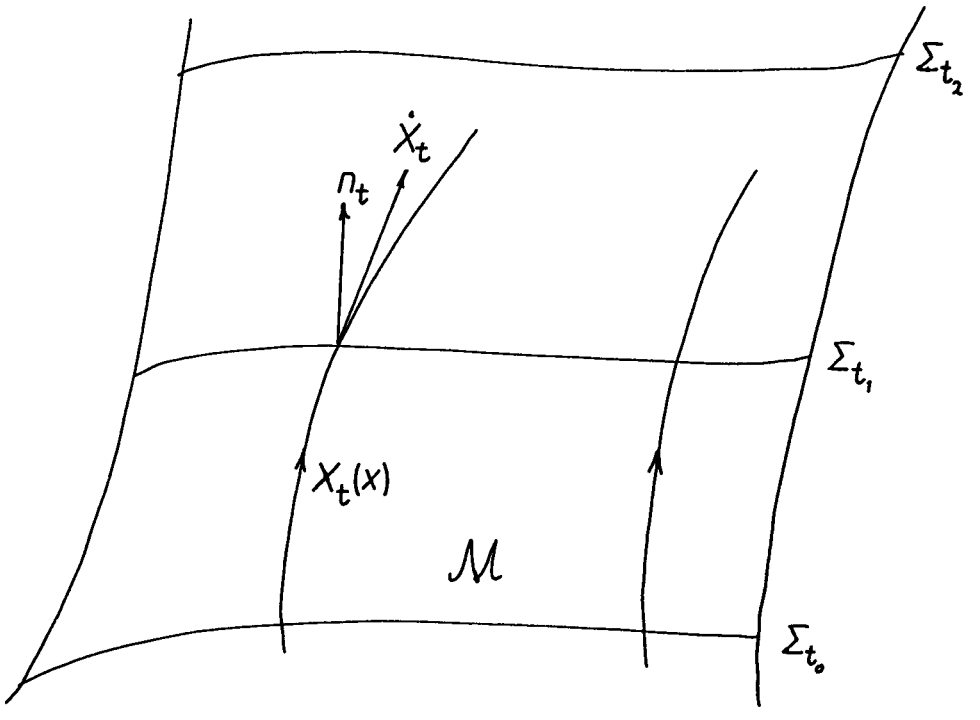


Fig. 4.2. The flow lines of the foliation

4.3.2 The Deformation Vector

The canonical analysis of general relativity proceeds as follows. We fix a foliation $X : \Sigma \times \mathbb{R} \approx \mathcal{M}$ and consider the situation in which $\mathcal{M}$ carries a Lorentzian metric g that satisfies the vacuum Einstein field equations $G_{\mu\nu} = 0$ and is such that each leaf $\Sigma_t := X_t(\Sigma)$ is a submanifold of $\mathcal{M}$ that is spacelike with respect to g. The problem is to find a set of canonical variables for this system and an associated set of first-order differential equations that determine how the variables evolve from one leaf to another.

The map $t \mapsto X_t$ is a curve in $\mathrm{Emb}(\Sigma, \mathcal{M})$ and therefore, for each $t \in \mathbb{R}$, it has a tangent vector, denoted $\dot{X}_t$, which is called the ***deformation*** vector of the foliation. This vector belongs to the tangent space $T_{X_t}\mathrm{Emb}(\Sigma, \mathcal{M})$ which is the set of all tangent vectors to the infinite-dimensional manifold $\mathrm{Emb}(\Sigma, \mathcal{M})$ at the particular embedding X_t.

From another perspective, for each $x \in \Sigma$ the map $t \mapsto X_t(x)$ is a curve in $\mathcal{M}$. As such, it also has a tangent vector, this time belonging to the tangent spaces of $\mathcal{M}$ (Figure 4.2). Note that, for any particular t, the tangent vector associated with the point $x \in \Sigma$ lies in the tangent space $T_{X_t(x)}\mathcal{M}$ at the point $X_t(x) \in \mathcal{M}$.

This is characteristic of elements that are tangent to function spaces, and in general we have

$$T_f C^\infty(\Sigma, \mathcal{M}) = \{v : \Sigma \to T\mathcal{M} \mid v(x) \in T_{f(x)}\mathcal{M}\} \tag{4.3.3}$$

where $f : \Sigma \to \mathcal{M}$. Note that an element v in $T_f C^\infty(\Sigma, \mathcal{M})$ is a "hybrid" object in the sense that to each point $x \in \Sigma$ it associates a vector $v(x)$ in the tangent space to $\mathcal{M}$ at the point $f(x) \in \mathcal{M}$. In component form this vector would be written as $v^\mu(x)$ where $\mu = 0 \ldots 3$ refers to a coordinate system on $\mathcal{M}$.

A function space has cotangent vectors as well as tangent vectors. Specifically, the space of cotangent vectors at the map $f : \Sigma \to \mathcal{M}$ is defined to be

$$T^*_f C^\infty(\Sigma, \mathcal{M}) = \{\ell : \Sigma \to T^*\mathcal{M} \mid \ell(x) \in T^*_{f(x)}\mathcal{M}\}. \tag{4.3.4}$$

In coordinate form, an element $\ell \in T^*_f C^\infty(\Sigma, \mathcal{M})$ associates to each point $x \in \Sigma$ a covariant vector $\ell_\mu(x)$ at the point $f(x) \in \mathcal{M}$.

An important example of an object of this type is the *normal vector* to the foliation at the embedding $X_t \in \mathrm{Emb}(\Sigma, \mathcal{M})$. This is defined to be the unique vector $n_t \in T^*_{X_t}\mathrm{Emb}(\Sigma, \mathcal{M})$ which satisfies the equations

$$n_\mu(x,t) X^\mu{}_{,a}(x,t) = 0 \tag{4.3.5}$$

$$g^{\mu\nu}(X(x,t)) n_\mu(x,t) n_\nu(x,t) = -1 \tag{4.3.6}$$

where $a = 1, 2, 3$ refers to some coordinate system on Σ. Note that the cumbersome expression $(n_t)_\mu(x)$ has been written as $n_\mu(x,t)$. Equation (4.3.5) defines what it means to say that $n_t(x)$ is normal to the hypersurface Σ_t at the point $X_t(x)$ in $\mathcal{M}$. It does not depend on the choice of coordinate system. [17] Equation (4.3.6) is a normalization condition on n and emphasises that this vector is *timelike* with respect to the Lorentzian metric on $\mathcal{M}$.

The normal vector is a functional of both the Lorentzian metric $g_{\mu\nu}$ and the foliation X. This can been seen clearly in the explicit expression

$$n_\mu(x,t) = \left. \frac{T_{,\mu}}{g^{\kappa\lambda} T_{,\kappa} T_{,\lambda}} \right|_{y = X(x,t)} \tag{4.3.7}$$

that relates n to $g_{\mu\nu}$ and the global time function T.

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[17] In a more abstract notation, (4.3.5) is equivalent to the equation $X_t^*(n_t(x)) = 0$ where $X_t^* : T^*_{X_t(x)}\mathcal{M} \to T^*_x\Sigma$ is the usual pull-back on cotangent spaces.

4.3.3 The Lapse and Shift functions

Any vector in $T_{X_t}\mathrm{Emb}(\Sigma, \mathcal{M})$ can be decomposed into two components, one of which lies along the surface Σ_t, and one that is parallel to n_t. In particular, we can write

$$\dot{X}^\mu(x,t) = N(x,t)g^{\mu\nu}(X(x,t))n_\nu(x,t) + N^a(x,t)X^\mu{}_{,a}(x,t) \tag{4.3.8}$$

where N and N^a are known respectively as the *lapse* function and the *shift* vector. We see that $N_t \in C^\infty(\Sigma, \mathcal{M})$ where $N_t(x) := N(t,x)$, and $N_t \in T_{X_t}\mathrm{Emb}(\Sigma, \mathcal{M})$ where $N_t^a(x) := N^a(x,t)$, $a = 1,2,3$.

Note that, like the normal vector, the lapse and shift are functionals of both the spacetime metric $g_{\mu\nu}$ and the foliation. It is important to realise that this relationship can be partially inverted with, for a fixed foliation, the lapse and shift functions being identified as parts of the metric tensor g on $\mathcal{M}$. This can be seen most clearly by studying the "pull-back" of g by the foliation $X : \Sigma \times \mathbb{R} \to \mathcal{M}$. This gives a Lorentzian metric X^*g on $\Sigma \times \mathbb{R}$ whose components are

$$(X^*g)_{\alpha\beta}(x,t) := g_{\mu\nu}(X(x,t))X^\mu{}_{,\alpha}(x,t)X^\nu{}_{,\beta}(x,t) \tag{4.3.9}$$

where g, β=0–3 denotes a coordinate system on $\Sigma \times \mathbb{R}$.

In particular, coordinates can be chosen that are "adapted" to the foliation in the sense that $x^{\alpha=0}(x,t) := t$, and $\alpha = 1,2,3$ refers to some coordinate system on Σ. It can then be shown that (4.3.9) is equivalent to the system of equations

$$(X^*g)_{00}(x,t) = N^a(x,t)N^b(x,t)\,{}_t\gamma_{ab}(x) - \big(N_t(x)\big)^2 \tag{4.3.10}$$

$$(X^*g)_{0a}(x,t) = N^b(x)\,{}_t\gamma_{ab}(x) \tag{4.3.11}$$

where

$${}_t\gamma_{ab}(x) := g_{\mu\nu}(X(x,t))X^\mu{}_{,a}(x,t)X^\nu{}_{,b}(x,t) \tag{4.3.12}$$

is the pull-back by $X_t : \Sigma \to \mathcal{M}$ of the Lorentzian metric g on $\mathcal{M}$ to give a Riemannian metric ${}_t\gamma$ on Σ.

It is clear from (4.3.10-11) that the shift vector and lapse function are essentially the g_{0a} and g_{00} parts of the spacetime metric. It was in terms of this coordinate approach that the original ADM approach was formulated. As we will see shortly, the Einstein field equations do not lead to any dynamical development for these variables that appear as Lagrange multipliers.

4.3.4 The Canonical Form of General Relativity

The variable conjugate to γ.

The Riemannian metric on Σ defined in (4.3.12) plays the role of the canonical configuration variable. Geometrically, we can say that the metric ${}_t\gamma$ measures the "intrinsic" curvature of the spacelike surface Σ_t.

The variable conjugate to ${}_t\gamma$ is closely related to the *extrinsic curvature* of Σ_t which is a measure of how Σ_t looks when seen from the perspective of the Lorentzian metric on $\mathcal{M}$. In coordinate terms, this object is defined as

$$\begin{aligned} K_{ab}(x,t) :&= -n_{\nu;\mu}(x,t)X^{\nu}{}_{,a}(x,t)X^{\mu}{}_{,b} \qquad (4.3.13)\\ &= \frac{1}{2N}\left(-\frac{\partial \gamma_{ab}}{\partial t} + N_{a|b} + N_{b|a}\right)\\ &= \frac{1}{2N}\left(-\frac{\partial \gamma_{ab}}{\partial t} + (L_{N\,{}_t\gamma})_{ab}\right) \qquad (4.3.14)\end{aligned}$$

where $n_{\nu;\mu}$ and $N_{a|b}$ denote covariant differentiation with respect to $g_{\mu\nu}$ on $\mathcal{M}$ and ${}_t\gamma$ on Σ. The symbol $(L_{N\,{}_t\gamma})_{ab}$ means the Lie derivative of the metric ${}_t\gamma$ along the vector field N^a.

The key step in deriving the canonical form of the action principle is to express the pull-back of the Einstein lagrangian density by the foliation $X : \Sigma \times \mathbb{R} \to \mathcal{M}$ as a function of the extrinsic curvature, the metric γ and the lapse and shift functions. The result is

$$\begin{aligned} X^*\big(R(g)(-\det g)^{\frac{1}{2}}\big) &= N(\det\gamma)^{\frac{1}{2}}\big(K_{ab}K^{ab} - (K^a_a)^2 + R^{(3)}(\gamma)\big)\\ &\quad - 2\frac{d}{dt}\big((\det\gamma)^{\frac{1}{2}}K_a{}^a\big) - \big((\det\gamma)^{\frac{1}{2}}(K_a{}^aN^b - \gamma^{ab}N_{,a})\big)_{,b} \qquad (4.3.15)\end{aligned}$$

where $R^{(3)}(\gamma)$ is the curvature scalar of the metric γ on Σ and the t label on γ is implicitly understood. Note that the extrinsic curvature K is to be thought of as the explicit functional (4.3.14) of γ_{ab}, $\dot{\gamma}_{ab}$ and N.

The spatial divergence term vanishes since Σ is assumed to be compact. However, the term with the total time derivative must be removed to produce a genuine action principle. The resulting Einstein lagrangian for matter-free gravity can be written as

$$L(t) = \frac{1}{\kappa^2}\int_{\Sigma} N(\det\gamma)^{\frac{1}{2}}\big(K_{ab}K^{ab} - (K_a{}^a)^2 + R^{(3)}(\gamma)\big)\,d^3x \qquad (4.3.16)$$

where $\kappa^2 := 16\pi G/c^2$.

The canonical analysis now proceeds in the standard way. The variable conjugate to γ_{ab} (with respect to the natural time parameter t associated with the foliation) is

$$\pi^{ab} := \frac{\delta L}{\delta \dot{\gamma}_{ab}} = -\frac{(\det \gamma)^{\frac{1}{2}}}{\kappa^2}(K^{ab} - \gamma^{ab}K_c{}^c) \tag{4.3.17}$$

which can be inverted in the form

$$K^{ab} = -\kappa^2(\det \gamma)^{-\frac{1}{2}}(\pi^{ab} - \frac{1}{2}\gamma^{ab}\pi_c{}^c). \tag{4.3.18}$$

However, the dependence of (4.3.16) on the lapse function and shift vector is such that

$$\pi_N := \frac{\delta L}{\delta \dot{N}} = 0; \quad \pi_{N^a} := \frac{\delta L}{\delta \dot{N}^a} = 0 \tag{4.3.19}$$

which are called the *primary constraints* of the theory. They justify the claim made earlier that the time-behaviour of N and N is not determined by the Einstein equations.

The constraint equations

The Einstein field equations can be usefully separated into the equations $G_{0\mu}(g) = 0$ and $G_{ab} = 0$. Expressed in terms of the canonical variables defined above, the $G_{ab} = 0$ equations are genuine first-order dynamical equations that equate $\dot{\gamma}_{ab}$ and $\dot{\pi}^{ab}$ with specific functions of γ, π, N and N. However, the equations $G_{0\mu} = 0$ are very different. They are equivalent to the equations

$$\mathcal{C}_a(\gamma, \pi) = 0 \tag{4.3.20}$$

$$\mathcal{C}_\perp(\gamma, \pi) = 0 \tag{4.3.21}$$

where

$$\mathcal{C}_a(x) = -\pi_a{}^b{}_{|b}(x) \tag{4.3.22}$$

$$\mathcal{C}_\perp(x) = \frac{\kappa^2}{(\det \gamma(x))^{\frac{1}{2}}}(\gamma_{ab}(x)\gamma_{cd}(x) - \frac{1}{2}\gamma_{ac}(x)\gamma_{bd}(x))\pi^{ac}(x)\pi^{bd}(x) - \frac{(det\gamma(x))^{\frac{1}{2}}}{\kappa^2}R(\gamma(x)) \tag{4.3.23}$$

and are therefore *constraints* on the canonical variables $\gamma_{ab}(x)$ and $\pi^{ab}(x)$.

It is useful to note that the constraints and the true equations of motion can all be derived from the first-order action

$$S[\gamma,\pi,N,N] := \int dt \int_\Sigma d^3x \big(\pi^{ab}\dot{\gamma}_{ab} - NC_\perp(\gamma,\pi) - N^a C_a(\gamma,\pi)\big) \tag{4.3.24}$$

in which γ_{ab}, π^{cd}, N and N are to be varied as independent variables. This shows clearly that N and N are Lagrange multipliers with no genuine dynamical status.

The importance of the constraints

The constraints (4.3.20-21) are of major importance in both the classical and the quantum theories of gravity. In particular, the initial value problem for general relativity is distinctly more complicated than for a normal field theory since the Cauchy data on an initial hypersurface must satisfy the constraints. Of the various results that have been obtained, the following are the most significant for our purposes:

1. If a Lorentzian metric $g_{\mu\nu}$ on a manifold $\mathcal{M} \simeq \Sigma \times \mathbb{R}$ satisfies the vacuum Einstein field equations then, as discussed above, the constraint equations (4.3.20-21) are satisfied on any spacelike hypersurface in $\mathcal{M}$.
2. Conversely, if a Lorentzian metric $g_{\mu\nu}$ on a manifold $\mathcal{M} \simeq \Sigma \times \mathbb{R}$ satisfies the constraint equations (4.3.20-21) on all spacelike hypersurfaces then $g_{\mu\nu}$ necessarily satisfies all ten Einstein field equations $G_{\mu\nu}(g) = 0$.

This last result is very significant since it implies that there is a sense in which the dynamical aspects of the Einstein equations are already contained in the constraints. This plays a crucial role in the quantum theory.

4.3.5 The Dirac Algebra of the Constraints

The classical canonical algebra of the system is expressed by the basic Poisson brackets

$$\{\gamma_{ab}(x), \gamma_{cd}(x')\} = 0 \tag{4.3.25}$$

$$\{\pi^{ab}(x), \pi^{cd}(x')\} = 0 \tag{4.3.26}$$

$$\{\gamma_{ab}(x), \pi^{cd}(x')\} = \delta^c_{(a}\delta^d_{b)}\delta(x,x') \tag{4.3.27}$$

which leads to the fundamental relations (Dirac, 1965)

$$\{\mathcal{C}_a(x), \mathcal{C}_b(x')\} = -\mathcal{C}_b(x)\partial^{x'}_a\delta(x,x') + \mathcal{C}_a(x')\partial^x_b\delta(x,x') \tag{4.3.28}$$

$$\{\mathcal{C}_a(x), \mathcal{C}_\perp(x')\} = \mathcal{C}_\perp(x)\partial^x_a\delta(x,x') \tag{4.3.29}$$

$$\{\mathcal{C}_\perp(x), \mathcal{C}_\perp(x')\} = \gamma^{ab}(x)\mathcal{C}_a(x)\partial^{x'}_b\delta(x,x') - \gamma^{ab}(x')\mathcal{C}_a(x')\partial^x_b\delta(x,x'). \tag{4.3.30}$$

Using the smeared variables

$$C(\eta) := \int_\Sigma C_\perp(x)\eta(x)\,d^3x, \quad C(\xi) := \int_\Sigma C_a(x)\xi^a(x)\,d^3x \tag{4.3.31}$$

where η and ξ are a scalar function and vector field on Σ, these equations can be written as

$$\{C(\xi_1), C(\xi_2)\} = C([\xi_1, \xi_2]) \tag{4.3.32}$$

$$\{C(\xi), C(\eta)\} = C(L_\xi \eta) \tag{4.3.33}$$

$$\{C(\eta_1), C(\eta_2)\} = C(\xi) \tag{4.3.34}$$

where, in (4.3.34), $\xi^a := \gamma^{ab}(\eta_1\eta_{2,b} - \eta_2\eta_{1,b})$.

The geometrical interpretation of these expressions is as follows.

1. The Lie algebra of the diffeomorphism group Diff(Σ) is generated by the vector fields on Σ with minus the commutator of a pair of vector fields

$$[\xi_1, \xi_2]^a := \xi_1^b\xi_{2,b}^a - \xi_2^b\xi_{1,b}^a \tag{4.3.35}$$

   playing the role of the Lie bracket. Thus (4.3.32) shows that the map $\xi \to -C(\xi)$ is a homomorphism of the Lie algebra of Diff(Σ) into the Poisson bracket algebra of the theory. Calculation of the Poisson brackets of $C(\xi)$ with $\gamma_{ab}(x)$ and $\pi^{cd}(x)$ confirms the interpretation of $-C(\xi)$ as a generator of spatial diffeomorphisms. For this reason, $C_a(x)$ are sometimes known as the "transverse" constraints.
2. Similarly, $C_\perp$ can be interpreted as generating deformations of a hypersurface *normal* to itself as it is embedded in $\Sigma \times \mathbb{R} \simeq \mathcal{M}$. Thus $C_\perp(x)$ are known as the "longitudinal" constraints.

Two important things should be noted about the "gauge" algebra (4.3.32-34).

1. It is *not* the Lie algebra of Diff($\mathcal{M}$) even though this was the invariance group of the original theory.
2. The presence of the γ^{ab} factor in the right hand side of (4.3.34) means that is not a genuine Lie algebra at all.

These two features are closely related since the Dirac algebra (4.3.32-34) is essentially the Lie algebra of Diff($\mathcal{M}$) *projected* along, and normal to, a spacelike hypersurface. The significance of this in the quantum theory will emerge later.

4.4 Canonical Quantisation

4.4.1 Reduction to True Canonical Form

We will now consider the quantisation of the system with the first-order action principle (4.3.24) in which the canonical variables $\gamma_{ab}(x)$ and $\pi^{cd}(x)$ are subject to the constraints (4.3.20-21) where $\mathcal{C}_a$ and $\mathcal{C}_\perp$ are given by (4.3.22-23).

The quantisation of any constrained system is a complicated matter and several different approaches can be taken. The two key questions are

1. How are the constraints to be handled?
2. What is the quantum status of Lagrange multipliers like N and N?

One approach is to reduce the theory to a true canonical form (*i.e.*, with no constraints and associated Lagrange multipliers) *before* the theory is quantised. The main steps are as follows:

1. Solve the constraints classically [18] for $4\infty^3$ of the $12\infty^3$ variables $\left(\gamma_{ab}(x), \pi^{cd}(x)\right)$.
2. Impose a suitable gauge. This can be done in several ways but it will always lead to
   a) the elimination of $4\infty^3$ of the $8\infty^3$ variables $\left(\gamma_{ab}(x), \pi^{cd}(x)\right)$ that remain after the first step;
   b) a set of equations that enable N and N to be determined as explicit functions of the remaining canonical variables.
3. Find a canonical action that reproduces the dynamical equations for the remaining $4\infty^3$ "true" canonical variables.
4. Impose canonical commutation relations on these variables and then proceed the usual way.

This programme is not easy to implement and it is unattractive for several reasons. For example,

1. The constraints (4.3.20–21) cannot be solved in a closed form. [19] It is possible to use weak-field perturbative methods but this merely takes us back to the framework we are trying to avoid.

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[18] The notation "$n\infty^3$" means n variables at each spatial point x in the three-manifold Σ.

[19] The use of the Ashtekar variables (§4.5) might change this situation. They lead to powerful new techniques for tackling the constraints at a *quantum* level and perhaps they could also be used effectively in the classical theory.

2. Choosing a gauge breaks the gauge invariance of the theory and, as usual, the task is then to show that the final quantum results are independent of the choice. However, the Diff($\mathcal{M}$) invariance of general relativity is so important that many people prefer a formalism in which a choice of gauge is deferred as long as possible and, in particular, to *after* the quantisation has been performed. Arguably, this makes it easier to discuss the overall significance of the diffeomorphism group in the quantum theory.
3. The programme violates the geometrical structure of general relativity by, for example, removing parts of the metric tensor. This makes it very difficult to explore the significance of spacetime geometry in the quantum theory.

We will return to some of these issues in §5.1 in our discussions of the problem of time in quantum gravity.

4.4.2 Quantisation Before Constraining

The canonical commutation relations

For these reasons, much interest has been devoted to the alternative approach in which all $12\infty^3$ variables $(g_{ab}(x), \pi^{cd}(x))$ are assigned quantum status, and questions of gauge fixing and/or solving the constraints are tackled at the quantum level. The starting point is the operator version of the Poisson algebra (4.3.25-27) in the form of the canonical commutation relations

$$[\widehat{\gamma}_{ab}(x), \widehat{\gamma}_{cd}(x')] = 0 \tag{4.4.1}$$

$$[\widehat{\pi}^{ab}(x), \widehat{\pi}^{cd}(x')] = 0 \tag{4.4.2}$$

$$[\widehat{\gamma}_{ab}(x), \widehat{\pi}^{cd}(x')] = i\hbar\delta^{c}_{(a}\delta^{d}_{b)}\delta(x, x'). \tag{4.4.3}$$

Several things need to be said about this algebra:

1. We are working in the "Schrödinger" representation in which no time dependence is carried by the canonical variables. As we will see later, there is more to this than meets the eye.
2. Like the Poisson brackets (4.3.25-27), these operator relations appear to employ the components of γ and its conjugate π in some coordinate system. However, this dependence is superficial and can be removed by rewriting the expressions in the smeared form

$$[\widehat{\gamma}(h), \widehat{\gamma}(h')] = 0 \tag{4.4.4}$$

$$[\widehat{\pi}(k), \widehat{\pi}(k')] = 0 \tag{4.4.5}$$

$$[\widehat{\gamma}(h), \widehat{\pi}(k)] = i\hbar \int_{\Sigma} h^{ab}(x) k_{ab}(x)\, d^3x \tag{4.4.6}$$

where h and k are a tensor density and a tensor respectively. The right hand side of (4.4.6) is coordinate independent even though I have chosen to express it in coordinate form.

3. Equation (4.4.1) is a form of microcausality. However, since we have made no *specific* choice of a time parameter (this would involve a partial gauge fixing), it is not clear what this means in terms of the usual ideas of an "equal-time" hypersurface, or indeed, how the notion of spacetime structure (as opposed to spatial structure) appears at all. The most that can be said at this stage is that, whatever the final spacetime interpretation will be, (4.4.1) guarantees that the points of Σ can be regarded as being spacelike separated. We will return later to this issue in the context of the problem of 'time' in quantum gravity.

The affine commutation relations

Another problem with (4.4.1-3) is of a more geometrical nature and arises from the fact that the classical object $\gamma_{ab}(x)$ is not merely a symmetric covariant tensor: it is also a *metric* tensor *i.e.*, , at each $x \in \Sigma$ it is an invertible matrix with signature (1,1,1). It is very relevant to ask:

1. Is there a well-defined *operator* analogue of this requirement?
2. If so, do we expect the canonical operator $\widehat{\gamma}_{ab}(x)$ to be such an object?
3. Is such a requirement compatible with the commutation relations (4.4.1-3)?

The first question is easy to answer. A natural analogue of the positive-definiteness of the classical metric is to require that the smeared quantum operator satisfies the condition

$$\widehat{\gamma}(v \otimes v) > 0 \tag{4.4.7}$$

for all non-vanishing vector-density fields v^a (of an appropriate weight) on Σ.

However, whether or not we expect this to be true is a different matter. There have been several suggestions in recent years that some degree of degeneracy should be allowed, *i.e.*, there may exist vector fields v^a that are non-vanishing but for which the smeared operator in (4.4.7) is nevertheless 0. For example:

1. Semi-definite operators of this type arise naturally in certain representations (Isham and Kakas 1984b) of the affine commutation relations (see below).
2. Ashtekar's formulation of general relativity admits degenerate metrics, even at the classical level (Ashtekar 1987).
3. Witten (1988) suggested that there might exist a sector of quantum gravity in which $g_{\mu\nu} = 0$ whilst maintaining Diff($\mathcal{M}$) invariance. This is an extreme case of degeneracy!
4. Horowitz (1990) has looked at a first-order form of general relativity in which the field equations admit a Lorentzian signature metric but with changes in the spatial topology. These solutions involve a degenerate metric.

The answer to the third question is unequivocal: the conventional commutation relations (4.4.4-6) are manifestly incompatible with (4.4.7) or, indeed, with the weaker requirement that the smeared metric operator be only *semi*-definite. This is analogous to the problem of constructing the quantum theory of a particle that is constrained to move on the positive real line $\mathbb{R}_+$. It is easy to show that the conventional commutation relations $[\hat{x}, \hat{\pi}] = i\hbar$ imply that the spectrum of $\hat{x}$ is the *whole* real line, which is inconsistent with the requirement $x > 0$.

In this case, it is well known (Klauder and Aslaksen, 1970) that the answer is to replace the conventional relations with the *affine* algebra defined by

$$[\hat{x}, \hat{p}] = i\hbar\hat{x}. \tag{4.4.8}$$

The appropriate affine algebra for general relativity has the form (Klauder 1970, Pilati 1982, 1983, Isham 1984, Isham and Kakas 1984a,b)

$$[\hat{\gamma}_{ab}(x), \hat{\gamma}_{cd}(x')] = 0 \tag{4.4.9}$$

$$[\hat{p}_a{}^b(x), \hat{p}_c{}^d(x')] = i\hbar\big(\delta_a^d\hat{p}_c{}^b(x) - \delta_c^b\hat{p}_a{}^d(x)\big)\delta(x, x') \tag{4.4.10}$$

$$[\hat{\gamma}_{ab}(x), \hat{p}_c{}^d(x')] = i\hbar\delta^d_{(a}\hat{\gamma}_{b)c}(x)\,\delta(x, x'). \tag{4.4.11}$$

Representations of this algebra can be constructed in which the spectrum of the smeared metric operator is guaranteed to satisfy (4.4.7), at least in a semi-definite form.

4.4.3 The Imposition of the Constraints

Dirac constraint quantisation

The key question now is how the constraint equations (4.3.20-21) are to be handled. The essence of the Dirac approach is to impose them as constraints on the physically allowed states in the form

$$\mathcal{C}_a(\widehat{\gamma},\widehat{\pi})\Psi = 0 \tag{4.4.12}$$

$$\mathcal{C}_\perp(\widehat{\gamma},\widehat{\pi})\Psi = 0. \tag{4.4.13}$$

We recall that, in the classical theory, the constraints are *equivalent* to the dynamical equations in the sense that if they are satisfied on all spatial hypersurfaces of a Lorentzian metric $g_{\mu\nu}$ then $g_{\mu\nu}$ necessarily satisfies the Einstein field equations. This is reflected in the quantum theory by the assumption that the operator constraints (4.4.12-13) are the *sole* technical content of the theory of quantum gravity.

This is closely related to the following fundamental observation. According to the first-order action (4.3.24), the canonical hamiltonian associated with general relativity is

$$H[N,\mathcal{N}] := \int_\Sigma (N(x)\mathcal{C}_\perp(x) + \mathcal{N}^a(x)\mathcal{C}_a(x))\, d^3x \tag{4.4.14}$$

where N and $\mathcal{N}$ are regarded as external c-number functions. However, (4.4.14) has a rather remarkable implication for the putative "Schrödinger equation"

$$i\frac{d}{dt}\Psi_t = \widehat{H}[N,\mathcal{N}]\Psi_t \tag{4.4.15}$$

since if the state Ψ_t satisfies the constraint equations (4.4.12-13), we see that it has no time dependence at all! Similarly it is not meaningful to speak of a "Schrödinger" or a "Heisenberg" picture since the matrix elements between physical states of a Heisenberg-picture field will be the same as for the field in the Schrödinger picture.

This so-called "frozen formalism" caused much confusion when it was first discovered since it seems to imply that nothing happens in a quantum theory of gravity. However, this is now understood to reflect the absence of any intrinsic definition of 'time' in general relativity; indeed, since we have not gauge-fixed the theory, no such coordinate has been selected. It *is* possible to speak of time evolution in quantum gravity, but the subject is subtle (not to say contentious) and is discussed at length in the next chapter.

4.4.4 The Dirac Operator Algebra

Many problems arise when attempting to implement the Dirac scheme, one of the most serious of which is knowing to what extent the classical Poisson bracket algebra (4.3.28-30) can, or should be, maintained in the quantum theory. The principle problems are

1. The constraints (4.3.22) and (4.3.23) are highly non-linear functions of the canonical variables and involve non-polynomial products of field operators evaluated at the same point. Such formal expressions are ill-defined and require significant regularisation and renormalisation. This is the canonical analogue of the ultraviolet divergences which mar the particle physics approach and, as in that case, the problem is very non-trivial.
2. Severe operator-ordering problems arise since the constraints contain products of $\widehat{\gamma}$ and $\widehat{\pi}$. There have been many discussions of whether a consistent ordering can be found, but the problem is complicated by a number of factors. For example:
   a) The problem is not merely one of defining the constraint operators. It is also necessary to decide where to place the $\widehat{\gamma}^{ab}(x)$ operator that appears on the right hand side of the operator version of (4.3.30). One important condition is that we do not wish the commutators of the given first-class constraints to generate *further* constraints on the physical state vectors.
   b) It is natural to assume that the constraint operators $\widehat{C}_{\perp}(x)$ and $\widehat{C}_{a}(x)$ should be hermitian. However, this has been challenged several times (Komar 1979, Kuchar 1986, 1987, Hajicek and Kuchar 1990a,b) and the issue clearly has important implications for the operator ordering problem. The possibility of using a non-hermitian operator can be partly justified by noting that the Hilbert space structure on the space that carries the representation of the canonical (or affine) algebra may be only distantly related to the Hilbert space structure which ought to be imposed on the *physical* states (*i.e.,* those that satisfy the constraints).
   c) The operator products in the constraints are so singular that any suggested ordering is anyway likely to be very ambiguous (Friedmann and Jack 1988).
   d) It is possible that an anomaly *should* be present in the theory as a genuine Planck length effect. But very little is known about, for example, central extensions of the Dirac algebra.

3. As mentioned above, it is very unclear how the inner product on the physical states is to be constructed from the Hilbert space structure on the original space $\mathcal{H}$ that carries the representation of the canonical, or affine, commutation relations. There are various aspects to this question.
   a) The set of physical states $\mathcal{H}_{\mathbf{phys}}$ is unlikely to be simply a subspace of $\mathcal{H}$. This is because, if the constraint operators *are* hermitian, the operator constraints (4.4.12-13) can be read as saying that the physical states are simultaneous eigenstates of $\widehat{\mathcal{C}}_{\perp}(x)$ and $\widehat{\mathcal{C}}_{a}(x)$ with a zero eigenvalue. However, if (as seems likely) the spectra of these operators are *continuous* then the resulting generalised eigenvectors will have an infinite norm and hence will not belong to the Hilbert space $\mathcal{H}$ itself.
   b) If the constraint operators are *not* hermitian the problem becomes even worse since very little can be said about the eigenvectors of an arbitrary non-hermitian operator.
   c) One way of trying to solve this, and related, problems, is to insist that, in some appropriate sense, the results obtained by quantising and then constraining are equal to those in which the constraints are solved before quantising (Kuchar 1986a,b). However, it is arguable whether it is either necessary or desirable that the acts of quantising and constraining *should* commute in this way.
4. From one perspective it is easy to see why the original Diff($\mathcal{M}$) invariance translates into the Dirac algebra in the canonical decomposition. Indeed, it is difficult to see *a priori* how the diffeomorphism group of the spacetime $\mathcal{M}$ could ever be represented by fields that are defined purely on a three-manifold Σ. However, it is natural to wonder if the original group *can* be recovered in some way, and this has been the subject of several different studies.
   a) In the approach of Bergmann and Komar (1972), the group is extended enormously by allowing the generating vector fields to be functionals of the canonical data.
   b) More recently, Isham and Kuchar (1985a,b) suggested a method which involves allowing the embedding functions $X_t : \Sigma \to \mathcal{M}$ to become canonical variables. This leads to a genuine representation of Diff($\mathcal{M}$). We will return to this idea in our discussion of 'time' in quantum gravity.

4.4.5 Representations on Functionals $\Psi[\gamma]$.

The general idea

In attempting to find concrete representations of the canonical algebra (4.4.1-3) it is natural to try an analogue of the scalar field representation (4.2.15-16). Thus the state vectors are functionals $\Psi[\gamma]$ of Riemannian metrics γ on Σ and the canonical operators are defined as

$$(\widehat{\gamma}_{ab}(x)\Psi)[\gamma] := \gamma_{ab}(x)\Psi[\gamma] \tag{4.4.16}$$

$$(\widehat{\pi}^{cd}(x)\Psi)[\gamma] := -i\hbar\frac{\delta\Psi}{\delta\gamma_{cd}(x)}[\gamma]. \tag{4.4.17}$$

These equations have been widely used in the canonical approach to quantum gravity. However, they are deceptive in several respects and it is important to note the following points:

1. As we have remarked already, the canonical algebra (4.4.1–3) is incompatible with the positive (or semi-positive) definiteness of a classical Riemannian metric. In particular, it *not* correct to say that the domain space of the state functionals is Riem(Σ): that is only possible if the affine commutation relations are used.
2. There are precise analogues of the general problems mentioned in connection with the scalar field operators in (4.2.15-16). In particular:
   a) There are no Lebesgue measures on, or associated with, Riem(Σ). Thus a weighted measure is needed with a compensating term added to (4.4.17) as in (4.2.19).
   b) If any measures do exist one would expect them to be concentrated on *distributional* versions of metrics. However, this concept has no obvious meaning since the space Riem(Σ) is *not* a vector space and therefore there is no natural way of defining its dual. One of the great virtues of using the affine commutation relations is that the representation theory of this algebra leads to an appropriate definition of what might be meant by a distributional metric (Isham 1984, Isham and Kakas 1984b).
3. Some of the representations of the affine relations (4.4.9-11) involve state vectors that are functionals of γ or their appropriate distributional analogues. However, several additional possibilities arise that are not present in the representations (4.4.16-17) of the naive commutation relations. For example,

a) There are whole classes of representations in which the state vectors are concentrated on distributional analogues of *degenerate* metrics (Isham and Kakas 1984b).

b) There are representations in which the state vector $\Psi[\gamma]$ has an *index* which acts rather like the internal spin for a relativistic particle.

The $\widehat{\mathcal{C}}(\xi)\Psi = 0$ constraint

Let us now consider the constraints (4.4.12-13) in the context of this representation on functionals $\Psi[\gamma]$. From a physical perspective it is easy to see the need for these constraints. Modulo the remarks above, the domain space of the state functionals is Riem(Σ), and to specify a metric $\gamma_{ab}(x)$ at a point $x \in \Sigma$ requires six numbers (the components of the metric in some coordinate system). However, the true gravitational system should have only *two* degrees of freedom per spatial point (the two directions of circular polarization of a gravitational wave, the two helicities of a massless graviton, *etc*) and therefore four of the six degrees of freedom need to be lost. This is precisely what is achieved by the imposition of the constraints (4.4.12–13). (Of course, the same counting argument applies if the system is reduced to true canonical form before quantising).

The easiest constraints to handle are those in the first set

$$\big(\widehat{\mathcal{C}}(\xi)\Psi\big)[\gamma] = 0. \tag{4.4.18}$$

We saw earlier that the classical functions $\mathcal{C}(\xi)$ are the infinitesimal generators of the diffeomorphism group of Σ, and the same might be expected to apply here. The key question in this context is whether the operator ordering problems can be solved so that the algebra (4.3.32) is preserved at the quantum level. Experience suggests that this is usually fairly straightforward and, indeed, one powerful way of *solving* the operator ordering problem for the $\widehat{\mathcal{C}}(\xi)$ generators is to insist that they form a hermitian representation of the Lie algebra of Diff(Σ).

It is therefore reasonable to expect the quantum theory to carry a unitary representation of Diff(Σ). In this language, the constraint equation (4.4.18) becomes

$$\big(D(f)\Psi\big)[\gamma] = \Psi[\gamma] \tag{4.4.19}$$

where $D(f)$ is the unitary operator representing $f \in \mathrm{Diff}(\Sigma)$.

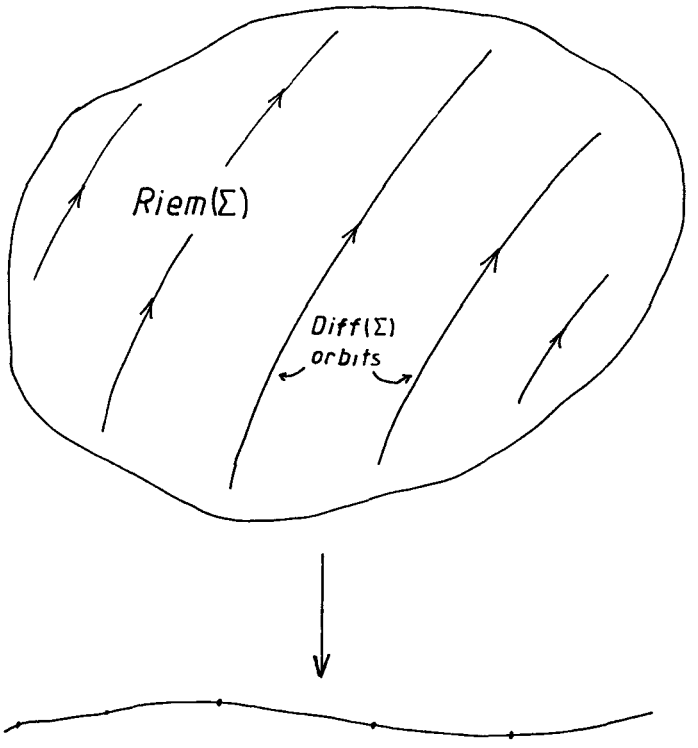


Fig. 4.3. Superspace as the space of orbits of Diff(Σ) on Riem(Σ)

The naive representation (4.4.16-17) suggests that the unitary operator $D(f)$ acts on Ψ as

$$(D(f)\Psi)[\gamma] := \Psi[f^*\gamma] \tag{4.4.20}$$

where $f^*\gamma$ is the usual pull-back of γ by the diffeomorphism $f : \Sigma \to \Sigma$. Then the constraint equation (4.4.19) is equivalent to

$$\Psi[f^*\gamma] = \Psi[\gamma] \tag{4.4.21}$$

for all $f \in \text{Diff}(\Sigma)$ and $\gamma \in \text{Riem}(\Sigma)$.

The implications of (4.4.21) are analogous to those in conventional Yang-Mills gauge theory. The group Diff(Σ) acts as a group of transformations on the space Riem(Σ) of Riemannian metrics on Σ with $f \in \text{Diff}(\Sigma)$ sending $\gamma \in \text{Riem}(\Sigma)$ to $f^*\gamma$. This leads to a bundle [20] picture in which Diff(Σ) is fibered by the orbits of the Diff(Σ) action (Figure 4.2).

[20] This is not quite true since the existence of metrics with isometry groups means that the action of Diff(Σ) is not strictly free. This problem can be solved by, for example, restricting attention to those diffeomorphisms that leave fixed some particular frame at a base point in Σ.

John Wheeler (1964, 1968) called the base space $\mathrm{Riem}(\Sigma)/\mathrm{Diff}(\Sigma)$ of this bundle *superspace* and it has played an important role in canonical quantum gravity. This is because the constraint equation (4.4.21) can be interpreted as saying that the state functional Ψ is constant on the orbits of $\mathrm{Diff}(\Sigma)$ and therefore passes to a function on superspace. Thus, according to Wheeler, it is superspace that is the true domain space of the quantum gravity state vectors. However, as usual, several cautionary remarks are in order.

1. Equation (4.4.18) only really implies that Ψ is constant under *infinitesimal* transformations and therefore it could transform non-trivially under "large" diffeomorphisms that cannot be connected continuously to the identity map. This leads to a gravitational analogue (Friedman and Sorkin 1980, Isham 1981) of the θ states that have been much studied in Yang-Mills theory.
2. The action in (4.4.20) is only unitary if the Hilbert space measure $d\mu[\gamma]$ on the domain space is itself invariant under $\mathrm{Diff}(\Sigma)$. It is most unlikely that any such measures exist. The best that could be hoped for is that the measure is *quasi-invariant* under the action of $\mathrm{Diff}(\Sigma)$ (*i.e.*, the transformed and untransformed measures have the same sets of measure zero), in which case a Jacobian factor is needed on the right-hand side of (4.4.20). This will alter the form of the constraint (4.4.21) and the simple idea of the state functional being a constant on the $\mathrm{Diff}(\Sigma)$ orbits is no longer correct.
3. If the domain space of Ψ is really a space of distributional metrics then the picture changes rather dramatically. The action $\mathrm{Diff}(\Sigma)$ on a space of this type is likely to be very non-free and a bundle picture is not appropriate.

4.4.6 The Wheeler-DeWitt Equation.

The basic equation

We must now consider the final constraint (4.4.13). Unlike the $C(\xi)$ constraint, this has no simple group-theoretic interpretation since, as we have remarked already, the presence of the explicit γ factor on the right hand side of (4.3.30) means that (4.3.28-30) is not a genuine Lie algebra. Thus the operator ordering problem becomes much harder. However, if we choose as a simple example the ordering in which all the π^{cd} variables are placed to the right of the γ_{ab} variables, the constraint (4.4.13) becomes

$$-\hbar^2\kappa^2\mathcal{G}_{abcd}(\gamma)\frac{\delta^2\Psi}{\delta\gamma_{ac}\delta\gamma_{bd}}[\gamma]-\frac{(det\gamma)^{\frac{1}{2}}}{\kappa^2}R(\gamma)\Psi[\gamma]=0 \tag{4.4.22}$$

where

$$\mathcal{G}_{abcd}(\gamma):=(\gamma_{ab}\gamma_{cd}-\frac{1}{2}\gamma_{ac}\gamma_{bd})(\det\gamma)^{-\frac{1}{2}} \tag{4.4.23}$$

is the DeWitt metric on Riem(Σ) (DeWitt 1967a).

Questions and problems

Equation (4.4.22) is the famous Wheeler-DeWitt equation. It is the heart of the canonical theory of quantum gravity from which everything must be extracted. Needless to say, a number of problems and questions arise that need to be considered early on. For example:

1. The ordering chosen in (4.4.22) is a simple one but there is no particular reason why it should be correct. One popular alternative is to write the "kinetic energy" term as a covariant functional Laplacian using the DeWitt metric (4.4.23). This has the advantage (?) that this part of the operator is then invariant under redefinitions of coordinates on Riem(Σ). Of course, a key issue in discussing the ordering of $\widehat{\mathcal{C}}_\perp(x)$ is whether one expects these operators to be *hermitian.*
2. The Wheeler-DeWitt equation contains products of functional differential operators evaluated at the same spatial point and is therefore likely to produce $\delta^3(0)$ singularities when acting on a wide variety of possible state functionals. (*cf* the remarks made in §4.2 concerning a scalar field theory). Thus regularisation will almost certainly be needed.
3. A major question is how to approach the problem of solving the Wheeler-DeWitt equation. One obvious tactic is to deal with it as a functional differential equation *per se.* However, whether or not this is really valid depends on the general interpretation of the constraint equations $\widehat{\mathcal{C}}_\perp(x)\Psi=0$.
   a) If this equation means that Ψ is an eigenvector of the hermitian operators $\widehat{\mathcal{C}}_\perp(x)$ with eigenvalue 0 then, as with standard eigenfunction problems for differential operators, some sort of boundary value conditions need to be imposed on Ψ, and the theory itself is not too informative about what these might be.
   b) In fact, there is a tendency to solve the constraint equation as a functional differential equation without checking that 0 is a genuine eigenvalue. This can

lead to highly misleading results. For example, consider the equation in one-dimensional quantum mechanics

$$(\widehat{p}^2 + \widehat{x}^2 + \lambda\widehat{x}^4)\psi = 0. \tag{4.4.24}$$

Under the substitution $\widehat{p} = -i\hbar d/dx$ this becomes an ordinary differential equation in x that is second order and which therefore has a two-parameter family of solutions. Now suppose the momentum representation is employed with $\widehat{x} = i\hbar d/dp$. Then (4.4.24) becomes a fourth-order differential equation in p and, as such, it has a *four*-parameter family of solutions. But the x and p representations of quantum theory are supposed to be equivalent! So what is going on?

The answer is that 0 is *not* a genuine eigenvalue of the hermitian operator $\widehat{p}^2 + \widehat{x}^2 + \lambda\widehat{x}^4$, and therefore the solutions to the differential equation (4.4.24) have no intrinsic significance. Unfortunately, this simple fact is often overlooked in the quantum gravity literature, especially in the context of minisuperspace (see below) approximations to the theory. Thus to get a true eigenvalue problem it might first be necessary to *renormalise* the Wheeler-DeWitt operator.

4. One of the hardest problems is to decide what the Wheeler-DeWitt equation means in physical terms. In particular, the notions of 'time' and 'time-evolution' must be introduced in some way. The central idea is one we have mentioned several times already: time must be defined as an *internal* property of the gravitational system (plus matter) rather than being identified with some external parameter in the universe. Indeed, the "frozen formalism" interpretation of (4.4.15) is one way of re-emphasing this idea. We will return to this important topic in the next chapter.

Perturbative approaches to the Wheeler-DeWitt equation

Practical considerations dictate the need to develop effective perturbation methods for solving the Wheeler-DeWitt equation, and in a way which preserves as much of the geometrical structure as possible. For example:

1. One could introduce a background metric on Σ and perform a weak-field perturbative expansion. However, this is a very non-geometrical approach and takes us back to the type of framework we are trying to avoid.

2. The Wheeler-DeWitt equation (4.4.22) looks like the sum of a kinetic energy plus a potential energy. This analogy is deficient in several respects but it does suggest a form of perturbation theory in which the coefficient in front of the $R(\gamma)$ term is regarded as the coupling constant. This corresponds to a strong-coupling expansion in powers of $1/G$ and has been the subject of some study.
3. WKB approximations have been used quite widely, especially in the context of the problem of time in quantum gravity. We will discuss this approach in the next chapter.
4. One of the most popular forms of perturbation theory is the so-called "minisuperspace" technique. This involves freezing all but a finite number of the infinite degrees of freedom in Riem(Σ) and quantising the small number that remain. The Wheeler-DeWitt equation becomes a second-order partial differential equation that can be tackled using conventional means. This approach is particularly popular in studies of quantum cosmology since the finite degrees of freedom can be chosen in a way that is adapted to the classical models of cosmology. A simple example will be given shortly.

   However, it should be emphasised that this cannot be regarded as a genuine approximation scheme since there is no way of estimating the effect of the infinite number of degrees of freedom that are dropped. On the other hand, minisuperspace models are very useful for discussing conceptual problems, including the interpretation of the state vector, the problem of time, and related issues.

Other approaches to solving the Wheeler-DeWitt equation

Several techniques have been suggested for finding solutions to the Wheeler-DeWitt equation other than tackling it as a functional differential equation. Some of these involve projecting out a solution $\Psi_{\mathbf{phys}}$ from a "seed" state Ψ. For example, one can try a formal trick (Barvinsky and Ponomariov 1986, Barvinsky 1986) and define

$$\Psi_{\mathbf{phys}} := \int \mathcal{D}[N]\,\mathcal{D}[N^a] \exp\left(i \int_{\mathcal{M}} (N(x)\widehat{\mathcal{C}}_{\perp}(x) + N^a(x)\widehat{\mathcal{C}}_a(x)) d^4x\right) \Psi. \qquad (4.4.25)$$

Note that this procedure does not specifically require the use of state vectors that are functionals of γ.

Another example is the suggestion of Woodward (1989) that a gauge-fixed formalism should be used with

$$\Psi_{\rm phys} := U(N|\mathcal{C})\Psi \tag{4.4.26}$$

where

$$U(N|\mathcal{C}) := \sum_{n=0}^{\infty} \frac{i^n}{n|} \int d^4x_1\, N^{\mu_1}(x_1) \ldots \int d^4x_n\, N^{\mu_n} \mathcal{C}_{\mu_1}(x_1) \ldots \mathcal{C}_{\mu_n}(x_n) \tag{4.4.27}$$

in which $N^0 \equiv N$ and N^a are q-number functions of the canonical variables that depend on the gauge chosen.

The use of functional integrals

A technique that has been much explored in recent years is the use of functional integrals to generate solutions to the Wheeler-DeWitt equation. This is motivated by the well-known fact that, in ordinary quantum mechanics, a solution to the time-dependent Schrödinger equation can be generated in the form

$$\psi(x,t) := \int \mathcal{D}x[s]\, \exp \frac{i}{\hbar} \int_{t_0}^{t} L(s)\, ds \tag{4.4.28}$$

where the path-integral is over all paths that end at the point x at time t. The solution thus obtained to $i\hbar\partial\psi/\partial t = \hat{H}\psi$ depends on the value $x(t_0)$ of the path at the initial time t_0.

The analogue of this for the Wheeler-DeWitt equation was first studied in some detail by Teitelboim (1982, 1983a,b). More recently, it has played an important role in the Hartle-Hawking approach to quantum cosmology in which the starting point is a four-manifold $\mathcal{M}$ whose boundary is the disjoint union of a pair of three-manifolds, one of which is the space Σ. A state function is then constructed in the form

$$\Psi[\gamma] := \int \mathcal{D}[g]\, e^{-S_E[g]} \tag{4.4.29}$$

where $S_E[g]$ is the "Euclideanised" form of the Einstein-Hilbert action and the functional integral is over all Riemannian four-metrics g on $\mathcal{M}$ which (i) induce the metric γ on Σ, and (ii) satisfy some other boundary conditions on the second three-manifold. It can be shown (Halliwell 1988, 1990) that $\Psi[\gamma]$ satisfies the Wheeler-DeWitt equation (at least, formally).

The solution obtained depends on the boundary conditions on the second three-manifold. The Hartle-Hawking (1983) ansatz for quantum cosmology is that the four-manifold $\mathcal{M}$ should have just a *single* boundary Σ with the integral being over all metrics $g_{\mu\nu}$ which induce the metric γ_{ab} on Σ.

4.4.7 A Simple Minisuperspace Model

Further discussion of the Wheeler-DeWitt equation is assisted by having access to a simple minisuperspace model of canonical gravity. We will use a standard cosmological model in which the spacetime geometries are restricted to be one of the usual homogeneous and isotropic Robertson-Walker metrics

$$ds^2 := -N(t)^2 dt^2 + a(t)^2 \omega_{ab} dx^a dx^b \tag{4.4.30}$$

where ω_{ab} is the metric for a three-space of constant curvature k. The case $k = 1$ is associated with Σ being a three-sphere, while $k = 0$ and $k = -1$ correspond to the flat and hyperbolic cases respectively. Note that N^a does not appear in this expression: in effect, we are employing a Diff(Σ) gauge in which $N^a = 0$. This is a common feature of minisuperspace models and does not impose any significant limitations on what we can do.

Spacetimes of this type produce a non-vanishing Einstein tensor $G_{\mu\nu}$ and therefore require a source of matter. Much work in recent times has employed a scalar field ϕ with the Lagrangian (4.2.1), and we will use this here. After some calculation, it can be shown that the coupled equations for the variables a, ϕ and N can all be derived from the first-order action [21] (cf (4.3.24))

$$S[a, \pi_a, \phi, \pi_\phi, N] := \int (\pi_a \dot{a} + \pi_\phi \dot{\phi} - NC) dt \tag{4.4.31}$$

with

$$C := \frac{-\pi_a^2}{24a} - 6ka + \frac{p_\phi^2}{2a^3} + a^3 V(\phi) \tag{4.4.32}$$

and where π_a, a, π_ϕ, ϕ and N are all to be varied independently.

To derive the Wheeler-DeWitt equation for this system we need first to confront some of the problems we have mentioned already. In particular:

1. the classical variable a satisfies the inequality $a \geq 0$;
2. the classical expression $\pi_a^2/24a$ will lead to operator ordering problems in the quantum theory.

The first problem can be tackled in several different ways:

1. Ignore the problem and impose standard commutation relations

[21] For simplicity, I have chosen units in which $16\pi G/c^4 = 1$.

$$[\widehat{a}, \widehat{\pi}_a] = i\hbar \tag{4.4.33}$$

even though we know that this leads to a spectrum for $\widehat{a}$ which is the entire real line. The Hilbert space will be $L^2(\mathbb{R}, da)$ with the operators defined in the usual way as

$$(\widehat{a}\psi)(a) := a\psi(a) \tag{4.4.34}$$

$$(\widehat{\pi}_a\psi)(a) := -i\hbar\frac{d\psi}{da}(a). \tag{4.4.35}$$

The problem in this approach is to give some physical meaning to the negative values of a.

2. Insist on using the Hilbert space $L^2(\mathbb{R}_+, da)$ of functions which are concentrated on $\mathbb{R}_+$ but keep the definitions (4.4.34-35). The conjugate momentum $\widehat{\pi}_a$ is no longer self-adjoint but nevertheless it is possible to arrange for the operator $\widehat{\mathcal{C}}_\perp$ to be a self-adjoint function of $\widehat{a}, \widehat{\pi}_a, \widehat{\pi}_a^\dagger, \widehat{\phi}$ and $\widehat{\pi}_\phi$.
3. Do a canonical transformation at the classical level to a new variable Ω defined by $a = e^\Omega$. This new variable ranges freely over the entire real line and can therefore be quantised as part of a conventional set of commutation relations using the Hilbert space $L^2(\mathbb{R}, d\Omega)$. The conjugate variable is $\pi_\Omega := e^{-\Omega}\pi_a$ and the constraint becomes

$$\mathcal{C} := e^{-3\Omega}\left(\frac{-\pi_\Omega^2}{24} + \frac{\pi_\phi^2}{2}\right) - 6ke^\Omega + e^{3\Omega}V(\phi). \tag{4.4.36}$$

4. Use the affine relations (4.4.8). The affine momentum p_a is related classically to the canonical momentum π_a by $p_a := a\pi_a$ and the constraint becomes

$$\mathcal{C} := \frac{1}{a^3}\left(\frac{-p_a^2}{24} + \frac{\pi_\phi^2}{2}\right) - 6ka + a^3V(\phi). \tag{4.4.37}$$

A self-adjoint representation of the affine commutation relations $[\widehat{a}, \widehat{p}_a] = i\hbar\widehat{a}$ can be defined on the Hilbert space $L^2(\mathbb{R}_+, da/a)$ with

$$(\widehat{a}\psi)(a) := a\psi(a) \tag{4.4.38}$$

$$(\widehat{p}_a\psi)(a) := -i\hbar a\frac{d\psi}{da}(a). \tag{4.4.39}$$

Note that the transformation $a := e^\Omega$ sets up an equivalence between this approach and the previous one.

To get a sample Wheeler-DeWitt equation, let us choose method 3 with the Ω variable satisfying standard commutation relations. Then, ignoring the operator ordering problem, we get

$$\left(\hbar^2 e^{-3\Omega}\left(\frac{1}{24}\frac{\partial^2}{\partial\Omega^2} - \frac{1}{2}\frac{\partial^2}{\partial\phi^2}\right) - 6ke^{\Omega} + e^{3\Omega}V(\phi)\right)\psi(\Omega,\phi) = 0. \tag{4.4.40}$$

This simple model illustrates many of the features of the full canonical theory of general relativity. We will consider the model further in the chapter dealing with the problem of time in quantum gravity. For a recent treatment using path integral techniques see Linden and Perry (1991).

4.5 The Ashtekar Variables

4.5.1 The Basic Ideas

One of the potentially most important ways of trying to solve the Wheeler-DeWitt equation is via the use of the Ashtekar canonical variables. The constraints (4.3.22–23) are much simpler when expressed in terms of these new variables, and a whole range of possibilities is opened up for constructing non-perturbative quantisations of gravity. The Ashtekar programme is complex and many-faceted, and is fully worthy of a course of lectures in its own right. I can only touch on a few of the more relevant features here.

The introduction of triads

The basic canonical variable in the Ashtekar programme is not a metric tensor γ_{ab} on Σ but a *triad*. A triad $e^{(i)}(x)$, $i = 1,2,3$ is a set of three linearly-independent cotangent vectors at the point $x \in \Sigma$ with components $e_a^{(i)}(x)$, $a = 1,2,3$. The metric tensor γ_{ab} can be written in terms of these triad fields as

$$\gamma_{ab}(x) = \sum_{i,j=1}^{3} e_a^{(i)}(x)e_b^{(j)}(x)\delta_{ij} \tag{4.5.1}$$

$$= \mathrm{Tr}\big(e_a(x)e_b(x)\big). \tag{4.5.2}$$

In (4.5.2) the index i on $e_a^{(i)}(x)$ is viewed as belonging to the Lie algebra $\mathcal{L}(SO3)$ of the Lie group $SO3$. Thus $e_a(x) := e_a^{(i)}(x)\lambda_i$ where λ_i, $i = 1,2,3$ are generators of $SO3$ normalised so that $\mathrm{Tr}(\lambda_i\lambda_j) = \delta_{ij}$.

Triads are used extensively whenever fermions are introduced into general relativity and result in an extra gauge group $C^\infty(\Sigma, SO3)$. The action of $\Lambda \in C^\infty(\Sigma, SO3)$ on a triad is

$$e_a^{(i)}(x) \mapsto e_a^{(j)}(x)\Lambda_j{}^i(x) \tag{4.5.3}$$

which leaves invariant the metric tensor γ_{ab}.

In the canonical formalism, each triad has a conjugate variable $\pi^a_{(i)}(x)$ and the analogue of the canonical commutation relations (4.4.1–3) is

$$[\widehat{e}_a^{(i)}(x), \widehat{e}_b^{(j)}(x')] = 0 \tag{4.5.4}$$

$$[\widehat{\pi}^a_{(i)}(x), \widehat{\pi}^b_{(j)}(x')] = 0 \tag{4.5.5}$$

$$[\widehat{e}_a^{(i)}(x), \widehat{\pi}^b_{(j)}(x')] = i\hbar\delta^i_j\delta^b_a\delta(x,x'). \tag{4.5.6}$$

with appropriate affine equivalents. In Dirac constraint quantisation, the presence of the extra gauge invariance is reflected in the existence of the constraints

$$\mathcal{G}_i(\widehat{e}, \widehat{\pi})\Psi = 0 \tag{4.5.7}$$

which supplements the constraints (4.4.12-13).

The Ashtekar-Sen connection

The heart of the Ashtekar programme is a connection which is the triadic version of one first introduced by Sen (1982). This object has the two possible forms (Ashtekar 1986,1987) [22]

$$^{\pm}A_a(x) := \Gamma_a(x) \pm iK_a(x) \tag{4.5.8}$$

where the $\mathcal{L}(SO3)$-valued connection $\Gamma_a(x)$ is the usual Christoffel symbol written in a triadic form, and K_a is the triadic form of the extrinsic curvature (4.3.18) with $K_a := K_a{}^{(i)}\lambda_i$.

One of the first key steps in Ashtekar's programme is his proof that the variables $\tilde{e}^a_{(i)} := e^a_{(i)}(\det\gamma)^{\frac{1}{2}}$ and $^{\pm}A_a$ form a classical canonical pair with the Poisson brackets

$$\{\tilde{e}^a_{(i)}(x), \tilde{e}^b_{(j)}(x')\} = 0 \tag{4.5.9}$$

$$\{^{\pm}A^{(i)}_a(x), {}^{\pm}A^{(j)}_b(x')\} = 0 \tag{4.5.10}$$

$$\{\tilde{e}^a_{(i)}(x), {}^{\pm}A^{(j)}_b(x')\} = \pm i\delta^j_i\delta^a_b\delta(x,x'). \tag{4.5.11}$$

[22] I am being somewhat cavalier here by dropping various dimensional coupling constants and assorted normalisation factors.

The constraints

The constraints of general relativity become dramatically simpler when expressed in terms of these new variables. Specifically,

$$\mathcal{G}_i = {}^{\pm}\mathcal{D}_a \tilde{e}^a_{(i)} \tag{4.5.12}$$

$$\mathcal{C}_a \approx \mathrm{Tr}(\tilde{e}^b {}^{\pm}F_{ba}) \tag{4.5.13}$$

$$\mathcal{C}_\perp \approx \mathrm{Tr}(\tilde{e}^a \tilde{e}^b {}^{\pm}F_{ab}) \tag{4.5.14}$$

where $\approx$ means equal modulo the constraints, ${}^{\pm}\mathcal{D}$ denotes the covariant derivative using ${}^{\pm}A$, and ${}^{\pm}F_{ab}$ are the usual field strengths for the connections ${}^{\pm}A$

$${}^{\pm}F_{ab} := \partial_a {}^{\pm}A_b - \partial_b {}^{\pm}A_a - [{}^{\pm}A_a, {}^{\pm}A_b]. \tag{4.5.15}$$

Various things can be said about (4.5.12-14).

1. The constraints are now polynomial functions of the canonical variables. This is the central feature that sustains the hope that the quantum version of the constraints may be more tractable than in conventional canonical quantum gravity.
2. From a spacetime perspective, ${}^{+}A$ is the restriction to the three-surface Σ of the *self-dual* part of the usual spacetime connection. Similarly, ${}^{-}A$ comes from the anti self-dual part. This has lead to new ways of constructing "half-flat" solutions to the classical Einstein equations (Ashtekar *et al* 1988).
3. I have presented the ideas in a strictly canonical form but there is also a covariant lagrangian formalism whose canonical reduction leads directly to the Ashtekar variables (Jacobson and Smolin 1988b). This has recently been extended in several ways to throw new light on the structure of generally covariant gauge theories (*e.g.*, Capovilla *et al* 1989).
4. The formalism introduces *complex* general relativity in an essential way. There are two options:
   a) e and A are both complex, corresponding to a fully complexified form of general relativity;
   b) e is real and A is complex, corresponding to a hybrid theory in which e and A are not true conjugates. In either case it is necessary to restore the reality of the theory at some stage. This is non-trivial step in the quantum theory.
5. The Ashtekar formalism allows *degenerate* triads/metrics and is hence a genuine extension of conventional general relativity.

4.5.2 Quantisation in the A-Representation

In setting up the quantisation of the theory it is necessary to make a choice of either ${}^{+}A$ or ${}^{-}A$ as the canonical variable. It is not helpful to try to use *both* since the Poisson bracket of ${}^{+}A$ with ${}^{-}A$ does not have any simple form. We will select the ${}^{+}A$ variable and drop the ${}^{+}$label in what follows.

The canonical commutation relations for the pair $(\tilde{e}, A)$ are

$$[\widehat{\tilde{e}}^{a}_{(i)}(x), \widehat{\tilde{e}}^{b}_{(j)}(x')] = 0 \tag{4.5.16}$$

$$[\widehat{A}^{(i)}_{a}(x), \widehat{A}^{(j)}_{b}(x')] = 0 \tag{4.5.17}$$

$$[\widehat{\tilde{e}}^{a}_{(i)}(x), \widehat{A}^{(j)}_{b}(x')] = -\hbar\delta^{j}_{i}\delta^{b}_{a}\delta(x, x'). \tag{4.5.18}$$

Note that, even if we use the hybrid form (so that $\widehat{\tilde{e}}$ is hermitian), the operator $\widehat{A}_a$ will not be hermitian, not even in a formal sense. This means the representation theory of (4.5.16-18) is rather ill-defined.

Nevertheless, proceeding by analogy with ordinary quantum field theory and conventional canonical quantum gravity, one can use the commuting algebra (4.5.17) of the A-variables to study representations on functionals Ψ of A in the form (cf. (4.4.16-17))

$$(\widehat{A}^{(i)}_{a}(x)\Psi)[A] = A^{(i)}_{a}(x)\Psi[A] \tag{4.5.19}$$

$$(\widehat{\tilde{e}}^{a}_{(i)}(x)\Psi)[A] = -\hbar\frac{\delta\Psi}{\delta A^{(i)}_{a}(x)}[A]. \tag{4.5.20}$$

Furthermore, to compensate for the complex nature of A, Ψ is required to be a *holomorphic* functional of A, thus restoring the correct count of degrees of freedom.

Note that the variable which is diagonalised is $\widehat{A}_a(x)$—which is like the canonical momentum in the normal theory—whereas the triad variable acts with a functional derivative. Thus the A-representation is analogous to the functional Fourier transform of the normal representation (4.4.16–17), and hence the geometrical interpretation of $\Psi[A]$ is very different from that of the familiar functional $\Psi[\gamma]$.

Imposition of the constraints

In order to study the Dirac constraint quantisation of this system it is first necessary to select an ordering for the operator versions of the constraints (4.5.12-14). Jacobson and Smolin (1988) chose the expressions

$$\widehat{\mathcal{G}}_i(x) = \mathcal{D}_a \frac{\delta}{\delta A_a^{(i)}(x)} \tag{4.5.21}$$

$$\widehat{\mathcal{C}}_a(x) = F_{ab}^{(i)}(x) \frac{\delta}{\delta A_b^{(i)}(x)} \tag{4.5.22}$$

$$\widehat{\mathcal{C}}_\perp(x) = \epsilon^{ij}{}_k F_{ab}^{(k)}(x) \frac{\delta^2}{\delta A_a^{(i)}(x) \delta A_b^{(j)}(x)} \tag{4.5.23}$$

which have the virtue that the operators $\widehat{\mathcal{G}}_i$ and $\widehat{\mathcal{C}}_a$ correctly generate the Lie algebras of the gauge group $C^\infty(\Sigma, SO3)$ and Diff(Σ) respectively.

The remarkable thing about this form for the constraints is that the Wheeler-DeWitt equation $\widehat{\mathcal{C}}_\perp(x)\Psi = 0$ has a number of *exact* solutions (Jacobson and Smolin 1988a):

1. The functional $\Psi[A] = 1$ satisfies all the constraints. $\Psi = 1$ may not seem very exciting, but it is the first exact solution to the equations of quantum gravity! Of course, any understanding of the physical significance of this result requires a knowledge of the measure on the space of connections A, and no serious study of this has yet been made.
2. A large class of holomorphic gauge-invariant functionals (*i.e.*, invariant under the $C^\infty(\Sigma, SO3)$ gauge group) is given by

   $$H_\eta[A] := \mathrm{Tr}\left(\mathcal{P} \exp \oint_\eta A\right) \tag{4.5.24}$$

   where the path-ordered line integral is over a smooth loop η in Σ.

   Then a formal solution to the Wheeler-DeWitt equation is provided by the product

   $$\Psi_{\{\eta_a\}}[A] := \prod_{a \in I} H_{\eta_a}[A] \tag{4.5.25}$$

   over the finite index set I where $\{\eta_a \mid a \in I\}$ is a set of smooth, non-intersecting loops.
3. Solutions can also be obtained using intersecting curves. They involve particular linear combinations of the states that correspond to the different ways in which such curves can be spliced.

These results are impressive, but there are various problems.

1. The proof that the functionals (4.5.25) satisfy the Wheeler-DeWitt equation exploits only the algebraic properties of F_{ab}, specifically the fact that it is skew-

symmetric in the indices a and b. This suggests that these particular solutions cannot capture the full content of the Wheeler-DeWitt equation.

2. The operator products in (4.5.21-23) are ill-defined and contain factors of $\delta^3(0)$. The problem of how to regularise these operators is a difficult one and is currently the subject of some debate. For example, to define a point-split operator it is necessary to introduce a background metric and curve, and the ultimate effect of this unwanted background structure is not clear.
3. The explicit dependence of the solutions on loops in Σ means they are not Diff(Σ) invariant. It is ironic that, in this formalism, the Wheeler-DeWitt equation can be solved fairly easily but the Diff(Σ) constraints—which are regarded as being relatively innocuous in the conventional approach—should now appear so intractable!

4.5.3 The Rovelli-Smolin Variables

One of the most important recent advances in the use of the Ashtekar variables is the development by Rovelli and Smolin (1988) of a new canonical algebra for general relativity. The basic objects are gauge-invariant functions defined on the phase space of the system by

$$T_\eta[\tilde{e}, A] := \mathrm{Tr}\Big(\mathcal{P}\exp\oint_\eta A\Big) \tag{4.5.26}$$

$$T^a_{\eta,s}[\tilde{e}, A] := \mathrm{Tr}\Big(\mathcal{P}\tilde{e}^a(\eta(s))\exp\oint_\eta A\Big) \tag{4.5.27}$$

where $\eta : S^1 \to \Sigma$ is a smooth loop in Σ and $s \in S^1$ is a particular parameter value. There are analogues for any number of points on the loop. For example, the function with two insertions is

$$T^{a_1 a_1}_{\eta;s_1,s_2}[\tilde{e}, A] := \mathrm{Tr}\big(\tilde{e}^{a_1}(\eta(s_1))\mathcal{P}_\eta(s_1, s_2)\tilde{e}^{a_2}(\eta(s_2))\mathcal{P}_\eta(s_2, s_1)\big) \tag{4.5.28}$$

where $\mathcal{P}_\eta(s_1, s_2)$ is the parallel transport operator from the point $\eta(s_1)$ to the point $\eta(s_2)$ along the loop η.

The first important property of these "T-variables" is that they generate *all* functions on the phase space. The second, and crucial, property is that they have a closed Poisson bracket algebra which can be written somewhat symbolically as

$$\{T^n, T^m\} \sim T^{n+m-1} \tag{4.5.29}$$

where the superscript represents the number of insertions (thus (4.5.26), (4.5.27) and (4.5.28) are T^0, T^1 and T^2 functions respectively). Note that the T^0 and T^1 variables generate a closed subalgebra of the complete T-algebra.

The central idea of the Rovelli-Smolin programme is to use this algebra as the basic algebra whose representations determine the quantum theory. In particular, they construct a type of Fock space quantisation in which the analogue of a "n-particle" state is a function $\psi(\eta_1, \eta_2 \ldots \eta_n)$ of n loops. The T^0 variable acts like a creation operator, for example

$$(T_\eta \psi_2)(\xi) := \psi_2(\eta, \xi), \tag{4.5.30}$$

whereas the T^1 operators map each n-loop sector into itself. By this means Rovelli and Smolin succeed in constructing a deformation (with powers of $\hbar$ as coefficients) of the classical T-algebra.

The next major step is to construct the constraints (4.5.13-14) as a limit of sequences of these T-variables. Rovelli and Smolin then show that the longitudinal constraints $\widehat{\mathcal{C}}_\perp(x)\Psi = 0$ (*i.e.*, the Wheeler-DeWitt equation) can be satisfied provided the Fock-space functions $\psi(\eta_1, \eta_2 \ldots \eta_n)$ are concentrated on smooth, non-intersecting loops.

Of course, these states are not Diff(Σ) invariant because the diffeomorphism group moves the loops around. However, a state that *is* Diff(Σ)-invariant can be found by requiring the n-loop functions $\psi(\eta_1, \eta_2 \ldots \eta_n)$ to be constant on the Diff(Σ) orbits, which are the *link* classes of the manifold Σ. Thus, by these means, Rovelli and Smolin claim to have constructed exact solutions to *all* the constraints of quantum gravity.

This is a most impressive result but, as always, there are various difficulties tucked away in the formalism.

1. The Fock space construction is rather formal. In particular, it is not known what inner product should be placed on the states.
2. The classical T-variables are not real and, correspondingly, there is no reason to expect the quantum operators to be hermitian. This has several drawbacks:
   a) The problem of deciding what inner product to choose is made much harder.
   b) It is difficult to construct any general representation theory of the T-algebra. This means it is also hard to assess the significance of the particular Fock-like representation that has been found.

3. The constraints are singular limits of the T variables and generate the usual unpleasant problems of operator-ordering and regularisation.
4. It is difficult to relate the T-variables to anything in conventional general relativity. This makes it difficult to interpret the quantum results.

Much more could be said about this approach and the best ways of tackling the problems which arise. The latter are certainly non-trivial but, nevertheless, the Rovelli-Smolin algebra is one of the most intriguing ways yet suggested of solving the constraints of canonical quantum gravity.

5. The Problem of Time in Quantum Gravity

5.1 General Remarks

5.1.1 The Major Issues

The question of 'time' is one of the deepest issues that must be faced in a quantum theory of gravity. It is intimately related to a whole variety of interpretational problems, including

1. the status of the notion of *probability* and the extent to which it can be said to be *conserved*;
2. the associated concepts of *causality* and *unitarity*;
3. the extent to which *spacetime* is a meaningful concept;
4. the way in which our "classical" world *emerged* from some primordial quantum event (in a theory of quantum cosmology).

These difficulties stem from the variety of roles which time plays in the technical and conceptual frameworks of conventional physics:

1. Time is "Heraclitean" in the sense that "time is that which allows contradictory things to occur" (Unruh and Wald 1989). For example, a particle cannot be at different positions at the same time, but it can be at the same position at different times.
2. Time is the parameter with respect to which change is measured.
3. In the case of quantum theory we have in addition that

a) the scalar product of states is defined at a single time;

b) complete sets of commuting observables are defined at a single time;

c) measurements are carried out at a single time, and states specify the probabilities of the results of such measurements.

Several important questions arise concerning these properties of time in the case of a quantum theory of gravity.

1. To what extent can time continue to play these roles?
2. In particular, is the concept of time *exact*, or can it only be implemented in some *approximate*, perhaps semi-classical, sense? If so, the notions of probability, unitarity, causality and spacetime will also only be approximate.

   But if 'probability' is not an exact concept does this not mean that quantum theory itself is only approximate? If so, all existing approaches to quantum gravity seem doomed from the outset!
3. In classical general relativity, time is only a local coordinate, and a manifold may need to be covered by more than one coordinate chart. How, if at all, is this reflected in the quantum theory of gravity?

5.1.2 The Role of Background Structure

In conventional quantum physics, 'time' is not an observable in the normal sense since it is not represented by any operator but is instead treated as a background parameter which, as in classical physics, is used to mark the evolution of the system. This is true not only of non-relativistic quantum theory but also of relativistic particle dynamics and quantum field theory. Thus 'time' is arguably part of the classical background which plays such a crucial role in the "Copenhagen" interpretation of the theory.

It is not surprising therefore that, within any particular approach to quantum gravity, the nature of the problem of time depends strongly on the type of background structure that is present. For example, there is no particular difficulty in the particle physics schemes since time can be measured with respect to the background Minkowski metric, thus taking us back to the familiar realm of special relativity and quantum field theory.

Of course, whether or not such a measure of time is *physically* correct is another matter. This is part of the general question of the extent to which the spacetime

concepts of classical general relativity can be properly captured by a weak-field perturbation around a Minkowskian background. For example, the peculiar behaviour of the lightcones (and therefore time) at the event horizon of a black hole is not the easiest thing to reproduce in a graviton-based picture.

Similar remarks apply to extensions of the particle physics approach in which the background is a more general spacetime manifold $\mathcal{M}$ with the gravitational fields—real or virtual—being required to be asymptotically flat. In such a situation it may be possible to define an asymptotic time which can be used to discuss a graviton S-matrix but, once again, it is not clear that this limited notion of time is sufficient to address all the questions we might wish to raise in a quantum theory of gravity.

In the canonical approach to quantum gravity the question of 'time' is very different. The background is now the *three*-dimensional manifold Σ. If Σ is non-compact then, as above, the asymptotic structure might be used to define an absolute time that is linked to the surface terms ignored in the discussions in §4.

However, in the "cosmological" case in which Σ is compact, the technical heart of the theory is the constraints on the $12\infty^3$ canonical variables $\left(\gamma_{ab}(x), \pi^{cd}(x)\right)$ defined on Σ, and the notion of time (and therefore spacetime) has to be *extracted* in some way from these variables, or from any matter fields or particles that may be added to the system. This is the question with which we will be concerned in the rest of this chapter.

The way in which time is related to the canonical variables depends rather on the approach to canonical quantisation that is being adopted.

1. In the scheme in which the system is reduced to true canonical form before it is quantised (§4.4.1), the definition of time is linked to the imposition of a gauge. This can take the form of
   a) an explicit definition of time as some functional of the canonical variables;
   b) a more implicit relation in which a gauge is chosen by requiring some function of the canonical variables to vanish, or
   c) by specifying the lapse function (or its time derivative) to be some specific functional of the canonical variables.
2. In the alternative approach, all $12\infty^3$ variables $\left(\gamma_{ab}(x), \pi^{cd}(x)\right)$ are quantised and the constraints are imposed as restrictions on the state vectors (§4.4.2). This leads

to the Wheeler-DeWitt equation whose interpretation (which we have studiously ignored so far) is deeply related to the meaning of time. Two conflicting views arise at this point:

a) It is possible to extract a concept of time from the Wheeler-DeWitt equation that is exact and which applies at all levels of the theory.

b) The notion of time is only valid in some semi-classical sense which can be extracted from WKB solutions to the Wheeler-DeWitt equation.

5.2 Time as a Classical Variable

5.2.1 The Main Idea

The idea is to employ the approach to canonical quantum gravity discussed in §4.4.1 in which the theory is reduced to a true canonical form before it is quantised. This involves gauge-fixing which, in the present context, means defining a time variable in terms of the gravitational or matter content of the theory.

For example, in a cosmological model, one might like to define time as the functional

$$T[\gamma] := \int_{\Sigma} \left(\det \gamma(x)\right)^{\frac{1}{2}} d^3x \tag{5.2.1}$$

which says that

1. the manifold Σ is always a surface of equal time;
2. the value of this time is equal to the volume of the universe.

Thus, rather than referring to, for example, the simultaneous value of a scalar field ϕ and the volume of the universe at some background time t, one speaks instead of the value of ϕ when the volume of the universe takes on various values.

Such a choice works in a minisuperspace model in which the Diff(Σ) action has already been gauge-fixed (*e.g.*, by choosing $N^a = 0$), but in general we need to remove $1\infty^3$ variables, which involves dropping the requirement that Σ is automatically a hypersurface of equal time. For example, one could define [23]

$$T(x;\gamma] := \left(\det \gamma(x)\right)^{\frac{1}{2}} \tag{5.2.2}$$

[23] The notation $T(x;\gamma]$ means that T is a function of x and a functional of γ.

which says that the value of time at a point $x \in \Sigma$ is equal to $\left(\det\gamma(x)\right)^{\frac{1}{2}}$. This "multi-fingered" time (Kuchar, 1972) is analogous to the Tomonaga multi-time formulation of conventional quantum field theory.

The most general definition of a time function is a functional $T(x;\gamma,\pi]$ on the *phase space* of the system with the interpretation that the value of 'time' at the point $x \in \Sigma$ is equal to $T(x;\gamma,\pi]$. The canonical variables of matter fields can also appear as arguments of T.

5.2.25 Reduction to Canonical Form

The key step is to reduce the constrained system to canonical form. In the Hamiltonian framework we are adopting this involves the following steps.

1. Find the momentum π_T which is conjugate to T so that

$$\{T(x), \pi_T(x')\} = \delta(x, x') \tag{5.2.3}$$

   for each point $x \in \Sigma$.

2. At each $x \in \Sigma$, solve the classical constraint $\mathcal{C}_\perp(x) = 0$ for π_T as a functional of T and the canonical variables. This functional becomes the Hamiltonian density $H_T(x;\gamma,\pi,T]$ for a conventional canonical system. [24]

3. This system can then be quantised in a standard way and leads to the Tomonaga-Schrödinger equation

$$i\hbar\frac{\delta\Psi}{\delta T(x)} = \widehat{H}_T(x)\Psi \tag{5.2.4}$$

One of the real advantages of this approach is the relative absence of any ambiguity about what the final quantum theory is supposed to mean. The state-vectors give the probability distributions of the results of measurements on the variables that remain after the "gauge" variables have been removed. In particular, there should be no difficulty in imposing a genuine Hilbert space structure on the quantum states.

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[24] Strictly speaking, it is first necessary to handle the Diff(Σ) constraints $\mathcal{C}(\xi) = 0$. This can be done by choosing three functionals $S^a(x;\gamma,\pi]$ to serve as coordinates on Σ, or a hybrid scheme can be used in which the spatial constraints are imposed in the form $\widehat{\mathcal{C}}(\xi)\Psi = 0$.

5.2.3 A Minisuperspace Model

Unfortunately, a number of problems arise with this scheme which can be illustrated with the aid of the simple superspace model introduced at the end of §4 (Blyth and Isham 1975). We will choose the cosmological case $k = 1$ where Σ is a three-sphere. Note that there is no explicit x-dependence in this model since, in effect, we have already chosen the Diff(Σ) gauge $N^a(x) = 0$. Thus the question of a multi-fingered time does not arise.

One natural choice for a global "cosmological" time is the radius a of the universe. Thus, according to the discussion above, the reduced Hamiltonian for this system can be found by solving the constraint

$$\frac{-\pi_a^2}{24a} - 6a + \frac{\pi_\phi^2}{2a^3} + a^3 V(\phi) = 0 \tag{5.2.5}$$

for the variable π_a. Hence

$$H_a = \pm\sqrt{24}\big(-6t^2 + \frac{\pi_\phi^2}{2t^2} + t^4 V(\phi)\big)^{\frac{1}{2}}. \tag{5.2.6}$$

Another possible choice is the "conjugate" variable $t := \pi_a$ which (in the simple case $V(\phi) = 0$) leads to the Hamiltonian

$$H_{\pi_a} = \pm\left(\frac{-\frac{1}{12}t^2 \pm (\frac{1}{144}t^4 + 48\pi_\phi^2)^{\frac{1}{2}}}{24}\right)^{\frac{1}{2}}, \tag{5.2.7}$$

while a natural "matter" definition is $t = \phi$ which yields

$$H_\phi = \pm\big(\frac{a^2\pi_a^2}{12} + 12a^4 - 2a^6 V(t)\big)^{\frac{1}{2}}. \tag{5.2.8}$$

A number of general problems can already been seen from these simple examples.

1. The constraint equation is typically an algebraic equation for the momentum π_T conjugate to the time variable, and therefore has more than one solution. In effect, we are factorising the classical constraint in the form $\mathcal{C}_\perp = (\pi_T - \alpha_1)\ldots(\pi_T - \alpha_r)$ and then choosing just one of the roots α_i to be the reduced Hamiltonian. This choice is partly determined by various physical requirements: for example, the reduced Hamiltonian should be a positive, hermitian operator. However, there may still be some residual ambiguity in the procedure.
2. The reduced Hamiltonian can be a very non-linear function of the canonical operators and severe operator-ordering problems occur.

3. The reduced Hamiltonian is time dependent. This has several effects.
   a) A time-dependent Hamiltonian means that energy can be fed into, or taken out of, the quantum system. In a normal physical situation this arises because the system is not closed, and then the time-dependence of the Hamiltonian is determined by the environment. However, a compact three-manifold Σ (the "universe") has no external environment, and so the time-dependence is perhaps a little odd.
   b) If $H(t)$ is time-dependent, the Schrödinger equation

$$i\hbar\frac{d\psi_t}{dt} = \widehat{H}(t)\psi_t \tag{5.2.9}$$

   does not lead to the simple second order equation

$$-\hbar^2\frac{d^2\psi_t}{dt^2} = (\widehat{H}(t))^2\psi_t \tag{5.2.10}$$

   since there is an extra term involving $\frac{d}{dt}\widehat{H}(t)$. In particular, this means that the Schrödinger equations above do *not* imply the Wheeler-DeWitt equation associated with the constraint (5.2.5). This is not necessarily a bad thing but it does show the inequivalence of the different approaches to the canonical quantisation of gravity.
3. There is no natural choice for the time variable and no reason to suppose that different choices will give equivalent quantum theories. This contrasts strongly with the situation in conventional quantum field theory where different global Lorentz frames lead to unitarily equivalent canonical quantum theories.
4. The square roots can be handled as operators if the object whose square root is being taken is hermitian and positive. However, in general this will only be true for a limited range of t (which depends on $V(\phi)$). It might still be possible to take the square-root outside this range but the result will certainly not be hermitian, resulting in a time evolution operator that is non-unitary.

This last problem has been analysed in depth in a series of papers by Hajicek (1986, 1988, 1989, 1990a, 1990b) which explore the general question of the construction of time functions on the phase space of a constrained system including, for example, the existence of solutions to (5.2.3).

The failure of time functions to be global seems to be a generic property of systems of this type but it is not necessarily a disaster. Much work has been done in recent

years on quantum theories of the creation of the universe and, in any such theory, something peculiar must necessarily happen to time near the origination event. For example, in the Hartle-Hawking (1983) and Vilenkin schemes (1988) there is a sense in which time becomes imaginary, and the non-unitarity of the canonical picture may sometimes be an analogue of this effect.

5.3 Time and the Wheeler-DeWitt Equation

5.3.1 Internal Time and the Meaning of $\Psi[\gamma]$

For the reasons above most of the discussion of time in canonical quantum gravity has been within the framework in which all $12\infty^3$ variables $(\gamma_{ab}(x), \pi^{cd}(x))$ are quantised and with the constraints imposed on the state vectors as in (4.4.12-13). In the representation on functionals $\Psi[\gamma]$ this leads to the Wheeler-DeWitt equation , and the problem is to understand the relation of this equation to the concept of time.

An appropriate starting point is to consider the physical meaning of a state-functional Ψ. The operators $\widehat{\gamma}_{ab}(x)$ and $\widehat{\pi}^{cd}(x)$ are defined as in (4.4.16–17) in order that they be *hermitian* with respect to the inner product defined using the "measure" [25] $\mathcal{D}[\gamma]$ on the space $\mathrm{Riem}(\Sigma)$ of all Riemannian metrics on Σ (for our present purposes, the technical problems mentioned in §4.4.5 that are associated with this assumption can be ignored). Similarly, in discussing whether or not the constraint operators $\widehat{\mathcal{C}}_{\perp}(x)$ and $\widehat{\mathcal{C}}_{a}(x)$ are, or should be, hermitian, it is the inner product

$$\langle \Psi_1 | \Psi_2 \rangle := \int_{\mathrm{Riem}(\Sigma)} \Psi_1^*[\gamma] \Psi_2[\gamma] \, \mathcal{D}[\gamma] \tag{5.3.1}$$

on $L^2\big(\mathrm{Riem}(\Sigma), \mathcal{D}[\gamma]\big)$ which is assumed.

In analogy with normal canonical quantum field theory, this suggests an interpretation of Ψ in which, if a measurement is made of the metric-tensor on the three-manifold Σ, [26] the probability of finding the result in a measureable subset B of $\mathrm{Riem}(\Sigma)$ is

$$\mathrm{Prob}(\gamma \in B; \Psi) := \int_B \big|\Psi[\gamma]\big|^2 \, \mathcal{D}[\gamma]. \tag{5.3.2}$$

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[25] The notation $\mathcal{D}[\gamma]$ refers to the non-existing Lebesgue measure on $\mathrm{Riem}(\Sigma)$. I will keep the symbol $d\mu[\gamma]$ for a genuine measure.

[26] Of course, it is not really feasible to measure the gravitational field *everywhere* on Σ, since that is the entire universe! However, (5.3.2) is still meaningful if a trace is taken over the fields on the regions of Σ where no measurements are made. This produces an effective density matrix for the state in the restricted region where measurements *are* made.

If a matter field ϕ is present then the state vector is of the form $\Psi[\gamma, \phi]$, with an analogous interpretation.

This view of $|\Psi[\gamma]|^2$ as a probability density for three-metrics may seem plausible and, indeed, an interpretation of this type has been used a number of times, especially in the context of quantum cosmology. However, as mentioned in §4.4.5, the counting of degrees of freedom is wrong.

A metric tensor γ_{ab} corresponds to $6\infty^3$ degrees of freedom, *i.e.,* six degrees of freedom per spatial point in Σ. Three of these are associated with the action of $\mathrm{Diff}(\Sigma)$ and can be removed by an explicit gauge-fixing or by imposing the constraints $(\widehat{\mathcal{C}}_a(\xi)\Psi)[\gamma] = 0$ as in (4.4.18). In the latter case, the state vector is defined on superspace $\mathrm{Riem}(\Sigma)/\mathrm{Diff}(\Sigma)$, which we will denote by writing $\Psi({}^{(3)}\mathcal{G})$. The inner product (5.3.1) must be replaced by a measure on $\mathrm{Riem}(\Sigma)/\mathrm{Diff}(\Sigma)$ which, following the example of Yang-Mills theory, can be done in a variety of ways. The state-functional $\Psi({}^{(3)}\mathcal{G})$ then has the interpretation of a probability amplitude for three-*geometries*,*i.e.,* equivalence classes of metric tensors with respect to the action of $\mathrm{Diff}(\Sigma)$.

However, the counting is still not correct since a three-geometry ${}^{(3)}\mathcal{G}$ has $6\infty^3 - 3\infty^3 = 3\infty^3$ degrees of freedom whereas the true gravitational system should have a count of $2\infty^3$. The reason for this discrepency can be seen from two perspectives:

1. We still have to impose the constraints $\widehat{\mathcal{C}}_\perp(x)\Psi = 0$, *i.e.,* solve the Wheeler-DeWitt equation. This will remove a further $1\infty^3$ degrees of freedom, leaving $3\infty^3 - 1\infty^3 = 2\infty^3$ as required.
2. From our earlier discussion, we know that the concept of 'time' has to be extracted from the system in some way. In the present context this means that $1\infty^3$ of the $3\infty^3$ degrees of freedom are associated with a multi-fingered time $T(x;\gamma]$ which, once identified, will leave the correct count of $2\infty^3$.

The close relation between these two positions can been seen clearly with the aid of our minisuperspace model. The first view says we must impose the constraint leading to the Wheeler-DeWitt equation (4.4.40). However, the second view says that some (in this case, one) of the configuration variables should be identified with 'time'. For example, if Ω is chosen for this purpose (we recall that $\Omega := \log a$ where a is the radius of the three-sphere) then the Wheeler-DeWitt equation (4.4.40) is to be interpreted

as a partial differential equation which describes how the *physical* mode ϕ evolves as a function of the time variable Ω.

It is now clear what is wrong physically with the inner product (5.3.1) and the associated interpretation (5.3.2). In the minisuperspace model, the analogue of (5.3.1) is the usual inner product on the Hilbert space $L^2(\mathbb{R}^2, d\Omega\, d\phi)$, which means we are integrating over the *time* variable as well as the physical configuration variable ϕ, and this is something that is not done in conventional quantum theory. Thus the Hilbert space structure (5.3.1) is physically inappropriate because it includes an integral over an internal multi-fingered time.

5.3.2 The Analogue of the Klein-Gordon Equation

The moral of all this is clear. The physical Hilbert space structure should not be imposed on arbitrary functionals on $\mathrm{Riem}(\Sigma)/\mathrm{Diff}(\Sigma)$ but only on *solutions* to the Wheeler-DeWitt equation, and in such a way as to make explicit the role of internal time.

The standard procedure at this point is to explore the analogy between the Wheeler-DeWitt equation (4.4.22) and the Klein-Gordon equation of a particle moving in a curved space with an arbitrary, time-dependent potential (DeWitt 1967a). The validity of this analogy can be seen especially clearly in the simple minisuperspace model (4.4.40).

The key observation is that the DeWitt metric (4.4.23) on $\mathrm{Riem}(\Sigma)$ has a hyperbolic character in which, roughly speaking, the conformal modes of the metric play the role of time-like directions, *i.e.*, the transformation $\gamma_{ab}(x) \to F(x)\gamma_{ab}(x)$, $F(x) > 0$, is a "time-like" displacement in $\mathrm{Riem}(\Sigma)$. This suggests defining the formal scalar product

$$\langle \Psi_1 | \Psi_2 \rangle := i \prod_x \int_\Sigma d\Sigma^{ab}(x) \Psi_1^*[\gamma] \left(\mathcal{G}_{abcd}(\gamma(x)) \frac{\overrightarrow{\delta}}{\delta \gamma_{cd}(x)} - \frac{\overleftarrow{\delta}}{\delta \gamma_{cd}(x)} \mathcal{G}_{abcd}(\gamma(x)) \right) \Psi_2[\gamma] \tag{5.3.3}$$

where $d\Sigma^{ab}(x)$ is the directed surface element in $\mathrm{Riem}(\Sigma)$ at the point $x \in \Sigma$.

Of course, considerable care would be needed to make this expression rigorous. For example:

1. it necessary to take account of the $\mathrm{Diff}(\Sigma)$-invariance and then project the inner product down to $\mathrm{Riem}(\Sigma)/\mathrm{Diff}(\Sigma)$;

2. the precise form of the scalar product depends on how the operator-ordering problem in the Wheeler-DeWitt equation is solved.

However, the essential idea is clear. In particular, (5.5.3) has the important property that it is invariant under deformations of the "spatial" hypersurface in Riem(Σ). This is the multi-fingered time analogue of the requirement in the normal Klein-Gordon equation that the scalar product be time independent.

In the minisuperspace example (4.4.40) the Wheeler-DeWitt equation can be simplified by multiplying [27] both sides by $e^{3\Omega}$ to give

$$\left(\hbar^2\left(\frac{1}{24}\frac{\partial^2}{\partial\Omega^2}-\frac{1}{2}\frac{\partial^2}{\partial\phi^2}\right)-6ke^{4\Omega}+e^{6\Omega}V(\phi)\right)\psi(\Omega,\phi)=0. \tag{5.3.4}$$

The associated scalar product is simply

$$\langle\psi_1|\psi_2\rangle = i\int_{\Omega=\text{const}} d\phi\left(\psi_1^*\frac{\partial\psi_2}{\partial\Omega}-\psi_2\frac{\partial\psi_1^*}{\partial\Omega}\right) \tag{5.3.5}$$

which is conserved in Ω-time by virtue of (5.3.4).

Unfortunately, the right hand side of (5.3.3) cannot serve as a genuine Hilbert space inner product because it is not positive definite. There are two aspects to this:

1. The scalar product vanishes for real functions Ψ. This is not a trivial property since the Wheeler-DeWitt equation is a *real* functional differential equation and therefore introduces no intrinsic coupling between the real and imaginary parts of Ψ. This is fundamentally different from the situation with a genuine time-dependent Schrödinger equation where the explicit appearance of the complex number i enforces such a coupling. This has lead some authors to question the applicability of the inner product (5.3.3) in quantum gravity (Barbour and Smolin 1988; Barbour 1990).
2. Complex solutions Ψ to the Wheeler-DeWitt equation can exist for which $\langle\Psi|\Psi\rangle < 0$.

In the normal Klein-Gordon equation the latter problem is solved by dividing the states into those that have positive frequencies and those whose frequencies are negative, and then restricting the physical space to be just the former. This strategy also works in the

[27] This is a contentious step. It is true that a constraint equation $\widehat{C}\Psi = 0$ is not formally affected by multiplying on the left by any invertible operator, but this "renormalisation" of $\widehat{C}$ affects the total constraint algebra, and the implications of this need to be considered at some point. Of course, it also affects the hermiticity properties of the original constraint.

simple minisuperspace model (5.3.5) but, unfortunately, no such division is possible in the general case.

This problem arises already for a relativistic particle moving in a curved spacetime with a metric $g_{\mu\nu}$ and an arbitrary time-dependent potential V. A division into positive and negative frequencies can only be made if $g_{\mu\nu}$ possesses a time-like conformal Killing vector which scales V in an appropriate way (*e.g.*, Kuchar 1981a, 1991). Unfortunately, Kuchar (1981b) showed that Riem(Σ) admits no such vector, and therefore there is no possibility of defining physical states to be some sort of analogue of the positive frequency solutions of the normal Klein-Gordon equation.

5.3.3 Third Quantisation

There have been several different reactions to the failure of the Klein-Gordon approach to the Wheeler-DeWitt equation. In the case of a relativistic particle with an external spacetime-dependent metric and/or potential, the failure to isolate positive-frequency solutions is connected with a breakdown of the one-particle interpretation of the theory. The standard resolution is to second-quantise the system by turning the Klein-Gordon wave function into a quantum field.

It has been suggested several times that a similar process might be needed in quantum gravity with $\Psi[\gamma]$ becoming an operator in some bigger Hilbert space. This procedure is usually called "third" quantisation since the original Wheeler-DeWitt equation is already the result of a quantum field theory (*e.g.*, Coleman 1988, Giddings and Strominger 1988, McGuigan 1988,1989). However, it is unclear what this means, or if the problem of time can really be solved in this way.

1. The approach to third quantisation that is closest to conventional quantum field theory is to construct a Fock space whose "one-particle" sector is associated with the functionals $\Psi[\gamma]$. The problem is how to interpret the states that are tensor products of these basic vectors. In the case of particles, if $|x_1\rangle$ and $|x_2\rangle$ are one-particle states corresponding to a particle localised at points x_1 and x_2 respectively, then the tensor product $|x_1\rangle\,|x_2\rangle$ describes a pair of particles, both of which move in the *same* physical three-space.

   However, if $|\gamma_1\rangle$ and $|\gamma_2\rangle$ are metric states with $\Psi[\gamma]=\langle\gamma|\Psi\rangle$, it is not clear to what the product state $|\gamma_1\rangle\,|\gamma_2\rangle$ refers. The simplest thing might be to say that γ_1 and

γ_2 are both metrics on the same space Σ, but this has no obvious geometrical or physical interpretation. [28]

2. A more meaningful geometrical interpretation is to say that γ_1 and γ_2 are metrics on *different* copies of Σ or, equivalently, a single metric on the disjoint union of two copies of Σ. However this also raises a number of problems.
   a) The transition from a state $|\gamma\rangle$ to a state $|\gamma_1\rangle\,|\gamma_2\rangle$ corresponds to a topology change in which Σ bifurcates into two copies of itself. But this is unlikely to be compatible with the Wheeler-DeWitt equation.
   b) A normal Fock space construction involves Bose statistics but it is not clear physically what it means to say that $|\gamma_1, \gamma_2\rangle$ is symmetric in γ_1 and γ_2. The two copies of Σ are disjoint and presumably no causal connection can be made between them. So what is the operational significance of a Bose structure?
   c) Once the original space Σ has been allowed to bifurcate into a pair of copies of itself it seems logical to extend the topology change to include an *arbitrary* final three-manifold. Thus the scope of the theory is increased enormously.
   d) It is anyway difficult to see how the bifurcation of space helps with the problem of time. Presumably the Wheeler-DeWitt equation will continue to hold in each disconnected piece of the universe, and the problem of internal time reappears in each.

5.3.4 The Use of Extrinsic Time

It could be argued that the problem with the Klein-Gordon approach is that too much weight is being given to the hyperbolic nature of the Wheeler-DeWitt equation as the source of the concept of time. In §5.2 we have already considered the possibility of a classical time function $T(x; \gamma, \pi]$ that is defined on the full *phase* space of the system, and perhaps this could also be of assistance in discussing the operator constraints $\widehat{\mathcal{C}}_\perp(x)\Psi = 0$.

The model of parametrised particle dynamics

The central idea is often illustrated with the aid of a simple model of parametrised dynamics. Consider a particle with an action principle

[28] Private communication with Karel Kuchar.

$$S[q] = \int \mathcal{L}(q, \frac{dq}{dT})\, dT \tag{5.3.6}$$

where T is standard Newtonian time. Now let τ be an arbitrary parameter on the path along which the particle moves, subject only to the condition that $T(\tau)$ is a smooth, monotonically increasing function of τ. Furthermore, append $T(\tau)$ to the set of dynamical variables so that the action principle (5.3.6) becomes

$$S[q,T] = \int \mathcal{L}(q, \frac{1}{\dot{T}}\frac{dq}{d\tau})\dot{T}\, d\tau \tag{5.3.7}$$

where the new time parameter is now τ and $\dot{T} := dT/d\tau$.

This action is invariant under redefinitions of time $\tau \mapsto \tau'(\tau)$ and can therefore serve as a simple model of the diffeomorphism invariance of general relativity. In particular, a full canonical analysis (*e.g.,* Kuchar 1981, Unruh and Wald 1989) shows that the variable π_T conjugate to T satisfies the constraint

$$\mathcal{C} := \pi_T + H(q, \pi_q) = 0. \tag{5.3.8}$$

Furthermore, the entire set of equations of motion, including (5.3.8), can be derived from the first-order action

$$S[q, \pi_q, T, \pi_T, N] = \int (\pi_\alpha \frac{dq^\alpha}{d\tau} - N(\tau)\mathcal{C})\, d\tau \tag{5.3.9}$$

where $N(\tau)$ is a Lagrange multiplier added to enforce the constraint (5.3.8) and the q^α variables are the original configuration variables q in (5.3.6) plus the variable T. The analogy with the gravitational action (4.3.24) is obvious.

The Dirac quantisation of this system is straightforward. If the classical quantity π_T is represented by the operator $-i\hbar\frac{d}{dT}$ then the equation $\widehat{\mathcal{C}}\psi = 0$ becomes

$$i\hbar\frac{d\psi}{dT} = \widehat{H}\psi \tag{5.3.10}$$

which is just the conventional time-dependent Schrödinger equation for the original variables q. This illustrates particularly clearly the problem of trying to relate the Hilbert space structure on the original space—on which all the operators are defined—to the inner product on the physical states.

The canonical transformation

The ideal situation would be if the constraints $\mathcal{C}_\perp(x)$ in (4.3.23) were already in the form (5.3.8), *i.e., linear* in the momentum which is conjugate to the time variable.

Unfortunately, the momentum variables in (4.3.23) have a manifestly quadratic form and so the constraints are not of the desired type. Of course, this is why the Wheeler-DeWitt equation resembles a Klein-Gordon equation.

This means that a "good" time variable (in the sense of (5.3.8) cannot be found merely by restricting the time function T to be a functional $T(x;\gamma]$ of the metric alone. The idea of the "extrinsic" time approach is to extend this to include functionals of π by performing a canonical transformation

$$\big(\gamma_{ab}(x),\pi^{cd}(x)\big) \rightarrow \big(T(x),\pi_T(x);\gamma_A(x),\pi^B(x)\big) \tag{5.3.11}$$

to new variables in which the longitudinal constraint has the form

$$\pi_T(x) + \mathcal{H}(x;T,\gamma_A,\pi^B] = 0 \tag{5.3.12}$$

without the appearance of the difficult square roots, or more general many-valued solutions, which we saw earlier. A Dirac quantisation of this system will lead to the many-fingered time Schrödinger equation

$$i\hbar\frac{\delta\Psi}{\delta T(x)} = \widehat{\mathcal{H}}(x)\Psi. \tag{5.3.13}$$

The York time

This technique works well for linearised gravity (Arnowitt, Deser and Misner 1962) but it is very difficult to find such a canonical transformation in the full theory. The best developed decomposition of variables of this type is the York time in which the choice of time variable is essentially (York 1972)

$$T(x;\gamma\pi] := \frac{2}{3}(\det\gamma(x))^{-\frac{1}{2}}\pi_a{}^a(x) \tag{5.3.14}$$

$$-\pi_T(x;\gamma\pi] := \big(\det\gamma(x)\big)^{\frac{1}{2}} > 0 \tag{5.3.15}$$

and the rest of the canonical variables are unchanged (the full analysis also includes a discussion of the spatial variables).

Defining $\Phi := (-\pi_T)^{\frac{1}{8}} = (\det\gamma)^{\frac{1}{12}}$, the longitudinal constraint becomes

$$\kappa^2\left(\frac{1}{8}\pi^{ab}\pi_{ab}\Phi^{-7} - \frac{3}{64}T^2\Phi^5\right) - \frac{1}{\kappa^2}\left(\frac{1}{8}R(\gamma)\Phi - \nabla_\gamma\Phi\right) = 0 \tag{5.3.16}$$

where ∇_γ is the Laplacian operator constructed from the metric γ. This is a non-linear, elliptic, partial differential equation for Φ which, in principle, can be solved for Φ (and

hence for $\pi_T = -(\det \gamma)^{\frac{1}{2}})$ as a functional $-\mathcal{H}$ of the remaining canonical variables. [29] Thus the longitudinal constraints can be written in the form

$$\pi_T(x) + \mathcal{H}(x; T, \gamma, \pi] = 0 \tag{5.3.17}$$

which leads to a many-fingered time Schrödinger equation. The fact that the transformation to the York time is a genuine canonical map was shown by Isenberg and Marsden (1984).

The main problems with this scheme are as follows.

1. The elliptic PDE in (5.3.16) is very non-linear and hence the relation between the solution Φ and the parameters in the equation is only implicit. It is quite unclear if, or how, such a function can be translated into a quantum operator. In particular:
   a) It is important that the Hamiltonian in (5.3.17) should be a hermitian operator but this requires the solution Φ to be expressed as a hermitian function of the operators $\widehat{\gamma}_{ab}$ and $\widehat{\pi}^{cd}$. The associated operator-ordering problem is likely to be quite horrendous.
   b) It may also be very difficult to ensure that $\widehat{\mathcal{H}}$ is a positive operator.
2. Different classical canonical transformations that reduce the constraint to a quasi-linear form like (5.3.12) are likely to produce inequivalent quantum theories.

5.3.5 Quantum Clocks and Material Reference Systems

We have seen that it is difficult to produce a satisfactory definition of time using only the canonical variables of the gravitational field. However, in practice, time is not measured in this way: real clocks are made of matter with definite properties—an observation that has generated a recent flurry of interest in the idea of "quantum clocks" with the hope that they may lead to a more tractable interpretation of the Wheeler-DeWitt equation.

In a sense, the idea of matter clocks has already been implicit in what has been said so far. For example, one of the dynamical variables in our simple minisuperspace model is a homogeneous mode of a scalar field ϕ, and this can be used to define time. Indeed, (5.2.8) is the Hamiltonian obtained from the choice $t = \phi$ for a classical time

[29] This is true for a compact manifold Σ provided one avoids the maximal slices where $\pi_a{}^a = 0$ and other odd pathologies.

variable. However, a "quantum clock" does not mean an arbitrary ollection of particle or matter field variables, but rather a device whose self-interaction and coupling to the gravitational field are deliberately optimised to serve as a measure of time. The important question is the extent to which the ability to measure (or, more precisely, define) time using one of these systems is compatible with its realisation as a real object; for example, it must have a positive energy.

Two different types of system have been considered. The first is a clock that can only measure time at a single point in space along its worldline. The second is a cloud of clocks that fills the space Σ and which can therefore provide a global measure of time (and position in space too). This latter idea has its origin in the old classical notion of a reference fluid and was first applied to quantum gravity in a major way by DeWitt (1963).

The work of Kuchar *et al*

There have been a number of discussions of quantum clocks and related topics but I will concentrate here on some particularly interesting recent work by Kuchar. The key idea is to use a coordinate condition on the spacetime manifold but to treat it as something that is physically realisable rather than just being part of the background structure. For example, Kuchar and Torre (1990) use a Gaussian reference frame in the following way.

1. The Gaussian coordinate conditions are $g^{00}(y) = -1$, $g^{0a}(y) = 0$ but a more covariant-looking expression can be obtained by introducing a foliation $X : \Sigma \times \mathbb{R} \to \mathcal{M}$ with an inverse $X^{-1} : \mathcal{M} \to \Sigma \times \mathbb{R}$ that, as in (4.3.2), can be written in the form $X^{-1}(y) = (S(y), T(y))$ where $S : \mathcal{M} \to \Sigma$ and $T : \mathcal{M} \to \mathbb{R}$. Then the Gaussian condition on g with respect to this foliation has the form

$$g^{\mu\nu}(y)T_{,\mu}(y)T_{,\nu}(y) = -1 \tag{5.3.18}$$

$$g^{\mu\nu}(y)T_{,\mu}(y)S^{a}{}_{,\nu}(y) = 0 \tag{5.3.19}$$

   where $a = 1, 2, 3$ denotes any coordinate system on Σ (the condition (5.3.19) is independent of the choice).

2. The conditions (5.3.18-19) are now imposed on the system with the aid of Lagrange multipliers M and M_a, $a = 1, 2, 3$. The extra term in the action is

$$S_F[g,M,M] := \int_{\mathcal{M}} \left(-\frac{1}{2}M\left(g^{\mu\nu}T_{,\mu}T_{,\nu}+1\right) + M_a g^{\mu\nu}T_{,\mu}S^a{}_{,\nu}\right)(-\det g)^{\frac{1}{2}}\, d^4y. \tag{5.3.20}$$

3. Although (5.3.20) is covariantly defined, the Diff($\mathcal{M}$) invariance is lost since the functions T and S^a are fixed, external, objects in the theory, and the only dynamical variable is $g_{\mu\nu}$. However, invariance can be restored by changing the status of T and S^a to *bona fide* dynamical variables that must be varied in the action. This is analogous to the way in which reparametrisation invariance is added to (5.3.6) by the introduction of the function $T(\tau)$.
4. These four extra variables T, S^a can be interpreted as describing a material system—the "Gaussian reference fluid"—that interacts with the gravitational field and has its own energy-momentum tensor, which turns out to be that of a heat conducting fluid.
5. A complete canonical analysis shows that the final longitudinal constraint can be written in the form

$$\Pi_A(x) + \mathcal{H}_A(x;T,\gamma,\pi] = 0 \tag{5.3.21}$$

where $\Pi_A(x)$, $A = 0\ldots3$ are the momenta conjugate to the variables $T^A(x)$ (with $T^a \equiv S^a$). Note that the four quantities T^A are now functions on Σ and can be interpreted as describing an *embedding* of Σ in the spacetime manifold $\Sigma \times \mathbb{R}$.

The crucial property of (5.3.21) is that it is *linear* in the Π_A variables. Hence Dirac constraint quantisation gives the first-order functional differential equation

$$i\hbar\frac{\delta\Psi}{\delta T^A(x)}[\gamma,T] = \mathcal{H}(x;T,\gamma,-i\hbar\frac{\delta}{\delta\gamma}]\Psi[\gamma,T] \tag{5.3.22}$$

in the embedding variables T^A. The first-order form of this equation means there is no problem in interpreting $\Psi[\gamma,T]$ as a probability density for finding the metric γ on the hypersurface in $\Sigma \times \mathbb{R}$ specified by the four embedding variables T^A.

What goes wrong?

This work of Kuchar and Torre is very elegant and can be extended in a variety of ways, including a treatment (Kuchar 1991) of the "unimodular-gauge" approach to the problem of time suggested by Henneaux and Teitelboim (1989), Unruh and Wald (1989) and Brown and York (1989).

Equation (5.3.22) has the great virtue of leading to what is probably the most plausible-sounding interpretation so far of the various approaches to understanding

the Wheeler-DeWitt equation. Unfortunately, it also suffers from a significant defect: the energy-momentum tensor of the embedding variables fails to satisfy the famous energy-conditions of general relativity and therefore the system cannot be regarded as physically realistic.

This seems to be an endemic property of all attempts to produce realistic models for physical reference systems. For example, Unruh and Wald (1989) show that any quantum clock whose Hamiltonian is positive must necessarily sometimes "run backwards", and a similar conclusion was obtained by Hartle [30] in his investigations into the problem of time in the context of a path-integral approach to quantum gravity.

Thus it seems that the concept of 'time' may only be an approximate one—a proposition that has been suggested by a number of workers in a variety of different contexts. For example, it forms an important ingredient in Vilenkin's approach to quantum cosmology (Vilenkin 1988, 1989) as well as in Rovelli's recent work on material reference systems (Rovelli 1990, 1991). However, as was emphasised earlier, the thesis that the concept of time is not exact has many implications for the general applicability of conventional quantum ideas to quantum gravity. It is therefore appropriate to discuss next what is one of the most sustained developments of this position: the so-called "WKB" resolution of the problem of time.

5.3.6 The WKB Approach to the Problem of Time

The WKB Solutions of the Wheeler-DeWitt equation

The main idea is that the notion of 'time' is only meaningful in a quantum state that has some semi-classical component which can serve as its measure. Thus

1. time is an *approximate*, semi-classical concept;
2. the definition of time depends on the quantum *state* of the system.

In particular, time would have no meaning in a quantum cosmology in which the universe never emerges into a semi-classical regime. Thus this approach pays some deference to the general idea that time is part of the "classical background" assumed in the Copenhagen interpretation of quantum theory.

The starting point is the WKB technique for obtaining an approximate solution to the Wheeler-DeWitt equation (see Singh and Padmanabhan (1989) for a comprehen-

[30] Hartle's work is comprehensive and wide-ranging and deserves far more than the cursory reference given here. It is only lack of space that stops me from trying to give a proper account of it.

sive review). This involves looking for a solution in the form

$$\Psi[\gamma] = A[\gamma]e^{\frac{i}{\hbar}S[\gamma]} \tag{5.3.23}$$

where $A[\gamma]$ is a "slowly varying" function of γ in the sense that

$$\left|\frac{\delta A}{\delta \gamma_{ab}}\right| \ll \left|A\frac{\delta S}{\delta \gamma_{ab}}\right|. \tag{5.3.24}$$

Inserting (5.3.23-24) into the Wheeler-DeWitt equation shows that the phase S satisfies the Hamilton-Jacobi equation [31]

$$-\kappa^2 \mathcal{G}_{abcd}(\gamma(x))\frac{\delta S[\gamma]}{\delta \gamma_{ac}(x)}\frac{\delta S[\gamma]}{\delta \gamma_{bd}(x)} - \frac{(\det \gamma(x))^{\frac{1}{2}}}{\kappa^2}R(\gamma(x)) = 0 \tag{5.3.25}$$

of classical general relativity. The amplitude factor A satisfies the "conservation law"

$$\mathcal{G}_{abcd}(\gamma(x))\frac{\delta}{\delta \gamma_{ac}(x)}\left(A^2[\gamma]\frac{\delta S[\gamma]}{\delta \gamma_{bd}(x)}\right) = 0. \tag{5.3.26}$$

The concept of time

The recent surge of interest in the application of WKB methods to the problem of time began with an important paper by Banks (1985). A particularly useful discussion in the context of quantum cosmology is Halliwell (1987, 1990). The basic idea is a type of Born-Oppenheimer approach and has been developed further in papers by Zeh (1986, 1988), Brout *et al* (1987,1990), Englert 1989, Kiefer and Singh (1990), and others.

Banks considered a system of matter fields coupled to gravity such that the Wheeler-DeWitt equation for the combined system can be written as (*cf* (4.4.22))

$$-\hbar^2\kappa^2 \mathcal{G}_{abcd}(\gamma)\frac{\delta^2 \Psi}{\delta \gamma_{ac}\delta \gamma_{bd}}[\gamma] - \left(\frac{(det\gamma)^{\frac{1}{2}}}{\kappa^2}R(\gamma) - \kappa^2 \widehat{\mathcal{H}}_m\right)\Psi[\gamma] = 0 \tag{5.3.27}$$

where $\mathcal{H}_m$ is the Hamiltonian density for the matter.

The next step is to find solutions to (5.3.27) of the special form

$$\Psi[\gamma, \phi] = A[\gamma]e^{iS[\gamma]}\Phi[\gamma, \phi] \tag{5.3.28}$$

where A and S are as above and ϕ denotes the collection of matter variables. Inserting this ansatz into the Wheeler-DeWitt equation (5.3.27) and keeping just the lowest order terms in the expansion gives the equation

[31] We are essentially performing an expansion in powers of $\hbar\kappa^2 \simeq L_P$

$$-2i\hbar\mathcal{G}_{abcd}(\gamma(x))\frac{\delta S[\gamma]}{\delta\gamma_{ac}(x)}\frac{\delta\Phi[\gamma,\phi]}{\delta\gamma_{bd}(x)}+\left(\widehat{\mathcal{H}}_m(x)\Phi\right)[\gamma,\phi]=0. \tag{5.3.29}$$

The key step now is to find a functional $T(x;\gamma]$ such that

$$2\mathcal{G}_{abcd}(\gamma(x))\frac{\delta S[\gamma]}{\delta\gamma_{ac}(x)}\frac{\delta T(x';\gamma]}{\delta\gamma_{bd}(x)}=\delta(x,x'). \tag{5.3.30}$$

Then, if Φ depends on γ only through T, (5.3.29) becomes the multi-fingered time Schrödinger equation

$$i\hbar\frac{\delta\Phi}{\delta T(x)}=\widehat{\mathcal{H}}_m(x)\Phi, \tag{5.3.31}$$

which is the desired result.

There are various things to note about this construction.

1. The equation (5.3.30) shows clearly how the definition of the time function $T(x;\gamma]$ depends on the quantum state via the choice of the solution S to the Hamiltonian-Jacobi equation (5.3.25).
2. Only the gravitational field is treated in this semiclassical way. The matter fields are fully quantised, although the probability associated with (5.3.31) is only conserved to the same order in the L_P expansion to which (5.3.31) is valid.
3. Padmanabhan (1990) and Greensite (1990a,b,c) have suggested an extension of the idea above in which time is defined with respect to the phase of *any* solution of the Wheeler-DeWitt equation, not just one that is in WKB form. Their construction can also be regarded as a result of requiring quantum gravity to satisfy the Ehrenfest principle.
4. If S is a real solution to the Hamilton-Jacobi equation (5.3.25), the function $\Psi = Ae^{\frac{i}{\hbar}S}$ oscillates rapidly along paths in superspace. However, imaginary S solutions can also exist, and these produce an exponential type of behaviour. This is often interpreted as indicating a Riemannian rather than Lorentzian spacetime picture and has been much studied in quantum cosmology with the idea that the universe "tunnels" from an imaginary-time region (Halliwell 1987, 1990).

Problems with the WKB approach

The WKB approach to the definition of time is an interesting one, but it raises a number of difficult questions and problems. For example:

1. There is no evidence that the defining equation (5.3.30) has global solutions on superspace. Indeed, Hajicek's analysis of a minisuperspace model suggests otherwise (Hajicek 1986).

2. The simple WKB approximation breaks down at the turning points of S and a more careful treatment is needed near such regions.
3. The elegant first-order form of (5.3.29) is completely lost at the next order in the WKB approximation. This makes it difficult to assess the proper status of (5.3.29) and to understand how the physics of time changes as one gets nearer to the Planck regime.
4. The ansatz (5.3.28) is only one of a very large number of possible types of solution to the Wheeler-DeWitt equation. Why should it have such a preferred status?
5. Similarly, why should one not consider superpositions of WKB solutions to the Wheeler-DeWitt equation? Indeed, the original discussions of WKB solutions to the Wheeler-DeWitt equation assumed that the aim was to construct a coherent superposition of such solutions to produce a quantum state that approximates a single classical spacetime manifold (Gerlach 1969). However, if the state vector is a sum

$$\Psi[\gamma] := \sum_i A_i[\gamma] e^{\frac{i}{\hbar} S_i[\gamma]} \tag{5.3.32}$$

   then each solution S_i of the Hamilton-Jacobi equation will lead to its *own* definition of time! The ensuing difficulty is reminiscent of the Schrödinger cat problem in ordinary quantum theory where a single value for a macroscopic property has to be extracted from a quantum state that is a linear superposition of eigenstates.
6. The alternative is to keep to just a single WKB function, but this also raises several difficulties.
   a) As we remarked earlier in the context of the Klein-Gordon inner product (5.3.13), there is something a little odd about choosing a complex solution Ψ to a functional differential equation that has only real coefficients. As emphasised by Barbour and Smolin (1988) and Barbour (1990) it means that a correlation between the real and imaginary parts of Ψ is imposed in some way. Quantum cosmologists respond that this is precisely what is to be expected in a theory that describes the quantum creation of the universe via the medium of a unique solution to the Wheeler-DeWitt equation. The Vilenkin (1988) scheme does indeed produce a solution that is naturally complex. On the other hand, the Hartle-Hawking ansatz appears to lead to a *real* solution of the equation,

although this issue is clouded by ambiguities in deciding what the ansatz really means.

b) With the aid of Wigner functions and other techniques, Halliwell (1987,1990) interprets a single WKB solution as describing a whole *family* of classical spacetimes. But this again raises the "Schrödinger cat" problem in the guise of having to decide how any specific spacetime arises.

7. If time is only a semi-classical concept then the notion of probability is also likely to be valid only in an approximate sense. This throws doubt on the utility of quantum theory in general.

6. Quantum Cosmology

6.1 General Remarks

The idea of quantum cosmology was introduced by DeWitt in his fundamental work on the canonical quantisation of gravity (DeWitt 1967a). The subject was developed in a series of papers by Misner (*e.g.*, Misner 1969a,b,c) and generated considerable interest until around the middle of the 1970s. Activity then waned for a while but picked up strongly following the work of Hartle and Hawking on constructing a unique wave function for the universe (Hawking 1982b, Hartle and Hawking 1983), and the parallel approaches by Vilenkin (1982)—summarised in Vilenkin (1988)— and Linde (1984, 1989).

The techniques adopted in the second phase of activity (1982 to the present) differ in several respects from those used in the first phase (1967-1975) although there is much in common, especially in regard to the conceptual problems that arise. However, it is important to appreciate that the goals of the present programme are somewhat different from those of the earlier period. The original workers in quantum cosmology were mainly concerned with understanding the effects of quantum theory on the spacetime singularity that appears in the description of the big-bang given by classical general relativity. Thus one of the primary aims was to replace the deficient classical description with a picture that is physically and mathematically meaningful.

This remains one of the goals of the modern programme but it is supplemented with a desire to produce a theory that predicts a *unique* quantum state of the universe. This is far more ambitious than merely providing a believable picture of the big-bang. It is also potentially open to observational refutation or confirmation.

The subject of quantum cosmology generates a number of conceptual issues over and above those that are present in quantum gravity in general. I will only give a broad sketch of the subject here and refer to the literature for details on the assorted attempts to unravel the various strands of thought. An excellent recent review article is Halliwell (1990a) who has also performed a great service by compiling a very comprehensive bibliography of papers on quantum cosmology (Halliwell 1990b).

6.2 Problems With the Copenhagen Interpretation of Quantum Theory

Any attempt to apply quantum theory to the universe as a whole is bound to encounter severe problems with the conventional "Copenhagen" interpretation of quantum theory. Of course, this itself has been the subject of many lively debates in the past and it is still a topic that generates considerable interest. The problems that arise in the interpretation of quantum theory can usefully be categorised under four headings:

1. the role of measurement;
2. the collapse of the state vector;
3. the meaning of 'probability' as applied in quantum theory ;
4. quantum entanglement (the Einstein-Podolsky-Rosen paradox, Bell's inequalities, *etc*).

These topics are closely interlinked and can be visualised as the four vertices of a tetrahegon, each vertex being closely tied to the remaining three. They raise particularly difficult questions in the context of quantum cosmology, as we can see by considering first the meaning of 'probability'.

6.2.1 The Meaning of Probability

The concept of probability is used in several different ways in theoretical physics, and each poses peculiar problems for quantum cosmology. Let us consider them in turn.

The statistical interpretation

For a physicist, a natural approach to probability is the "statistical" interpretation in terms of relative frequencies of repeated experiments. But this raises obvious difficulties for quantum cosmology. For example, if such a theory predicts that, with probability 0.87, I will be typing these notes on January 27, 1991, what does the 0.87 mean? And is this prediction to be viewed as a great success for the theory? The statistical interpretation says merely that if the experiment is repeated many times, then in 87% of the cases I will indeed be found typing the notes. But how is the experiment to be repeated?

Opinions are sharply divided about the meaning of such a prediction. We are faced here with the old problem of how a probabilistic statement is to be understood if it is intended to apply to a single event. It might perhaps be possible to employ a subjective interpretation involving a "degree of belief" but, in practice, the usual tendency is towards some implementation of the idea of relative frequencies. In the case of quantum cosmology, one school of thought maintains that there really is a large ensemble of universes to which the usual statistical interpretation can apply. Various origins have been suggested for such an ensemble. For example:

1. The universe may continually expand and contract giving rise to an endless series of big-bangs and big-crunches. Each cycle is to be thought of as a member of the statistical ensemble.
2. The initial big-bang may have been such that many parts of the physical universe emerged in a causally disconnected way (although still part of a single, spatially-connected universe). The probabilistic statements of quantum cosmology then apply to the statistical distributions of properties in these separate regions.
3. The physical universe may be spatially *disconnected*, as a result perhaps of some quantum topology operation near the big-bang. The statistical predictions could then be applied to these separate "universes".

Note that one common feature of all these suggestions is the extreme difficulty of *testing* them!

The epistemic interpretation

In classical physics, the most common meaning attached to probability is an epistemic one. Thus it is assumed that probabilities measure the extent to which an (idealised) observer *knows* the precise details of the system. Our ignorance of these details arises

from the great complexity of the system (for example, a box of gas) but the underlying assumption is that the uncertainty in the actual behaviour can be made arbitrarily small—at least in principle—with the aid of more precise measurements (or a more detailed theoretical description) of the *possessed* properties of the constituents of the system.

However, unless hidden variables are posited (and these bring problems of their own), the situation in quantum theory is very different. It is generally no longer consistent to talk of individual entities 'possessing' a value for an observable: it is more as if the property concerned is *latent* in the system until such time as an observation "brings it into being". As John Bell has put it, "beables" are replaced by "observables", and the verb "to be" becomes "to be measured".

Needless to say, the situation is even worse in quantum cosmology. For who is the cosmic observer whose "knowledge" is reflected in the wave function of the universe?

Probability for a single system

It is clear that what is needed is an interpretation of probability that is meaningful for a single event and which therefore has a chance of being applied to the universe in its entirety.

One possibility is to stipulate that the only predictions which have any real meaning are those that affirm something with probability one (or zero). This is consistent with the observation that the only time in ordinary quantum theory when it is unequivocably meaningful to say that an individual system *possesses* a value for a quantity is if a measurement of the corresponding observable is guaranteed to yield that value with probability one. However, this is the exception rather than the rule, and to stick to just probability-one statements is to ignore most of the theory.

On the other hand, there are some intriguing "meta-theorems" that purport to show that if the actual universe is sufficiently complex to admit many physical copies of some system, then one of the probability-one statements is that measurements made on this *physical* ensemble will reproduce the statistical distributions of conventional quantum physics (Finkelstein 1963, Hartle 1968, Geroch 1984). This is comforting in its way but it does little to help in understanding the content of quantum cosmology as it applies to the universe as a whole.

6.2.2 The Role of Measurements

One of the problems with the conventional interpretation of quantum theory is the instrumentalist approach that it takes to the relation between theory and reality. In particular, it posits a radical dualism between a system and an observer, the role of the latter being two-fold:

1. the observer *prepares* states of the quantum system;
2. he/she/it then makes *measurements* of some observable pertaining to the system.

The emphasis on measurements (*i.e.*, observables rather than beables) is due mainly to the perceived lack of any quantum-theoretical analogue of the microstates of classical statistical physics. The gap which is thereby placed between the system being observed, and that which is making the observation, contrasts sharply with the situation in classical physics where all aspects of the physical world are deemed to be describable by the same set of physical laws. It poses the obvious problems of (i) when is an interaction between two systems to count as a measurement by one of a property of the other?, and (ii) what happens if an attempt is made to restore a degree of unity by describing the measurement process in quantum mechanical terms? There is no universally accepted answer to either of these questions.

In the context of quantum cosmology such a dualistic picture is particularly problematical. For example:

1. One again we encounter the question of who, or what, is the cosmic observer?
2. The act of measurement is often associated with a *reduction* of the state vector into an eigenvector of the observable measured. This poses no problems for a frequentist interpretation of probability (one simply selects the subsensemble on which the particular value has been obtained) but it raises real difficulties for a more realist, "single-system" interpretation such as we might like to employ in quantum cosmology. There are two quite different reactions to this situation.
   a) The first is to say that the state vector never collapses in quantum cosmology. Indeed, it is difficult to see how such an operation can be consistent with the Wheeler-DeWitt equation. This means that the notion in conventional quantum theory of state vector collapse is deemed to be merely a phenomenological way of describing what really happens in terms of an idealised separation into a system plus an observer. The no-collapse position is almost always associated

with some version of the infamous "many-worlds" interpretation of quantum theory.

b) The more radical view is to say that not only does the state-vector collapse, but this is actually *caused* by the effects of gravity. This is the non-linear picture of quantum theory espoused by Penrose (1986, 1987) to which we referred earlier (see also Karolyhazy *et al* 1986).

3. In conventional quantum mechanics, measurements are made at a fixed time. How is this idea to be reconciled with the many difficulties that arise in quantum gravity when trying to develop an exact concept of time (§5)?

6.3 Where Does a Unique Quantum State Come From?

There are several approaches that might be adopted in an attempt to construct a *unique* quantum state for the universe. For example:

1. Perhaps the post Planck-length quantum state is essentially *independent* of what happened before: a situation that is typical of the so-called "chaotic cosmologies" (Linde 1989). It is difficult to believe that this can be true at the level of fine detail, but it might work for some of the more coarse-grained features of the cosmos.
2. The requirements of *mathematical consistency* may be so restrictive that they single out just one quantum state. Such a view is implicit in DeWitt's seminal discussion of the canonical quantum gravity programme (DeWitt 1967a).
3. It may be possible to select a unique solution to the Wheeler-DeWitt equation by imposing some natural conditions on the values of $\Psi[\gamma]$ at the *boundary* of superspace $\mathrm{Riem}(\Sigma)/\mathrm{Diff}(\Sigma)$. For example, DeWitt's suggestion was that Ψ should be forced to vanish there. Note that what is hoped for is that a relatively *simple* condition on the boundary of $\mathrm{Riem}(\Sigma)/\mathrm{Diff}(\Sigma)$ may lead to a state vector that can describe the very *complex* world we see around us today.

   Vilenkin's scheme is of this type (Vilenkin 1988, 1989). He starts by dividing the boundary of superspace into a "singular part"—which corresponds to the limit of three-geometries that are the slices of a spacetime with a singularity, and which converge towards that singularity—and a regular part, which corresponds to the limit of three-geometries that are the slices of a four-manifold with no singular limit (for example, the equitorial slices of a four-sphere that tend towards a pole

on the sphere). Vilenkin requires that the solution to the Wheeler-DeWitt equation must be such that, at the singular boundary, it carries flux *out* of superspace. This suggestion has many attractive features. However, there are several difficulties:

a) The flux is measured using the Klein-Gordon inner product (5.3.3). In the present context, this gives a positive probability density only in the WKB approximation. Thus the prescription as stated may only work at a semi-classical level. A more inclusive version is not known.

b) In a few simple minisuperspace models the prescription does give a unique wave function. However, it is by no means clear that this would apply in the complete theory or, indeed, even in a superspace model with a larger number of degrees of freedom.

4. There may be a simple/natural *ansatz* for constructing a specific solution to the Wheeler-DeWitt equation. An example is the Hartle-Hawking scheme which we discussed briefly in the context of the Euclidean functional integral (4.4.29). Several points should be made about this idea.

   a) It is difficult to give the Hartle-Hawking ansatz a precise mathematical meaning although it is certainly very elegant in a rather formal sense. In practice, it tends to be used in a semi-classical way with the functional integral (4.4.29) being approximated by the value of $e^{-S_E[g]}$ at a classical solution g to the Riemannian field equations subject to the Hartle-Hartle boundary conditions.

   b) More detailed calculations suggest that the value of Ψ calculated using (4.4.29) may depend on the *contour* in the complexified Riem($\mathcal{M}$)-space that must be chosen to make the integral convergent (Halliwell and Hartle 1990).

   c) These problems might be overcome to some extent if the Hartle-Hawking ansatz could be rewritten in terms of boundary conditions on superspace. Unfortunately, this has not yet been possible in any definitive way.

In assessing any of these schemes it should not be forgotten that none of the traditional approaches to tackling the Wheeler-DeWitt equation can surmount the fierce operator divergences that arise. The recent developments in solving the Wheeler-DeWitt equation using Ashtekar's new variables are therefore of particular interest and could have important implications for the future development of quantum cosmology.

6.4 How Does the Classical Universe Emerge?

One of the major problems for any QCU theory is to explain how the world we see around us "emerged" from the primordial quantum soup of the creation region. In particular, the theory needs to explain why it is that (provided we do not probe too deeply) the universe has such a strikingly classical structure.

A unique state Ψ produced by a QCU theory needs to satisfy a number of requirements that culminate in the classical world we see around us.

1. A definition of 'time' must be provided (perhaps using the state Ψ in some way) so as to give meaning to the idea of the "beginning" of the universe and the creation of space and/or time.
2. The initial pure state must have properties that make it look like a mixed, thermal state shortly after the big-bang.
3. If the theory is to agree with modern ideas on early universe physics the state must lead to a period of inflationary expansion. This could happen in two different ways:
   a) An inflationary expansion could be a direct prediction of the QCU theory which generated the state Ψ.
   b) It could happen in a more "phenomenological" way if the QCU theory is blended into one of the, now-standard, ways of producing a quantum-induced inflationary phase.
4. When interpreted in terms of an appropriate definition of 'time', the state vector must have the property that the system that it now describes evolves according to an essentially classical set of dynamical equations.

It is a highly non-trivial technical task to produce a QCU theory that can satisfy these requirements. However, there is also the crucial conceptual problem of understanding what is really *meant* when we talk about "the" classical universe.

It is certainly possible in principle to construct wave-packet solutions $\Psi[\gamma]$ to the Wheeler-DeWitt equation that are "peaked" around a single spacetime in the sense that the support of Ψ is primarily on the set of all spacelike slices of a single Lorentzian metric on a manifold $\mathcal{M} \simeq \Sigma \times \mathbb{R}$. However, the QCU schemes that have been invented so far (*e.g.*, Hartle-Hawking or Vilenkin) produce states whose WKB approximation is a more like a single e^{iS}, or a finite sum of such, rather than a true wave packet.

And, as was mentioned in §5, state-vectors of this type seem to correspond to a whole *family* of classical spacetimes.

This is the problem of Schrödinger's cat writ large: "How can a single macroscopic world emerge from a state vector that looks like a superposition of states corresponding to different macroscopic possibilities?" This is often translated into the question of how a pure state can turn into what is, in effect, a mixed state so that the interference between the different "branches" is lost.

This problem has been extensively discussed in the context of the "measurement problem" of conventional quantum theory, and the view that has been repeatedly presented in one form or another is that the phase interference between the different branches of the superposition is essentially wiped out by the effects of the systems's environment. In recent years this idea has been developed to particulary good effect by Zurek (1981, 1982, 1991), and spirited attempts have been made to develop an analogous concept in the context of quantum cosmology.

The problem of course is that the universe does not have an environment, and therefore any "decoherence" that takes place must be understood purely in terms of the internal structure of the theory; in effect, some of the variables must serve to decohere the rest (*e.g.,* Halliwell 1989, Gell-Mann and Hartle 1990, Hartle 1990). A number of ideas have been developed in the cited references and the whole subject is most intriguing. However, one cannot help but feel a little uneasy about what is being done.

Firstly, there is still the question of what 'probability' is supposed to mean, even if the replacement of the pure state by a mixed state means that the probability under consideration is of a more classical type.

Secondly, decoherence is only an approximate concept. For example, in minisuperspace models that have been studied, the gravitational modes of the system are described using the WKB approximation. This may work in a purely phenomenological sense but it leaves a big question mark about what the state vector Ψ "really" means, especially in the deep quantum regime near the creation event itself.

Thus—as is so often the case in quantum gravity—we seem to end where we began, with the unequivocal statement:

Physics at the Planck length is something of a mystery!

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